Peace or Push? Understanding Global Patterns of Refugee Return

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Abstract

When do refugees return? A core premise of the refugee regime is that refugees remain abroad while home is unsafe and return as conditions improve. Yet using a novel global dataset, I show that most recorded returns occur during ongoing conflict, while many refugees remain abroad after wars end. I argue that existing research can miss these patterns because it focuses primarily on individual-level preferences measured through return intentions, while realized return is concentrated in episodes often shaped by population-level shocks. In particular, coercive policies and conflict in asylum can force large refugee populations to leave even while home remains unsafe, while UNHCR repatriation assistance can enable return by financing transport and rehabilitation. I introduce the Global Refugee Return Dataset, covering 293 origin–asylum dyads from 1980 to 2023, which improves measurement of realized return and provides new data on exclusion policies and UNHCR repatriation assistance. Within dyads, asylum-side shocks and repatriation assistance are associated with substantially higher return rates than in other years, while globally they coincide with far more return than war endings, largely because many more refugees are exposed to them.

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1 Introduction

When do refugees return? The expectation in both the international refugee regime and existing research is that refugees return once the conflict or persecution that caused their displacement has ended. The regime is designed to protect refugees in asylum for as long as they would face persecution or serious harm upon return. Its legal cornerstone is the principle of *non-refoulement*, which prohibits states from returning refugees under those conditions.[1] UNHCR treats voluntary repatriation as its preferred "durable solution."[2] Studies of return intentions likewise find that refugees become more willing to return as security in their country of origin improves (e.g., Alrababah et al., 2023; Beaman, Onder and Onder, 2022; Adema et al., 2025; Al Husein and Wagner, 2023).

Global historical patterns of refugee return, however, do not align with these expectations. The novel Global Refugee Return (GRR) dataset, covering 293 origin–asylum dyads from 1980 to 2023, reveals two striking patterns. First, more than three-quarters of recorded returns occurred while conflict was ongoing in the country of origin. Second, refugees often do not return when there is peace: only 8 of 24 wars that ended and remained at peace for five years saw even half the refugee population return within that period. Refugees therefore frequently return when prevailing accounts predict that they should remain abroad, while many remain displaced when those accounts predict that they should return.

Why do refugees return during war, yet often remain in asylum after peace? To answer these questions, I argue that we have to look beyond refugees' individual or household-level preferences over origin and asylum to the population-level conditions that determine whether those preferences can be acted upon. Most quantitative research examines return intentions within particular contexts, showing that willingness to return depends especially on security at home relative to conditions in asylum (e.g., Alrababah et al., 2023; Al Husein and Wagner, 2023; Adema et al., 2025). This helps explain where refugees would prefer to live, but not necessarily when large refugee populations actually move.

I show that realized return is concentrated around two population-level shifts that map onto constraints on refugees' choices: whether they are able to remain in asylum and whether they have the resources required to return. First, a large share of return occurs amid asylum-side push, particularly exclusion policies designed to force refugees out and conflict in the country of asylum. Both can make staying in asylum untenable

[1] The principle is codified in Article 33(1) of the 1951 Convention Relating to the Status of Refugees and its 1967 Protocol. It is also recognized as a rule of customary international law and therefore binds states regardless of whether they are parties to the Convention.

[2] The other two durable solutions are local integration and third-country resettlement. UNHCR has treated voluntary repatriation as its preferred durable solution since the mid-1980s (Barnett and Finnemore, 2004; Loescher, 2001).

even when conditions at home remain dangerous. 47.4% of recorded return occurs in years with either asylum shock, compared with 26.9% in the three years after war ends. These pressures are common because most refugees live in low- and middle-income countries, often in neighboring states where they may face new conflicts or governments that restrict their ability to remain (UNHCR, 2025).

Second, peace may fail to produce return not only because refugees prefer to remain in asylum, as existing research emphasizes, but also because those who want to return may lack the resources to do so. After years of restricted employment or camp residence, refugees may be unable to finance transportation and rebuild their lives at home. UNHCR repatriation programs can relax this constraint by providing transportation and reintegration assistance, but whether such assistance is available depends not only on conditions of displacement, but also on funding from donor countries and decisions made by international bureaucrats. Post-war return is heavily concentrated in years when this assistance is available: 85.4% of returns following war endings occur in years with UNHCR repatriation funding.

I evaluate this argument using novel cross-national data and historical comparisons. First, within-dyad models show that both asylum-side push and repatriation assistance are strongly associated with return. Severe exclusion policies are associated with nearly three times as much return, and low-intensity conflict in asylum with more than twice as much; these relationships are similar whether or not conflict continues at home. A UNHCR repatriation programme is associated with two times as much return, while a one-standard-deviation increase in funding per refugee is associated with a 36% increase. Post-war return is also substantially greater when repatriation funding is available. Because UNHCR assistance is endogenously allocated, I interpret these results as robust associations consistent with the proposed resource mechanism rather than as causal effects.

Second, the aggregate patterns show why these factors matter globally. Return is highly concentrated, consistent with population-wide shocks: the largest 5% of dyad-years contain approximately 94% of all return, and the median dyad experiencing return records half of its lifetime return in a single year. Asylum shocks coincide with substantially more return than war endings, largely because they affect many more refugees: they occur during 27.9% of refugee stock-years at risk, compared with just 3.4% for war-ending periods. Post-war return, meanwhile, is heavily concentrated in years with UNHCR repatriation assistance, and that assistance often arrives years after conflict ends. Thus, global realized return depends not only on how strongly different conditions are associated with return, but on how many refugees are exposed to them.

Existing explanations nevertheless remain important. War endings are associated with 2.71 times the return rate, and refugees are less likely to return to more repressive origin countries or from wealthier asylum countries. Other origin-country conditions, including income and the broader intensity of conflict, are less consistently associated with return. Thus, improving conditions at home matter, but they cannot by themselves explain when large refugee populations actually move.

I trace these mechanisms through two historical cases. Rwandan refugees returned at different times across neighboring asylum countries despite facing the same conditions at home, following conflict in the DRC and separate expulsion orders in Burundi and Tanzania. Eritrean refugees in Sudan, by contrast, were unable to return after independence because disputes between UNHCR and the Eritrean government delayed the repatriation program required to fund the travel and rehabilitation of returnees.

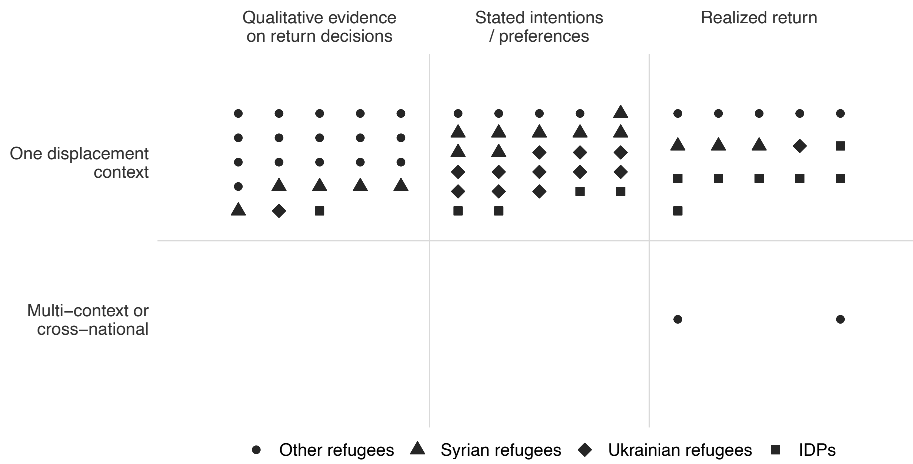
This research makes three contributions. First, despite the centrality of repatriation to the international refugee regime, macro-level patterns of return remain understudied. Existing quantitative work has taught us a great deal about where specific refugee groups would prefer to live, but much less about when displacement actually ends. I provide the first global analysis of realized refugee return using improved historical data and bring into quantitative study two factors long discussed in historical and legal scholarship but rarely examined systematically: coercive repatriation and the role of UNHCR in facilitating return.

Second, the paper makes a major data contribution. The principal existing UNHCR data systematically undercount coerced returns and therefore mischaracterize when and why refugees return. With 36 research assistants, I constructed the GRR dataset by digitizing and coding 7,526 dyad-years of archival country reports compiled through field visits, interviews, and media reporting. GRR enables systematic analysis by providing both a more accurate measure of return itself and new measures of the conditions that may explain it, including exclusion policies and UNHCR repatriation financing. It also records broader *de facto* asylum conditions, complementing *de jure* datasets such as the Dataset of World Refugee and Asylum Policies (DWRAP) (Blair, Grossman and Weinstein, 2022).

Third, the paper connects research on refugees and international security with IPE scholarship on international regimes. Relative to regimes governing trade, finance, the environment, and human rights, the international refugee system has received comparatively little systematic attention in quantitative IPE research.[3] The data and results here enable an assessment of compliance with its core legal principle of *non-refoulement*, revealing systematic violations in some cases and continued protection in others. This variation raises a question central to research on international regimes but rarely examined systematically in the refugee context: why are core rules upheld in some cases but violated at scale in others, and when do international organizations constrain rather than accommodate those violations (Chayes and Chayes, 1993; Simmons,

[3]Important policy, IR, historical, and legal work on the refugee regime includes Loescher (2001), Barnett (2001*a*), Chimni (2000), Skran (1995), Betts (2009), Betts (2010), Abdelaaty (2021), Arar (2017), and Davutoglu (2025).

Figure 1: Scope and outcomes in empirical research on refugee return decisions



Notes: Each point represents an empirical study of refugee return identified through a systematic search of nineteen political science, migration, and development journals from 2006–2026, supplemented by citation tracing, the 2024 World Bank–UNHCR Joint Data Center review, and targeted searches. A displacement context may include multiple origin populations in one host country or one origin population across multiple host countries; only studies spanning multiple contexts are classified as cross-national. See the Appendix for the full list of studies.

2000; Mitchell, 1994; Stone, 2011)?

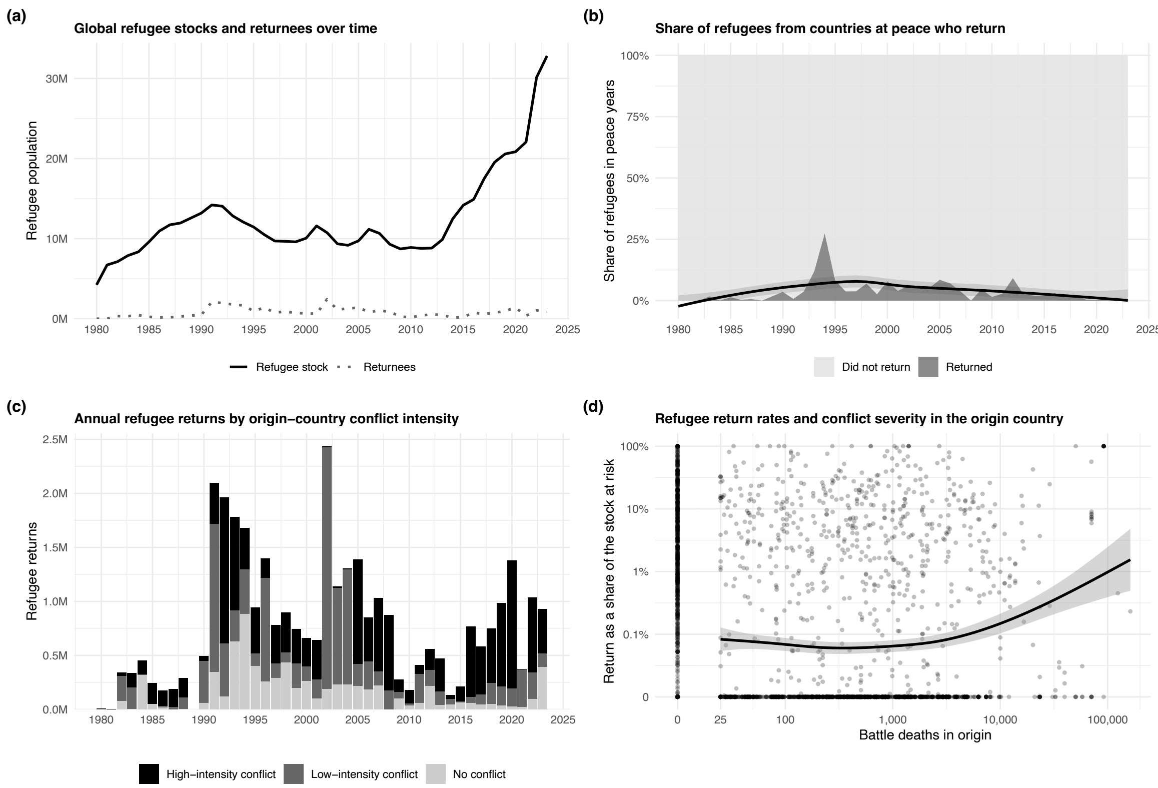
The article first reviews existing research and documents the prevalence of return during conflict. I then develop the argument, introduce the GRR dataset and its advantages over UNHCR data, and test the theory using descriptive evidence, within-dyad models, and a global decomposition, before tracing the mechanisms through Rwandan and Eritrean cases.

2 Puzzling Patterns of Refugee Return

Figure 1 shows that existing research on refugee return has focused primarily on stated return intentions within particular displacement contexts, especially among Syrian and Ukrainian refugees. Drawing on household models of migration choice (Borjas, 1987), this work generally asks whether refugees want to return once it is safe to do so, reflecting security concerns and attachment to their country of origin, or whether economic opportunities and living conditions in asylum make remaining abroad more attractive.

The consistent finding of this research is that origin security conditions are the primary determinant of return intentions (Alrababah et al., 2023; Beaman, Onder and Onder, 2022; Adema et al., 2025). The “threshold model” developed by Alrababah et al. (2023), for example, argues that refugees consider asylum-country economic factors only once a minimum level of safety at home has been reached: Syrian refugees in Lebanon were unlikely to consider returning while Syria remained unsafe, even when they faced poor living conditions in asylum. Political conditions and attach-

Figure 2: Return during conflict and non-return during peace



Notes: (a) Global refugee stock (solid) and recorded returns (dotted). (b) Share returning, among refugees who fled war, in dyad-years when the origin is at peace. (c) Annual returns by origin-country conflict intensity (UCDP: ≥ 25 and $\geq 1{,}000$ battle deaths). (d) Return rate against log battle deaths (dyad-year). Lines are LOESS smoothers; 1989 is omitted. *Sources*: GRR; UCDP/PRIO.

ment to home also shape return intentions (Kaya and Orchard, 2020; Ghosn et al., 2021). By contrast, evidence on asylum-country conditions is mixed: some studies find they shape return intentions, others find little relationship, and Beaman, Onder and Onder (2022) find that better conditions in exile are associated with greater realized return (Kayaoglu, Şahin-Mencütek and Erdoğan, 2022; Müller-Funk and Fransen, 2023; Alrababah et al., 2023).

This literature yields two expectations for realized return. First, return should be more common when the origin country is at peace. Second, large-scale return should be rare while intense conflict continues. To examine whether realized return follows these expectations globally, I use the Global Refugee Return (GRR) Dataset, which records annual refugee stocks and return flows for 293 origin–asylum dyads from 1980 to 2023 and is described in Section 4.2.

Figure 2 shows a striking pattern: most refugees return to war, not peace. Panel (a) first shows that return flows are generally small relative to the size of displaced populations. Across dyad-years with more than 1,000 refugees, the median return rate is zero and the mean is 2.99%; in 89% of dyad-years, fewer than 5% of refugees return. Panel (b) shows that peace rarely produces return. Among refugee populations displaced by war, the average annual return rate is only 3.03% when there is no conflict in the country of origin.[4] Only 8 of 24 wars that ended and remained at peace for five years saw even half the refugee population return within that period.

Panel (c) shows that most return occurs during war, not after it. Between 1980 and 2023, more than three-quarters of all recorded returns took place while conflict was ongoing in the country of origin, including 17 of the 20 largest episodes. Roughly two-fifths occurred during high-intensity conflict, more than one-third during low-intensity conflict, and only one-fifth during peace. Some of this pattern reflects exposure, since the largest refugee populations disproportionately come from countries experiencing protracted conflict.[5] But exposure does not resolve the puzzle: it explains why most refugees at risk come from countries at war, not why they return while war continues. Panel (d), which scales returns by the refugee stock at risk, reinforces this point, showing that returns do not decline as battle deaths increase.

3 Theory

Why do refugees often return during ongoing conflict, yet remain in asylum even after peace? To understand global patterns, we need to look beyond refugees' hypothetical preferences over where they would ideally live to the constraints that determine whether those preferences can be acted upon. Refugees who prefer to remain abroad may nevertheless return when conditions in asylum make staying infeasible, while those who want to return may lack the resources needed to move. These constraints are difficult to study through within-context surveys of individual intentions, which have limited leverage on the population-wide shocks that drive major return episodes.[6]

3.1 Constraints on Remaining in Asylum

Two types of shocks can make remaining in asylum infeasible: host-government coercion and armed conflict in the country of asylum. First, host governments may

[4] This calculation is restricted to origin–asylum dyads in which refugees generally fled war rather than broader non-conflict persecution.

[5] Dyad-years with conflict in the origin constitute 34.1% of observations but contain 72.5% of refugee stock-years at risk and account for 79.0% of recorded returns. Thus, conflict's share of return exceeds its share of the population at risk by only 6.5 percentage points. Calculations use 5,109 dyad-years with more than 1,000 refugees at risk, excluding 1989, when the *World Refugee Survey* was not published.

[6] The main cross-national exception, Zakirova and Buzurukov (2021), examines realized return, but similarly emphasizes the political and economic conditions shaping the relative attractiveness of origin and asylum.

pressure refugees to leave before conditions at home have improved. Despite the prohibition on forced return under the 1951 Refugee Convention,[7] states may circumvent or ignore *non-refoulement* (Schwartz, 2025; Chimni, 2004; Harrell-Bond, 1989; Long, 2013; Stein and Cuny, 1994). Deportation is the most severe and obvious violation. But governments may also revoke residence permits, restrict access to work and services, withdraw assistance, close camps, or threaten detention and expulsion. These measures can make remaining increasingly difficult while the resulting return is still recorded as "voluntary."[8] Consider the treatment of Afghan refugees in Iran in 2005:

> "UNHCR [complained] that Iran was pressuring [Afghan refugees] to return by suspending education and medical services and revoking their residence permits. UNHCR also accused Iran of levying a punitive tax on refugees to force them to repatriate. Afghan media reports accused Iranian police of beating refugees and demanding bribes to extend residence permits."
>
> *2006, Iran*

Second, armed conflict in the country of asylum can make remaining there more dangerous than returning home. Most refugees remain near their countries of origin, often in neighboring states that share similar structural risk factors for conflict and are vulnerable to the cross-border spread of war (UNHCR, 2025; Fearon and Laitin, 2003; Collier and Hoeffler, 2004; Gleditsch, 2007; Lischer, 2005). Refugees may therefore escape one conflict only to encounter another in asylum. Violence can directly harm refugees or disrupt the humanitarian assistance on which they depend. Under these conditions, refugees may return even while their country of origin remains at war. The 1996 return of Rwandan refugees from the DRC[9] illustrates this mechanism:

> "In October, an outbreak of civil war in Zaire [DRC], which engulfed Rwandan refugee camps there, produced what UNHCR described as the 'largest and swiftest' repatriation in memory. About a half-million Rwandan refugees returned home during a four-day period in mid-November. Most returnees repatriated on foot, jamming the main Rwandan highway from the border. The sea of humanity stretched nearly 100 miles during the height of the return."
>
> *1997, Zaire [DRC]*

[7] *Non-refoulement* is generally recognized as binding all states as customary international law, while states parties to the 1951 Convention or its 1967 Protocol are additionally bound by their treaty obligations.

[8] UNHCR may facilitate repatriation under these conditions, both to reduce the harm refugees would face if expelled without assistance and because it depends on host-government cooperation to maintain access to refugee populations (Barnett, 2001*b*,*c*; Barnett and Finnemore, 2004; Whitaker, 2003). Its participation does not necessarily establish that refugees would have returned in the absence of government pressure.

[9] The country was officially known as Zaire in 1996 and was renamed the Democratic Republic of the Congo in May 1997. For consistency with current usage, I refer to it as the DRC in the text; quotations and source labels retain the contemporaneous name.

When refugees are forced to leave asylum, onward movement to a third country is possible in principle but often infeasible. Onward migration may seem like a realistic alternative in high-income asylum countries, where refugees have often already traveled considerable distances. But most refugees live in low- and middle-income countries, often near the border they crossed and without the resources or legal permission to travel elsewhere. Many remain in peripheral regions because of community ties or government efforts to contain them there (Camarena, 2024); 79% of UNHCR camps in Africa lie within 50 kilometers of an international border (Fein and Van Den Hoek, 2022). Reaching a third country may therefore require a long, costly, and illegal journey that authorities can block, particularly when host governments are seeking to return refugees home (Whitaker, 2003). When conflict drives refugees from asylum, traveling farther into the host country may also be dangerous. For many refugees unable to remain in asylum, their country of origin is therefore the only destination they can realistically reach.

3.2 Constraints on Returning Home

Returning home can require far more resources than an ordinary international move. Refugees are often returning to places devastated by war, where homes and farmland have been destroyed, local economies have contracted, and there may be few public services. Return therefore requires not only paying for transportation across a border, but resources to rebuild livelihoods and to sustain families while finding new shelter and income sources in a post-conflict economy.

These costs are especially difficult to bear because displacement often limits refugees' ability to accumulate resources. Many spend years in camps or other settings where access to employment is restricted.[10] At the same time, remaining in asylum may offer a degree of material security, whether through humanitarian assistance or through informal work, businesses, and social networks established over years or even decades. Returning means giving up these relatively predictable sources of support for a much riskier environment. Refugees may therefore want to return while lacking the resources needed to rebuild their lives at home.

UNHCR repatriation programs can make such returns feasible. These are often large logistical operations that provide transportation, information, cash, food, shelter materials, legal assistance, and support for restoring housing and livelihoods. The repatriation of Sierra Leonean refugees from Guinea in the early 2000s illustrates the scale of this support:

> "About half of the returning refugees received assistance from UNHCR and other agencies in the form of transportation, cooked meals, and medical care during the

[10] Fifty-five percent of refugees live in countries that significantly restrict their right to work in practice (Ginn et al., 2022).

journey home... Returnees received blankets, cooking utensils, shelter materials, and a two-month food supply. UNHCR appealed for a nine-fold increase in its budget, to \$18 million, to support reintegration programs including schools, health clinics, and livelihood programs."

2002–2004, Guinea & Sierra Leone

These operations require substantial donor financing, and funding availability can determine how many refugees UNHCR is able to assist and the resources it can provide per person. In 2023, for example, more than 24,000 Burundian refugees had expressed a desire to return, but UNHCR reported a funding shortfall of more than \$13 million for transportation and assistance restarting their lives. At UNHCR's annual pledging conference, High Commissioner Filippo Grandi said "repatriation is possible, is demanded by refugees, but we can do much less than we would like because there is lack of resources" (Grandi, 2023).

3.3 Formalization

For clarity, I formalize the argument using a standard random utility model of migration choice (Borjas, 1987). Such models allow individual preferences to remain partly unobserved while deriving how changes in common conditions affect the probability of return. This is particularly useful here because major shocks need not induce every refugee to leave. Instead, they shift the relative attractiveness of return for many refugees at once, causing a larger share of the population to cross individual return thresholds.

I extend the standard framework in two ways. First, I incorporate asylum-country push factors, allowing refugees to return even when conditions at home remain unsafe. Second, I incorporate the transition costs of return, allowing refugees who would prefer living at home to remain in asylum because undertaking the move is costly.

Refugee i compares the expected utility of remaining in the asylum country a with that of returning to the origin country o. Utility from remaining in asylum is

$$U_{ai} = v_a(s_a, e_a) - \phi(p_a) + \varepsilon_{ai},$$

where $s_a \in [0, 1]$ indexes security from generalized armed conflict in the asylum country, $e_a \in [0, 1]$ economic conditions, and $v_a : [0, 1]^2 \to \mathbb{R}$ maps these general asylum-country conditions into utility and is strictly increasing in both. Host-government exclusion policies and refugee-targeted violence enters separately through $\phi(p_a)$. The term ε_{ai} captures unobserved individual-specific factors affecting the value of remaining in asylum. The function $\phi(p_a)$ maps the severity of an exclusion policy $p_a \in \mathbb{R}_+$ into the additional cost it imposes on remaining in asylum, with ϕ strictly increasing and

$\phi(0) = 0$.[11] Exclusion policies may raise the cost of remaining without eliminating choice; deportation is the limiting case in which remaining is no longer feasible.

Utility from returning to the origin country is

$$U_{oi} = v_o(s_o, e_o, \rho_o) - \tau_i(p_a, s_a) + \varepsilon_{oi},$$

where $s_o \in [0,1]$ indexes security in the origin country, $e_o \in [0,1]$ economic conditions, and $\rho_o \in [0,1]$ political repression. The function $v_o : [0,1]^3 \to \mathbb{R}$ is strictly increasing in s_o and e_o and strictly decreasing in ρ_o, and can implicitly capture an average value across refugees for living in their ancestral home country or locality. The term $\tau_i(p_a, s_a) \geq 0$ captures the transition cost of return, including transportation and the resources needed to re-establish housing and livelihoods. I assume

$$\frac{\partial \tau_i}{\partial p_a} < 0 \qquad \text{and} \qquad \frac{\partial \tau_i}{\partial s_a} > 0.$$

Stronger exclusionary pressure and worsening security in asylum therefore lower the transition-cost barrier to departure. Refugees planning an orderly return may wait until they can finance transportation, housing, and rebuilding a livelihood, whereas refugees facing coercion or conflict may leave with fewer resources and less preparation.

A refugee has a hypothetical preference for return when $v_o + \varepsilon_{oi} > v_a + \varepsilon_{ai}$. Whether refugee i decides to undertake return, however, depends on whether the total utility of returning exceeds that of remaining in asylum, accounting for exclusion policies and transition costs:

$$U_{oi} > U_{ai} \iff v_o(s_o, e_o, \rho_o) - v_a(s_a, e_a) + \phi(p_a) - \tau_i(p_a, s_a) > \varepsilon_{ai} - \varepsilon_{oi}.$$

A decision to return is not sufficient for realized movement. Let $r_i \in \mathbb{R}_+$ denote the resources available to refugee i, including private resources and external assistance. Realized return requires both

$$U_{oi} > U_{ai} \qquad \text{and} \qquad r_i \geq \tau_i(p_a, s_a).$$

Refugees who satisfy the first condition but not the second would choose to return but lack the resources needed to move. Additional resources relax this constraint, allowing

[11] The distinction concerns the source and target of the threat: s_a captures generalized conflict-related insecurity, while $\phi(p_a)$ captures host-government actions directed at refugees, including coercion or violence intended to induce departure.

more refugees to undertake return.

The idiosyncratic utility terms imply that refugees differ in their thresholds for return. Changes in common conditions shift the share of refugees who cross those thresholds. Greater security or economic opportunity at home increases the attractiveness of return, while greater repression reduces it. Exclusion policies and conflict in asylum operate through two channels: they reduce the value of remaining in asylum and lower the transition-cost barrier to departure. Greater resources affect a different margin, determining whether refugees can finance the transition.

This formulation therefore distinguishes preferences over destinations from realized movement. A refugee may prefer living in the origin country to living in asylum but nevertheless remain abroad because they cannot finance the transition. Conversely, sufficiently severe coercion or insecurity in asylum can both make remaining less attractive and induce refugees to leave with fewer resources and less preparation.

There are therefore two principal routes to realized return. The first is *origin-improvement return*: improving conditions at home raise U_{oi} relative to U_{ai}, but realized return also requires sufficient resources to cover τ_i. The second is *asylum-push return*: exclusion policies or deteriorating security in asylum reduce U_{ai} and lower τ_i, inducing return even without corresponding improvements at home.

3.4 Empirical Implications

The model generates three main hypotheses. First, asylum-side shocks increase return through two channels. Exclusion policies increase $\phi(p_a)$, while conflict in asylum reduces s_a, making remaining in asylum less attractive. Both also lower the resource threshold $\tau_i(p_a, s_a)$ at which refugees are willing to leave. Because neither channel requires improvement at home, asylum-side shocks can increase return even while origin-country conflict continues.

> ***H1 (Asylum-side push).*** *Refugee return increases following severe exclusion policies and armed conflict in the asylum country, including when conflict continues in the country of origin.*

Second, some refugees who prefer return cannot satisfy the feasibility constraint $r_i \geq \tau_i(p_a, s_a)$. Repatriation assistance increases the resources available for return, raising r_i and allowing more refugees to act on a preference for return.

> ***H2 (Repatriation assistance).*** *Refugee return is higher when a major UNHCR repatriation program is operating and increases with the resources available for repatriation assistance.*

Third, an end to conflict raises s_o and makes return more attractive, but does not itself relax the feasibility constraint. Where refugees who prefer return are resource constrained, repatriation assistance should therefore increase the extent to which peace translates into realized movement.

> ***H3 (Post-war feasibility).*** *The increase in refugee return following the end of conflict in the country of origin is greater when repatriation assistance is available.*

The model also generates a cross-host testable implication. Among refugees from the same origin, return should be higher from asylum countries experiencing higher p_a, lower s_a, or greater repatriation assistance, holding origin-country conditions constant.

These predictions complement rather than replace conventional explanations. When refugees can remain in asylum and have sufficient resources to move, higher s_o and e_o, lower ρ_o, and less attractive conditions in asylum should continue to increase return. Globally, the model does not imply that return responds more strongly to war endings, exclusion policies, conflict in asylum, or repatriation assistance. Assessing their aggregate importance instead requires measuring not only their effects on return, but how many refugees are exposed to each condition.

4 Data

4.1 Limitations of UNHCR Data

Testing the theory requires cross-national data on realized return, but UNHCR's population statistics, the principal existing source, have important limitations. They are designed primarily to measure voluntary repatriation, omitting forced return, and are vulnerable to systematic undercounting precisely when return is coerced or otherwise pushed.

One problem is that refugees may disappear from UNHCR's population statistics before they are returned. Host governments sometimes withdraw refugee status precisely to facilitate deportation; once individuals are no longer classified as refugees, their subsequent movement may not be recorded as refugee return. Forced returns may also occur where UNHCR has limited access. Governments hostile to refugee protection may restrict its activities, while conflict in the country of asylum may force personnel to evacuate. For example, UNHCR records no returns from South Africa to Mozambique despite contemporaneous reports of large-scale removals:

> "With no UNHCR presence in South Africa ... [n]either country recognizes them as refugees ... During 1985, South Africa was returning some 1,500 a month to the danger they had fled."
>
> *1986, South Africa*

Refugee population figures can also become politically contested. Host governments may understate return to preserve higher refugee counts and associated aid. In Somalia, for example:

> "Somalia refused to accept any reduced estimate ... UNHCR reluctantly continued to provide assistance for the supposed 840,000 refugees ... [before] publicly refut[ing] that figure, saying that the actual number is less than half that."
>
> *1989, Somalia*

These measurement problems are not random. Returns are especially likely to go unrecorded when refugees lose formal status, move outside assisted channels, or return from settings where humanitarian access is limited. Analyses based only on UNHCR data may therefore understate forced return precisely in the cases central to this paper.

4.2 The Global Refugee Return Dataset

To study realized return, I created the Global Refugee Return (GRR) Dataset. GRR includes origin populations with at least 25,000 refugees in exile in any year and uses a broad definition of refugees that includes formally recognized refugees, asylum seekers, and unregistered refugees. For each origin population, I include the principal asylum countries that collectively host at least 75% of that population.[12]

To address the measurement problems described above, GRR draws primarily on contemporaneous country reports that combine locally collected information with official and nonofficial sources. These reports often capture refugee dynamics missing from administrative data, including unregistered populations, spontaneous and forced returns, and disputes over official figures. Their coverage is also less dependent on the presence of any single international organization and, because the reporting organizations did not administer refugee assistance, less directly exposed to the political pressures surrounding aid-linked population estimates.

The principal source is the *World Refugee Survey* (WRS), published annually by the U.S. Committee for Refugees and Immigrants from 1980 through 2008. USCRI describes its reporting process as follows:

> "[A] rigorous method including, when possible, site visits to the countries on which we report. In most countries, we have Research Partners—local human rights NGOs that interview officials, refugees, and representatives of local civil society and international agencies based on our standard questionnaire ... We consult with and seek the input of representatives of each government ... We receive extensive input from the field offices of UNHCR and UNRWA. Finally,

[12] The 75% threshold is calculated using UNHCR data, supplemented with data on Palestinian refugees under UNRWA and Mozambican refugees in South Africa, whom the UNHCR data list under an unknown country of asylum.

> we crosscheck all of this with secondary sources in the media, on the internet, and in major governmental and nongovernmental human rights reports."
>
> *USCRI, 2008*

The *core* dataset covers Global South asylum countries from 1980 through 2008, where WRS reporting is most consistent.[13] The *extended* series adds high-income asylum countries and continues through 2023 using the U.S. Department of State's annual country reports on human rights practices. These reports contain country-level reporting on refugees and migrants, including forced return, exclusion policies, and unregistered populations (U.S. Department of State, 2024). Because the *State* reports do not always provide refugee stock or voluntary return estimates, the extended series uses UNHCR figures as a baseline and supplements them with additional evidence on forced returns and unregistered populations. The extended series therefore provides broader geographic and temporal coverage, while the core series offers more consistent measurement.

Constructing GRR required digitizing WRS reports, many available only as scans or physical volumes. I hand-coded the 25 largest origin populations, covering 83 dyads, and worked with 36 trained research assistants to code the remainder using a common protocol. I then reviewed and adjudicated every observation, returning to the original reports, adjacent years, and additional desk research when GRR and UNHCR figures differed substantially. To my knowledge, this is the first project to digitize and systematically code the full WRS corpus, producing 7,526 dyad-year observations.

A comparison with UNHCR data, reported in the Appendix, lends support to the measurement concerns raised above. Across the GRR core sample, UNHCR records 19.1 million returns compared with 19.5 million voluntary returns in GRR, indicating close agreement on voluntary repatriation. However, UNHCR data appears to underreport coerced returns. GRR records an additional 3.6 million forced returns that UNHCR does not capture, meaning that UNHCR records only 83% of total realized return. Nearly one-third of forced-return episodes in GRR are entirely absent from UNHCR data, and almost half of forced-return volume occurs in episodes that are either absent or only minimally recorded.

5 Empirical Analysis

The empirical analysis proceeds from within-dyad associations to aggregate patterns. I first evaluate H1 by examining whether return increases following asylum-side shocks, including while conflict continues in the country of origin, and then assess the timing and cross-host robustness of these associations. I then evaluate H2 using measures

[13] It contains 105 dyads across 58 origin populations and 61 asylum countries. The WRS was not published in 1990, so GRR has no observations for 1989.

of UNHCR repatriation assistance and H3 by testing whether assistance conditions post-war return, while also assessing conventional explanations for return. Finally, I examine how these factors account for aggregate patterns of realized return.

5.1 Explanatory Variables

Asylum-Country Push Factors. *Exclusion Policy* is an ordinal measure of coercive policy at the asylum–origin dyad level, hand-coded from the WRS and U.S. Department of State human rights reports. It equals 0 when there is no policy, 1 for moderate coercion, and 2 for severe coercion. Moderate policies include official threats or encouragement to return. Severe policies include implemented measures intended to produce widespread return, such as workplace raids or the revocation of residence permits. The coding is based on the policy imposed, not the amount of return subsequently observed; the Appendix provides the complete coding rules.[14]

Conflict in COA records the maximum UCDP conflict intensity in the country of asylum in a given year: no conflict, low-intensity conflict with 25–999 battle deaths, or high-intensity conflict with at least 1,000 battle deaths (UCDP, 2024). This is based on conflict location; a country participating in a war abroad is not coded as experiencing conflict in asylum. Both exclusion-policy variables enter as indicators, with no exclusion policy and no conflict as the respective reference categories.

UNHCR Repatriation Programs. *Repatriation Assistance* is the log of one plus UNHCR repatriation funding per member of the origin country's global refugee population, hand-coded from UNHCR Executive Committee documents and Yearbooks for 1980–1999 and 2011–2020. Because historical budgets cannot consistently be allocated across asylum countries, this measure varies at the origin-country-year level.

I also construct two dyad-level measures: *UNHCR Repatriation Programme*, indicating whether a UNHCR-assisted programme was operating, coded from the *World Refugee Survey* for 1980–2008, and *Tripartite Repatriation Agreement*, indicating an active agreement among UNHCR and the origin and asylum governments, coded from UNHCR archives and supplemented with Camarena (2022). Because these measures cover different periods and do not fully overlap with the funding measure, I estimate them in separate models.

Existing Explanations. For the country of origin, *Conflict in COO* distinguishes no, low-intensity, and high-intensity conflict, analogous to the COA measure, but is hand-coded to match the UCDP conflict that produced the refugee population rather than unrelated fighting in the same country. *War Ending* indicates the first three peaceful years after a war ends in the origin country, using an extension of Fearon's

[14] An additional external measure of exclusion policies from Amnesty International reports is underway.

(2017) civil war list through 2023. Years are coded as post-war only if conflict does not resume. *Civil Liberties Restrictions (COO)* is the Freedom House civil liberties index, which ranges from 1 for the greatest protection to 7 for the least, and *GDP (COO)* is logged GDP per capita from the World Bank.

For the country of asylum, *GDP (COA)* is logged GDP per capita, and *Refugee Rights* is the DWRAP index of de jure refugee and asylum policy, with higher values indicating more liberal policies (Blair, Grossman and Weinstein, 2022). *Conflict Duration* is the logged cumulative number of years of conflict in the origin country.[15] Because there is no data on when individual refugees entered asylum, this measure serves as a proxy for time in asylum.

The pooled model also includes six time-invariant measures absorbed by dyad fixed effects in the main specifications: whether the host is bound by the 1951 Convention or 1967 Protocol for the relevant origin population, accounting for states with geographic restrictions; distance and contiguity; common spoken language from CEPII GeoDist; religious similarity from the Correlates of War World Religion Project; and the share of the origin population with transnational ethnic kin in the asylum country from EPR-TEK (Vogt et al., 2015). The pooled model also includes log refugee stock as a covariate, in addition to its use as the offset, because it lacks dyad fixed effects to absorb persistent differences in population size across dyads.

5.2 Regression Specification

I estimate Poisson pseudo-maximum likelihood (PPML) models with dyad and year fixed effects. The unit of analysis is the directed origin–asylum dyad-year, and the dependent variable is the number of refugees who return during the year. PPML is well suited to a non-negative, highly skewed outcome with many zeros and does not require transforming or dropping zero-valued observations. Dyad fixed effects absorb time-invariant differences across refugee populations and origin–asylum relationships, while year fixed effects absorb shocks common to all dyads. Because the analysis is observational, I interpret them as conditional associations rather than causal effects.

I include the log refugee population at risk as an offset, so the coefficients describe proportional changes in the return rate rather than differences driven by population size. The population at risk is the previous year's refugee stock.[16] I restrict the analysis to dyad-years with a refugee stock of more than 1,000 refugees in the dyad. The baseline model is

[15] The counter runs while conflict is active at either intensity, tolerates a single quiet year, and resets after two consecutive years without conflict, so it measures the age of an ongoing war rather than a country's cumulative history of violence.

[16] In 21 dyad-years, a large inflow arrives and returns within the same year, causing return to exceed the lagged stock; for these observations, I use the estimated within-year peak population.

$$E[Y_{it} \mid \mathbf{X}_{it}] = \exp(\boldsymbol{\beta}'\mathbf{X}_{it} + \alpha_i + \gamma_t + \ln S_{it-1}), \quad (1)$$

where Y_{it} is the number of returnees in dyad i and year t, $\mathbf{X}_{it}$ is the vector of time-varying covariates, α_i and γ_t are dyad and year fixed effects, and $\ln S_{it-1}$ is the offset. Standard errors are clustered by dyad.

To examine the cross-host implication of H1, I estimate a specification with origin-by-year fixed effects, which absorbs conditions common to refugees from the same origin in a given year and identifies asylum-side coefficients from variation across the countries hosting them at the same time. I also estimate a complementary specification with asylum-by-year fixed effects, which absorbs conditions common to a host in a given year and identifies the exclusion policy coefficient from differences in how that government treats refugee groups from different origins. The latter absorbs asylum-country conflict by construction. Both specifications retain dyad fixed effects and are reported in the Appendix.

5.3 Regression Results

Table 1 presents the main estimates. Model 1 is the primary specification. Model 2 adds UNHCR repatriation assistance and therefore uses the smaller subset of origin-years for which funding is recorded. Because the refugee population at risk enters as an offset, exponentiated coefficients describe proportional differences in the return rate. The mean return rate in the Model 1 sample is 6.6%, which is lower than the broader estimation frame because fixed-effects PPML drops dyads with no variation in return.[17] Because this restriction excludes many refugee populations that never experience return, I also estimate pooled PPML models without fixed effects in Model 3.

Asylum-Country Push Factors. H1 predicts that return rises with push factors in asylum, even if conditions at home have not improved. The clearest evidence comes from coercive host-state policies. Relative to years without such a policy, a severe policy is associated with 2.97 times the return rate (coefficient 1.09, SE = 0.19), the largest association in Panel A of Figure 3. A moderate exclusion policy is associated with a 62% higher return rate (coefficient 0.48, SE = 0.22; return-rate ratio = 1.62).

Conflict in the country of asylum is also associated with more return. Low-intensity conflict is associated with 2.24 times the return rate (coefficient 0.81, SE = 0.21),

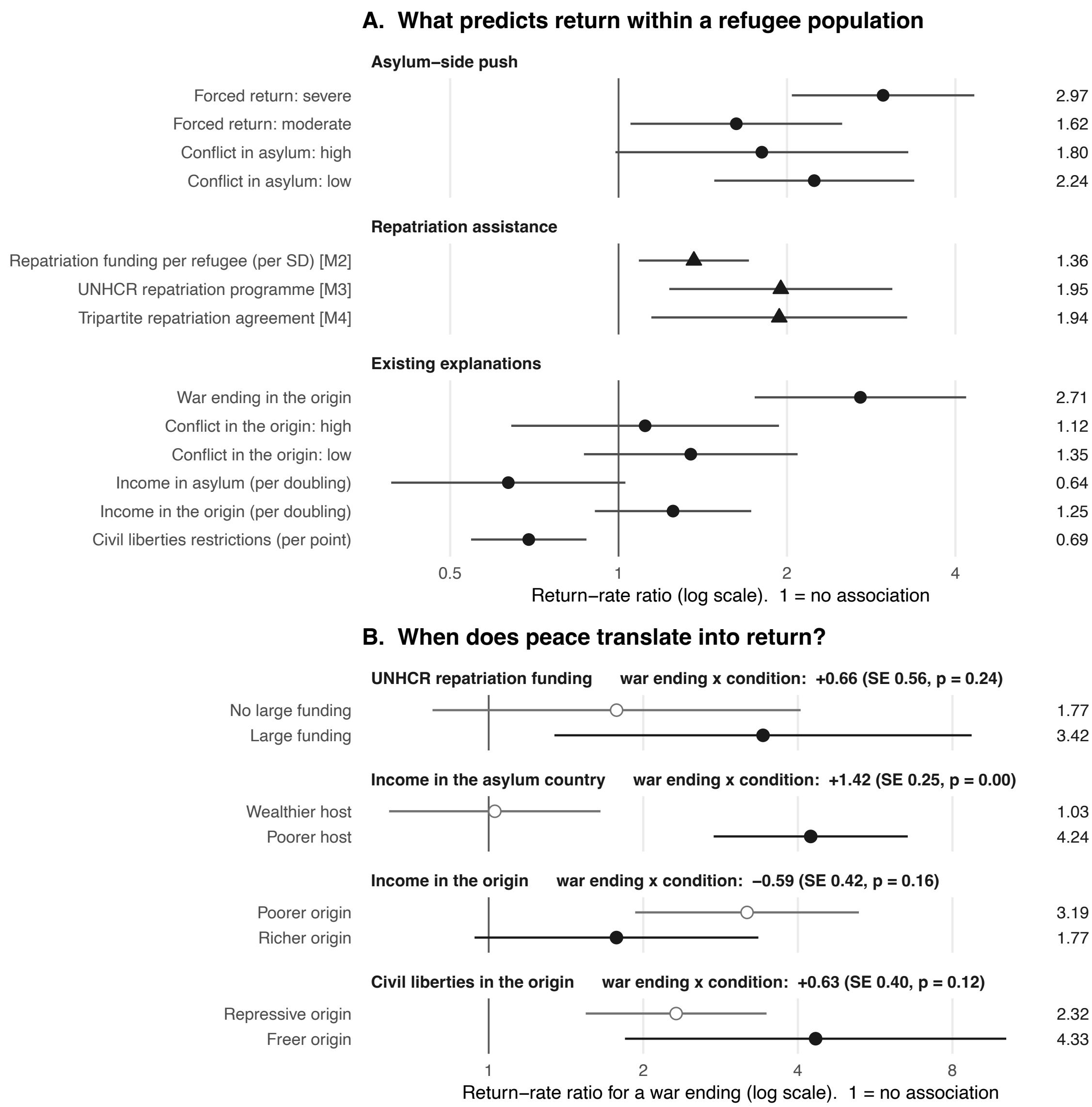
[17] Percentage-point translations below are anchored to the regression-sample mean of 6.6%. The estimation sample contains 2,327 dyad-years across 110 dyads, from an estimation frame of 5,131 dyad-years in 285 dyads. Of the 175 excluded dyads, 148 record no return in any year, 25 are lost to covariate missingness, and 2 are single-year dyads absorbed by the fixed effects. The fitted sample contains 75.6% of all return recorded in GRR and 58.7% of refugee stock-years in the estimation frame.

Table 1: Poisson pseudo-maximum likelihood estimates of refugee return

Term	(1) Main	(2) With assistance	(3) Pooled
Asylum-side push			
Forced Return: severe	1.09*** (0.19)	1.29*** (0.31)	0.98*** (0.23)
Forced Return: moderate	0.48* (0.22)	0.13 (0.28)	0.27 (0.17)
Conflict in COA: high	0.59† (0.31)	0.79† (0.43)	0.50 (0.31)
Conflict in COA: low	0.81*** (0.21)	0.95** (0.36)	0.59** (0.18)
Repatriation assistance			
UNHCR Repatriation Funding (COO)	–	0.31** (0.12)	–
Existing explanations			
War Ending	1.00*** (0.22)	1.05** (0.38)	1.16*** (0.25)
Conflict in COO: high	0.11 (0.28)	0.38 (0.35)	0.14 (0.26)
Conflict in COO: low	0.30 (0.22)	0.22 (0.25)	0.30 (0.28)
GDP (COO)	0.39 (0.28)	0.80 (0.50)	-0.46*** (0.13)
GDP (COA)	-1.09† (0.59)	-1.40 (0.86)	-0.80*** (0.17)
Civil Liberties Restrictions (COO)	-0.51** (0.17)	-0.10 (0.26)	-0.50*** (0.10)
Refugee Rights	-0.02 (0.17)	-0.19 (0.22)	-0.00 (0.08)
Conflict Duration	-0.03 (0.08)	-0.06 (0.12)	0.04 (0.10)
Factors in the pooled specification only			
Refugee Convention	–	–	0.25 (0.22)
Distance (log)	–	–	0.04 (0.15)
Contiguous	–	–	0.54 (0.43)
Refugee Stock at Risk (log)	–	–	-0.04 (0.06)
Common spoken language	–	–	-0.32† (0.18)
Religious similarity (z)	–	–	0.24** (0.08)
Kin share of origin population (z)	–	–	-0.06 (0.07)
Specification and fit			
Dyad and year fixed effects	Yes	Yes	No
Sample	Extended GRR	Funding coded	Extended GRR
Years	1980–2022	1980–1999, 2011–2020	1980–2022
Dyad-years	2,327	1,378	3,810
Dyads	110	100	249
Pseudo R^2	0.799	0.782	0.724

Notes: COO = country of origin; COA = country of asylum. All models are PPML with the log refugee stock at risk as an offset and standard errors clustered by dyad. Column 1 includes dyad and year fixed effects. Column 2 includes repatriation assistance and is estimated on a smaller sample. Column 3 is pooled and includes dyad-level characteristics directly. Continuous variables are standardized separately.

Figure 3: What predicts refugee return, and when peace translates into return



Notes: Return-rate ratios from PPML models with dyad and year fixed effects, log refugee stock at risk as an offset, and dyad-clustered standard errors. Bars show 95% confidence intervals; the vertical line at one indicates no association. Circles report the baseline specification in Table 1 (M1). Triangles report auxiliary models adding UNHCR repatriation funding (M2), a UNHCR repatriation programme indicator (M3, 1980–2008), or a tripartite repatriation agreement indicator (M4, 1980–2002), each estimated on the available sample. Income is scaled per doubling and civil liberties per Freedom House point. In Panel B, estimates show the return-rate ratio associated with a war ending at the indicated value of each condition, based on the corresponding interaction model.

while high-intensity conflict is associated with 1.80 times the rate (coefficient 0.59, SE = 0.31). The low-intensity estimate is statistically distinguishable from zero, whereas the high-intensity estimate is less precise. A Wald test does not reject equality of the two coefficients. The evidence therefore supports an association between conflict in asylum and return, but not a clear difference by conflict intensity. Leave-one-region-out checks in the Appendix suggest that this is driven by cases in Sub-Saharan Africa.

In the Appendix, I report a specification with origin-by-year fixed effects, comparing refugees from the same origin in the same year across different asylum countries. The same patterns hold: refugees are more likely to return from asylum countries experiencing exclusionary policies or conflict, while those from the same origin living in asylum countries without these conditions are more likely to remain. A complementary specification with asylum-by-year fixed effects yields similar results: when asylum-country governments implement exclusionary policies targeting specific groups, those groups are more likely to return home, while groups not targeted by such policies are more likely to remain.

Most importantly, I find no evidence that these associations depends on peace at home. Interaction models in the appendix show that the association between asylum-side shocks and return is similar whether or not conflict continues in the country of origin. The Appendix also repeats the analysis using UNHCR return counts over the core sample; the asylum-side associations largely disappear, consistent with the undercounting of forced and asylum-driven returns documented above.

UNHCR Repatriation Programs. H2 predicts that refugees who want to return may nevertheless remain abroad if they lack the resources to move, and that assistance reducing these costs should increase realized return. Consistent with H2, all three measures of UNHCR repatriation assistance are positively associated with return. In Model 2, a one-standard-deviation increase in logged repatriation funding per refugee is associated with a 36% higher return rate (coefficient 0.31, SE = 0.12). Appendix analyses show similar patterns for the other measures: the presence of a UNHCR repatriation programme is associated with 1.95 times the return rate, while an active tripartite repatriation agreement is associated with 1.94 times the rate.

Panel B of Figure 3 shows a pattern consistent with H3. A war ending is associated with 1.77 times the return rate when repatriation funding is low, compared with 3.42 times the rate when substantial funding is present. This difference is large, but the interaction is imprecisely estimated in the relatively small sample (0.66, SE = 0.56, $p = 0.24$). Descriptively, among post-war dyad-years, the aggregate return rate is 22.86% when repatriation funding is recorded and 5.18% when it is not.

An important concern is selection into repatriation assistance. UNHCR may devote resources to cases where it already expects substantial return, including places where conditions in the origin have improved. In that case, assistance could simply track returns that would have occurred anyway. The archival evidence, however, suggests that variation in repatriation funding also reflects donor priorities and bureaucratic funding constraints, rather than only expectations about return. I address the selection concern in two ways.

First, the association is driven by how much assistance is available per refugee, rather than simply whether any funding is recorded. Among observations with positive funding, greater funding per refugee is strongly associated with higher return (coefficient 0.82, SE = 0.21). By contrast, an indicator for whether any funding is recorded is small and imprecise (0.14, SE = 0.30). When the extensive and intensive margins enter jointly, the continuous funding measure remains strongly positive (0.64, SE = 0.16). This pattern is consistent with a resource mechanism: where more resources are available per person, assistance can cover a larger share of the costs of return.

Second, peace and repatriation assistance often occur at different times. Not all war endings are followed by a UNHCR repatriation programme: 60 post-war dyad-years have one, while 86 do not. Even when programmes do follow war endings, they often begin later. The median lag is one year, 24% begin at least two years later, and 10% begin five or more years later. These delayed cases help distinguish returns associated with improved conditions from those associated with the arrival of assistance.

Across fourteen post-war episodes in which assistance began at least three years after the war ended, return averaged just 0.8% in the two years before assistance and rose to 9.9% over the three years beginning with its arrival. Post-war situations that never received assistance show no comparable increase, averaging 1.4% during years four through eight after a war ending.[18] The timing of tripartite agreements points in the same direction. Return averages 3.7% in the four years before an agreement, rises to 8.1% in the year of signature, and reaches about 18% in each of the following two years. Together, these patterns suggest that assistance may increase return independently of the immediate effects of war ending.

These analyses are not causal. Repatriation programmes may still be targeted toward cases where return is especially feasible, and UNHCR programmes can also provide information about conditions at home. The evidence therefore cannot isolate a pure resource effect. It does, however, provide systematic cross-national evidence on a previously difficult-to-measure dimension of repatriation. Across several newly assembled archival measures, return is consistently higher when repatriation resources and institutional support are available, increases with the amount of funding available per

[18] These fourteen episodes occur across nine dyads, with six or seven dyad-years contributing to each relative-year cell, so the pattern is suggestive rather than precisely estimated. Return after assistance begins also accounts for only 15% of total recorded return in these episodes. For these delayed programmes, assistance therefore appears to restart return after an earlier wave has subsided rather than account for most post-war movement.

refugee, and often rises after assistance arrives even when the end of conflict occurred years earlier. Taken together, these patterns are consistent with H2's implication that limited resources can constrain refugees' ability to return.

Existing Explanations. Existing research suggests that return should respond to the relative attractiveness of origin and asylum, particularly security in the origin country. The results support some of these expectations more consistently than others. Most clearly, return tracks the end of war. The coefficient on the post-war window is 1.00 (SE = 0.22) in Model 1, implying a return rate 2.71 times higher during the first three peaceful years after a war ends.[19] The broader level of violence at home is less predictive, as neither low- nor high-intensity conflict in the origin differs significantly from peace.

Political conditions also matter. A one-standard-deviation increase in civil-liberties restrictions in the origin is associated with less return (−0.51, SE = 0.17). Return is also lower as asylum-country income rises (−1.09, SE = 0.59), although this estimate is imprecise. By contrast, origin-country income, refugee rights in asylum, and conflict duration are not distinguishable from zero in the fixed-effects model.

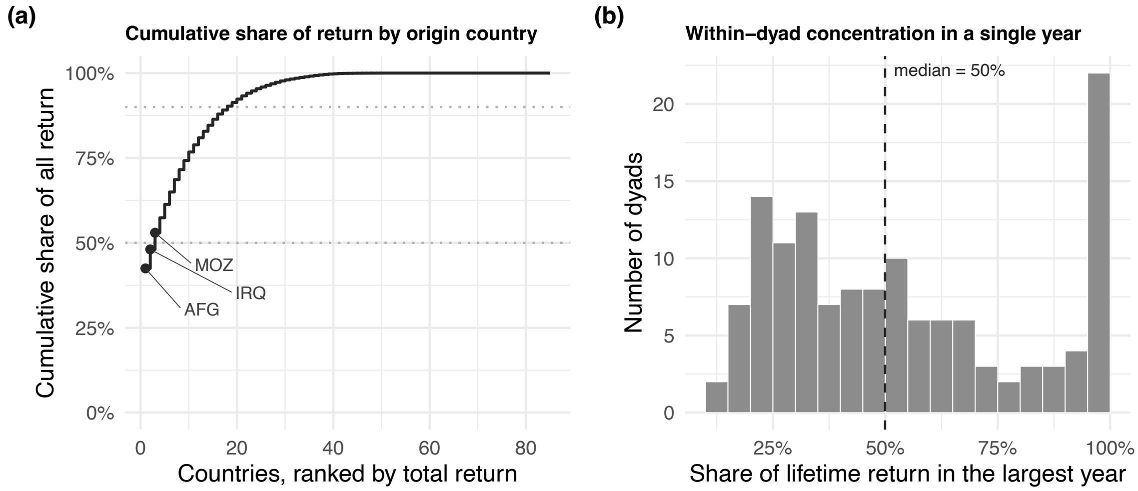
The pooled model broadly reproduces these patterns across cases. Income matters more in this comparison, with higher GDP in both origin (−0.46, SE = 0.13) and asylum (−0.80, SE = 0.17) associated with lower return. Among time-invariant characteristics, return is lower where origin and asylum share a widely spoken language (−0.32, SE = 0.18) and higher where their religious composition is more similar (0.24, SE = 0.08). I find little evidence of an association with transnational ethnic kin (−0.06, SE = 0.07) or Refugee Convention status (0.25, SE = 0.22).

Existing research also suggests that quality-of-life considerations should matter more once security at home improves. Panel B of Figure 3 examines whether the association between a war ending and return varies with conditions in origin and asylum. The clearest difference is by asylum-country income: a war ending has essentially no association with return in wealthier asylum countries, where the return-rate ratio is 1.03, but a large association in poorer ones, at 4.24 (interaction 1.42, SE = 0.25, $p < 0.001$). The corresponding differences by origin-country income and civil liberties point in the expected direction but are more modest and imprecisely estimated.

Taken together, these results support important parts of existing explanations. Refugees return at higher rates after wars end, remain abroad when repression at home is greater, and appear less likely to leave wealthier asylum countries. At the same time, I find little consistent evidence that the broader intensity of conflict in the origin, statutory

[19] When the ending year itself is added separately, its coefficient is 0.05 (SE = 0.25) and is not statistically distinguishishable from zero. Return is therefore substantially higher during the first full years of peace, rather than in the year the fighting stops.

Figure 4: Concentration of recorded refugee return



Notes: Panel (a) uses the full GRR panel of 7,498 dyad-years across 292 dyads, aggregated to the 85 countries of origin it covers; its dotted lines mark the 50th and 90th percentiles of cumulative recorded return. Panel (b) includes the 135 dyads observed for at least five years that record any return.

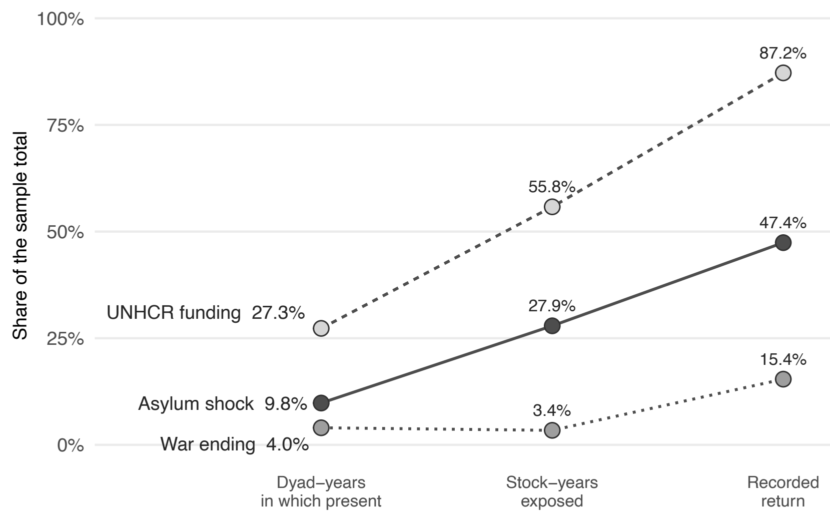
refugee rights, conflict duration, transnational ethnic kin, or Refugee Convention status predict return. More importantly, these explanations are incomplete. Coercive policies and conflict in asylum are also associated with substantial return, including while conflict continues in the country of origin.

5.4 Global Patterns of Return

The hypotheses concern different components of an individual refugee population's return decision. They do not, by themselves, establish which mechanism matters most for the global pattern documented at the beginning of the paper. Aggregate importance depends on exposure as well as the size of the conditional association: a factor that strongly predicts return can account for relatively little global movement if few refugees experience it.

Figure 4 shows that refugee return is concentrated in a small number of displacement contexts and, within those contexts, in a handful of years. The top 5% of dyad-years account for 93.8% of all recorded return, and the top 1% for 63.5%. Among dyads that record any return, the median experiences 49.9% of its lifetime return in a single year, and 19.3% experience at least 90% in one year. Most dyad-years with more than 1,000 refugees record no return, while Afghanistan alone accounts for 42.5% of recorded return, the three largest origins for 53.0%, and the ten largest for 76.8%. Realized return is therefore not a steady annual flow. In most dyad-years, few or no refugees return; in a small number, large portions of a population move together. This

Figure 5: Exposure and observed return across post-war, asylum-shock, and UNHCR program contexts



Notes: War ending = first three peaceful years after an origin-country war; asylum shock = severe exclusion policy or high-intensity conflict in asylum. Sample includes dyad-years with more than 1,000 refugees at risk, excluding 1989. UNHCR estimates use years with available repatriation expenditure data. Categories may overlap.

concentration motivates examining shocks that can affect many refugees at once.

Figure 5 shows that recorded return is concentrated in periods of UNHCR repatriation funding and asylum-side shocks. In years with available expenditure data, 87.2% of recorded return happens in dyad-years with UNHCR funding, while asylum shocks coincide with 47.4%, compared with 15.4% during war-ending windows across the full dataset. The larger aggregate importance of asylum shocks relative to war endings appears to reflect exposure: asylum-shock years contain 27.9% of refugee stock-years at risk, compared with just 3.4% in war-ending years. In the Appendix, I also illustrate how common these asylum-side shocks are in the twenty dyads with the greatest total return by plotting these dyads over time.

Because these conditions overlap, the Appendix repeats the comparison using mutually exclusive categories. Asylum-shock-only years account for 44.8% of recorded return and 27.1% of refugee stock-years at risk, with an aggregate return rate of 8.27%. Post-war-only years account for 12.8% of return and 2.6% of stock-years, with a substantially higher return rate of 24.61%. Repatriation funding is associated with substantial return beyond the immediate post-war period: in the smaller sample of the data where repatriation assistance data is available, funded years outside the post-war window account for 76.9% of recorded return and 53.5% of stock-years at risk, with a return rate of 7.29%. By contrast, the 2.7% of dyad-years that are both post-war and funded account for 10.3% of return, with a much higher return rate of 22.86%. Thus, return is especially high when repatriation funding coincides with the end of war, but most assisted return occurs outside the first three post-war years.

Because the descriptive comparisons do not adjust for compositional differences across exposed dyad-years, I also conduct model-adjusted aggregate contrasts. Using a pooled PPML model including repatriation assistance, I compare fitted return under the observed data with predictions obtained after setting each factor to its reference value wherever it occurs, holding the remaining covariates fixed. Setting repatriation funding to zero reduces fitted aggregate return by 47.5%, against 23.2% for severe exclusion policy, 11.0% for the post-war window and 6.9% for high-intensity asylum conflict.

6 Alternative Explanations and Robustness

I address three broad concerns about the results. The Appendix reports the principal robustness analyses.

A first set of robustness checks considers alternative explanations for the main results. One possibility is that the relationship between conflict and return reflects anticipatory return before wars end rather than asylum-country push factors. There is some evidence of this: return is modestly elevated before war endings, and coercion near an approaching ending predicts additional return. However, anticipation cannot account for the main result. Only 11 of 129 severe-policy dyad-years fall within three years before a war ending, and excluding all dyad-years within three years on either side of an ending leaves the severe-policy estimate essentially unchanged. Adding the war-ending year directly to the model produces the same result: return is not elevated in the year the war ends, while the severe-policy estimate remains essentially unchanged.

Similarly, exclusion policies could simply coincide with return already underway. In the Appendix, I estimate a stacked event study around policy onset, comparing treated dyads with those whose hosts impose no severe policy during the same window. Return shows little evidence of increasing before policy onset, but rises sharply at implementation and in the following year. A linear specification produces the same pattern.

Another possibility is that higher return during conflict reflects failed asylum claims: people flee, are rejected, and return, with this movement counted as repatriation. This is unlikely to explain the results. People turned away at the border are not included, and the return data and archival reports indicate that large return episodes generally involve recognized refugees or people who had been living in the asylum country without formal status. Moreover, large asylum-seeker populations are concentrated in the Global North, as many countries in the Global South automatically recognize people fleeing particular conflicts as *prima facie* refugees (United Nations High Commissioner for Refugees, N.d.). Severe policies also tend to target large, long-established refugee populations, a pattern supported by Appendix diagnostics linking lagged refugee population size to policy onset.

The second set of checks concern the validity of the exclusion policy measure. Because coding archival reports requires judgment, I make each classification auditable by providing the supporting text and coding rules in the GRR data file. Policies are coded based on their intent and implementation, not on whether forced return actually occurs. Consistent with this, 36% of moderate-policy and 8% of severe-policy dyad-years record no return. Severe policy remains statistically significant when the outcome is restricted to returns coded as voluntary. As a consistency check, exclusion policies are less common where DWRAP records stronger protections against refoulement. Finally, I apply the same coding rules to Amnesty International reports to assess whether the classifications can be reproduced using an independent source corpus.[20]

I examine the robustness of the findings across alternative measures of return, conflict, and war endings; different estimators and clustering strategies; lagged covariates and additional fixed effects; alternative samples, including core GRR; and the exclusion of refugee-warrior cases, particular regions, and the largest origins, asylum countries, dyads, and return episodes. The severe-policy association is the most robust, remaining positive across nearly all of these checks. The asylum-country conflict and repatriation funding results also generally persist. Among the existing explanations, restrictions on civil liberties at home are relatively robust, while the post-war association is more sensitive to the outcome, sample, and definition of a war ending; the remaining predictors show less consistent relationships with return.

7 Case Studies

To illustrate the mechanisms underlying the cross-national results, I examine two historical cases. The first compares Rwandan refugees across Burundi, the Democratic Republic of the Congo (DRC), and Tanzania in 1996. Because refugees from the same origin faced broadly similar conditions at home, differences in the timing of return across asylum countries help illustrate how coercion and conflict in asylum can produce return even without an improvement in origin-country conditions. The second examines Eritrean refugees in Sudan after the end of Ethiopia's civil war. In this case, peace did not immediately produce large-scale return, but return increased after UNHCR repatriation assistance became available.

7.1 Asylum-Side Push: Rwandan Refugees in Burundi, the DRC, and Tanzania, 1996

After the 1994 genocide and the subsequent victory of the Tutsi-led Rwandan Patriotic Front, hundreds of thousands of Hutu fled to neighboring countries, including members

[20] The data collection and analysis is underway.

of the former government and armed forces implicated in the genocide. Although the new Rwandan government encouraged return and UNHCR intensified repatriation efforts in 1995, most refugees remained in Burundi, the DRC, and Tanzania. Many feared violence or arrest in Rwanda, while former-regime leaders intimidated refugees and sometimes prevented them from leaving the camps. Efforts to promote voluntary repatriation therefore initially produced relatively little return.

Mass return instead followed distinct shocks in each country of asylum. In Burundi, the government increasingly pressured Rwandan refugees to leave during 1996. Officials told refugees they were no longer welcome, soldiers and local militias attacked and burned camps, and authorities forcibly transported refugees across the border. In July, soldiers expelled 15,000 refugees and temporarily blocked UNHCR access to two camps. By year's end, "about 98 percent of Rwandan refugees had left Burundi, due to fear for their safety and coercive policies by Burundian authorities" (1997, Burundi).

In the DRC, repatriation efforts likewise produced little movement until civil war reached the refugee camps in October 1996. Rebel attacks dispersed some former-regime leaders who had obstructed departure, while the fighting itself forced refugees from the camps. More than half a million returned within four days in what UNHCR described as the "largest and swiftest repatriation in memory" (1997, Zaire). Some returned willingly once freed from camp leaders' control, while others returned to escape the fighting or because they had no other option. At least 200,000 instead moved deeper into the DRC. Their attempt to remain outside Rwanda is important for interpreting the return that did occur: displacement from the camps did not uniformly reflect a new preference for return.

Return from Tanzania followed yet another asylum-side shock. Only 2% of Rwandan refugees returned during the first eleven months of 1996. On December 5, following the return from the DRC, the government proclaimed that "all Rwandese refugees can now return to their country in safety" and imposed a year-end deadline for departure (1997, Tanzania). Hundreds of thousands initially moved deeper into Tanzania, away from Rwanda, but security forces blocked their movement, closed the camps, and directed them toward the border. Return subsequently reached as many as 15,000 people per hour, while approximately 50,000 avoided repatriation by dispersing elsewhere in Tanzania or presenting themselves as Burundian.

The staggered timing across these three countries is difficult to explain through changes in Rwanda alone. Refugees faced the same Rwandan government and broadly similar origin-country conditions, yet mass return followed coercion in Burundi, conflict reaching the camps in the DRC, and an expulsion order in Tanzania. Refugees' efforts to move farther into the DRC and Tanzania show that many did not want to return. The episode therefore illustrates the central implication of H1. Realized return can rise sharply when remaining in asylum becomes infeasible, even without a corresponding improvement in refugees' country of origin.

7.2 Post-War Return and Repatriation Assistance: Eritrean Refugees in Sudan, 1991–2005

The experience of Eritrean refugees in Sudan illustrates a different constraint on realized return. Hundreds of thousands of Tigrayans and Eritreans had fled Ethiopia's civil war into Sudan beginning in the 1960s. In May 1991, the Mengistu regime fell and the EPLF took control of Eritrea, bringing the conflict to an end. If improved security at home were sufficient to generate return, this change should have produced rapid large-scale repatriation. Instead, only about 10,000 refugees returned in the year of independence.

One obstacle was the absence of a large repatriation operation. A dispute between the Eritrean government and UNHCR over the scale and financing of reconstruction assistance prevented the immediate launch of a return program. Eritrea sought more rehabilitation assistance than donors were willing to provide. A UNHCR repatriation program eventually began in 1994 and approximately 81,000 refugees returned over the following two years, but roughly 320,000 remained in Sudan. Return subsequently stalled amid "financial disputes over the scope of the repatriation program, and deteriorating relations between the governments of Sudan and Eritrea" (WRS, Sudan, 1997), alongside insecurity near the border.

A second repatriation effort around 2000 provided transportation, reintegration grants, and housing assistance. Approximately 130,000 refugees returned over the following five years. The timing is consistent with the mechanism in H2: after the war had already ended, return increased when institutional and financial support reduced the costs of moving and rebuilding a life in Eritrea.

Many refugees continued to remain in Sudan even once assistance became available. By this point, some had lived there for decades and had established livelihoods and social ties. By the early 1990s, half lived in camps with access to food aid, health and education programs, and technical training, often near relatives or co-ethnics and with access to agricultural land, while the remainder lived in urban areas. When encouraged to return in 2001, "many would-be returnees chose to remain longer in Sudan to harvest their crops" (WRS, Sudan, 2002). A survey reported that roughly 85 percent of refugee households considered assistance levels in Eritrea an important determinant of whether they would return.

Even the formal termination of refugee status did not produce universal repatriation. In 2002, UNHCR invoked the cessation clause, requiring Eritreans to "repatriate, file individualized asylum claims to remain in Sudan as refugees, or take steps to become permanent legal residents of Sudan" (WRS, Sudan, 2003). Only 25,000 repatriated in 2002 and 9,500 in 2003, while many remained in Sudan.

The contrast with the Rwandan case highlights the two constraints emphasized in the theory. Rwandan refugees returned in large numbers when coercion or conflict undermined their ability to remain in asylum, despite limited evidence that conditions at home had become more attractive at precisely those moments. Eritrean refugees, by contrast, often remained abroad after the war had ended, and return increased as organized repatriation assistance became available. Yet when refugees were given the option to remain in Sudan by seeking permanent legal residence, rather than being compelled to leave, many chose to stay. Together, the cases show why realized return cannot be inferred from origin-country security alone. Whether refugees actually move also depends on whether remaining in asylum is feasible and whether refugees have the resources needed to undertake return.

8 Conclusion

This paper revisits a foundational premise of the international refugee regime: that refugees receive protection while conditions at home remain unsafe and return once those conditions improve. Using the Global Refugee Return dataset, which covers 293 origin–asylum dyads from 1980 to 2023, I show that realized return often follows a different pattern. Most recorded returns occur while conflict is still ongoing in the country of origin, while many refugees remain abroad long after war has ended.

I argue that realized return depends not only on refugees' individual or household-level preferences over origin and asylum, but also on population-level conditions that determine whether remaining abroad and undertaking return are feasible. Host-government coercion and conflict in asylum can make staying infeasible, while repatriation assistance can make return possible for refugees who otherwise lack the resources to move. Two sets of findings support this argument.

First, both asylum-side push and repatriation assistance are strongly associated with when refugees return. Comparing the same refugee populations over time, severe exclusion policies and asylum-country conflict are associated with 3.0 and 2.2 times the annual return rate observed in their absence, respectively, compared with 2.7 for war endings. A one-standard-deviation increase in UNHCR repatriation funding per refugee is associated with a 36% higher return rate, and return following a war ending is substantially greater when repatriation funding is available. The asylum-side associations persist while conflict continues at home, while return also rises after the onset of severe exclusion policies and major repatriation programs.

Second, these constraints help explain the aggregate timing of return. Severe asylum-side shocks coincide with 47.4% of recorded return, compared with 15.4% in the first three years after a war ends, largely because far more refugees are exposed to them. Repatriation assistance is also present during much of realized return: in years with expenditure data, 87.2% of return occurs when UNHCR repatriation funding is available, much of it outside the immediate post-war period. War endings therefore remain an important predictor of return, especially when repression at home is lower and asylum countries are poorer, but peace alone neither guarantees movement nor accounts for most realized return.

These findings have several implications. Theoretically, these findings help reconcile existing research on return intentions with global patterns of realized movement. Individual preferences remain important, but they do not by themselves determine when large refugee populations move. Population-wide shocks can induce return without corresponding improvements at home, while limited resources can delay return even among refugees who would prefer to go back.

This distinction also matters for how repatriation is understood within the refugee regime. Voluntary return after peace can provide a durable solution when refugees want to return and can do so safely, but observed return is not necessarily evidence that these conditions have been met. Some of the largest return movements instead occur under coercive pressure and while conflict continues at home, while other returns depend heavily on international assistance to finance transportation and reintegration, which may be under threat with recent aid cuts. The conditions under which return occurs may therefore shape both returnees' welfare and reintegration (Fransen, Ruiz and Vargas-Silva, 2017) and the consequences of mass return for origin communities and post-conflict politics (Schwartz, 2019; Blair, Van Dijcke and Wright, 2026).

The analysis also underscores the difficulty of measuring refugee return. Existing UNHCR statistics omit many forced and spontaneous returns, including the coercive movements central to the argument here. GRR reconstructs refugee stocks and realized return and adds new data on exclusion policies, but it still relies on expert reporting and remains an imperfect measure of movements that are often difficult to observe precisely. Other poorly measured displacement dynamics remain beyond its scope and important for future research, including efforts to prevent access to asylum through maritime pushbacks and border closures (FitzGerald, 2019; Goodwin-Gill, 2011; Schwartz, 2025), as well as temporary and repeated movement between origin and asylum (Masterson and Vidarte, 2026). These dynamics remain difficult to study cross-nationally because existing data rarely track them systematically.

Several questions follow. First, when and why do governments impose exclusion policies? Analysis in the appendix suggests that severe coercion is concentrated in large, unresolved displacement crises. Population size is its most consistent correlate, while crisis duration, origin-country violence, and host characteristics are much less robust predictors. More detailed case analysis could examine the domestic pressures behind these decisions, the role of international aid and diplomacy, and whether host governments face meaningful costs for violating the refugee regime's core protection principle. Future research should also examine the role of origin governments, which may encourage return to signal normalization, but resist it when they lack the capacity to absorb returnees or regard particular groups as political or security threats.

More broadly, the paper brings the refugee regime into debates about the performance of international regimes. *Non-refoulement* is its core legal principle, yet the analysis reveals both continued protection and systematic violations at scale. UNHCR may even facilitate return under coercive conditions when refusing to participate would leave refugees without assistance or jeopardize its access. This variation raises a question familiar from research on other international institutions but rarely examined systematically in the refugee regime: why are core rules upheld in some cases but violated in others, and when do international organizations constrain rather than accommodate those violations?

Taken together, the findings show that repatriation does not automatically follow the restoration of peace. Displacement may end because protection in asylum erodes, because international assistance makes return possible, or it may persist long after the conflict that produced it has ended. Explaining when displacement ends therefore requires understanding the political processes that determine whether states continue to provide asylum or push refugees to leave, when donors finance repatriation, and when UNHCR facilitates return, as well as where refugees displaced by war, often for decades, want to live.

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Peace or Push? Understanding Global Patterns of Refugee Return

Mae MacDonald

Supplementary Materials

Table of Contents

1 Global Refugee Return (GRR) Dataset

The Global Refugee Return (GRR) dataset is an original global dataset on refugee stocks, returns, and the conditions shaping repatriation. It covers origin–asylum dyad-years from 1980 to 2023. The dataset makes two core contributions: (1) improved stock and repatriation data that corrects systematic errors in UNHCR figures and incorporates forced returns; and (2) new indicators of *de facto* refugee conditions in asylum, including coercive return policies.

Unit of analysis. Each observation records conditions for refugees from a specific country of origin residing in a specific country of asylum in a given year.

Geographic coverage. The dataset includes all **origin countries** with at least 25,000 refugees in exile in any given year, consistent with UNHCR's threshold for a protracted refugee situation. For each origin country, the dataset includes the principal **asylum countries** that together account for at least 75% of the origin's total refugee population, calculated using UNHCR refugee stock data. Dyad-years in which no refugee population is present are excluded.

Temporal coverage. The dataset spans 1980–2023. The year 1989 is missing for all variables coded from the World Refugee Survey (WRS), because the 1990 WRS report was not published.

Primary sources. GRR draws on three primary sources: (1) the World Refugee Survey (WRS), 1980–2008, providing annual country reports on refugee stocks and conditions in asylum; (2) the U.S. Department of State Country Reports on Human Rights Practices (USHR), 2009–2023, with data extracted from Section 2.d on freedom of movement; and (3) the UNHCR population statistics database, which supplies the stock and voluntary-return baseline for the years after 2008 and for the high-income asylum countries that enter only in the extended series. Additional variables are drawn from the UCDP Armed Conflict Dataset, the Ethnic Power Relations (EPR) Dataset, DWRAP (Blair et al., 2022), the World Bank World Development Indicators, and the CEPII GeoDist dataset. The UNHCR financing measures are compiled from UNHCR Executive Committee reports and archival documents for 1980–1997 and from UNHCR statistical yearbooks and financial reports for 1998–2023. Neither source contributes to the refugee stock or return series.

Core and extended versions. There are two versions of GRR. The **core** dataset covers Global South asylum countries from 1980 through 2008 and rests on the World Refugee Survey evidence in which I have the greatest confidence: 1,623 dyad-years across 105 dyads, 58 origin populations and 61 asylum countries. The **extended** series adds high-income asylum countries and continues through 2023 using the U.S. Department of State's annual country reports on human rights practices, giving 7,526 dyad-years across 293 dyads, 87 origin populations and 93 asylum countries. Because the State Department reports do not always provide stock or voluntary-return estimates, the extended series takes UNHCR figures as a baseline and supplements them with additional evidence on forced returns and unregistered populations. The extended series therefore offers broader geographic and temporal coverage, and the core series more consistent measurement.

1.1 Limitations of UNHCR Data

The case for reconstructing the return series from contemporaneous reporting rests on four sources of measurement error in UNHCR's population statistics, each of which pushes recorded return – and forced return in particular – below realized return. The illustrations below are drawn from the *World Refugee Survey*, the same corpus from which GRR is coded.

Unregistered refugees. The term "refugee" encompasses several legal and statistical categories. UNHCR's published definitions cover recognized refugees, asylum-seekers, people in refugee-like situations, and others in need of international protection. These data often miss unregistered refugees, including people who qualify as refugees but never apply for asylum and those whose refugee status is later revoked and who are subsequently treated as irregular migrants.

This gap matters especially for measuring forced return, because withdrawing formal status can itself facilitate removal. Governments may revoke refugees' status before deporting them, so those individuals may no longer be counted as refugees and their return may not appear in official refugee statistics. The Croatian government's treatment of Bosnian refugees in 1995 illustrates the problem:

> "Croatia revoked the refugee status of 100,000 Bosnian refugees … saying it was now safe for them to return. … The irony here is that the Croatian army, which is occupying many of those same towns, has refused to allow Muslims from Bosnia to come into those same places."
>
> *1996, Croatia*

Dependence on institutional presence. Accurate return estimates often depend on UNHCR's access and operational presence in the country. UNHCR may lack access because governments restrict its activities, because it has limited resources or no office, or because war and insecurity force personnel to evacuate. This creates a systematic measurement problem, since some of the same conditions that increase forced return – host-government reluctance to cooperate on refugee protection, and conflict in asylum – also make UNHCR less able to observe and record those movements. UNHCR data record no returns from South Africa to Mozambique, for example, despite reports of large-scale removals:

> "With no UNHCR presence in South Africa … [n]either country recognizes them as refugees … During 1985, South Africa was returning some 1,500 a month to the danger they had fled."
>
> *1986, South Africa*

The Somali civil war created a similar problem when it reached areas hosting Ethiopian refugees. UNHCR evacuated its personnel, and the resulting returns do not appear in its data:

> "Beginning in December 1990, as fighting in the Somali civil war intensified … UNHCR personnel were evacuated from Somalia … Large numbers of Ethiopian refugees streamed home to escape the fighting, many on foot. … More than 34,000 returned in January alone. … By the end of May, 64,000 returnees had reached Ethiopia."
>
> *1992, Ethiopia*

Measurement of forced and spontaneous returns. UNHCR's return statistics are designed to measure voluntary repatriation, not forced return; its own data-content documentation describes the solutions series in those terms. Deportations and refoulement therefore generally fall outside the data, while coerced movements may still be recorded as voluntary when they occur through formal repatriation channels. The data can thus both omit forced returns and obscure coercion among the returns they do record. Spontaneous returns are also difficult to observe: refugees may go uncounted if they leave without deregistering or cross at unmonitored border points.

The politics of population figures. UNHCR figures can also become objects of political negotiation. Host governments may understate returns in order to maintain higher refugee counts and attract more foreign aid. UNHCR may accept these figures when it cannot verify them independently, or when challenging them could jeopardize government cooperation or access to refugee populations. Somalia provides one example:

> "UNHCR had no firm estimate of the number of refugees in Somalia because the government did not permit a census. … During 1988, large numbers of Ethiopian refugees returned to Ethiopia … [but] [t]he Somali government denied that a mass repatriation had occurred. … The Somali government claimed that about 537,000 refugees remained in the country at year's end, but some voluntary agencies put the total at less than 400,000."
>
> *1989, Somalia*

These sources of error are systematic rather than random. Forced returns are especially likely to go unrecorded when refugees lose formal status, move outside assisted channels, return after humanitarian access has collapsed, or are classified as deportees rather than returnees. Analyses based only on UNHCR's return series may therefore understate forced return, particularly during ongoing conflict or following coercive policies in the country of asylum. Section 1.6 quantifies how much of GRR's recorded return the UNHCR series misses.

1.2 Codebook

Tables 1–5 describe all variables used in the main analysis. Variables with the prefix `a_` refer to asylum-country characteristics, `o_` to origin-country characteristics, and `d_` to dyad-level variables. Variable suffixes follow a consistent convention: `_z` denotes a standardized (z-scored) variable, `_l` a log-transformed variable, and `_zl` a standardized log-transformed variable. In the regression models, several variables are transformed for estimation. Binary high/low indicators (e.g., `a_conflict_high`, `a_conflict_low`) are derived from the ordinal variables by splitting them into separate dummies for the high and low categories, with zero as the reference. Continuous variables used in standardized or logged form in the regressions are noted below; standardization is performed on the estimation sample (dyad-years with more than 1,000 refugees in the previous year).

Table 1: Codebook: Identifiers

Variable	Description
`id`	Unique dyad identifier, formatted as COO–COA.
`year`	Year of observation.
`coo`	Country of origin name.
`coo_iso`	Country of origin ISO 3166-1 alpha-3 code.
`coa`	Country of asylum name.
`coa_iso`	Country of asylum ISO 3166-1 alpha-3 code.

Table 2: Codebook: Stock and Return Variables

Variable	Description	Source
`refugees`	Stock of forcibly displaced persons in the dyad at year-end, including refugees, asylum-seekers, individuals in refugee-like situations, and unregistered refugees. Primary stock variable.	WRS (1980–2008); UNHCR + USHR unregistered (2009–2023)
`refugees_formal`	Stock of forcibly displaced persons excluding unregistered refugees.	WRS (1980–2008); UNHCR (2009–2023)
`trepat`	Total refugees repatriated (voluntary + forced).	Derived: `vrepat` + `frepat`
`vrepat`	Voluntary repatriation count.	WRS (1980–2008); UNHCR (2009–2023)
`frepat`	Forced repatriation count.	WRS (1980–2008); USHR (2009–2023)

Table 3: Codebook: Asylum-Country Variables (prefix: `a_`)

Variable	Description	Source
`a_conflict`	Conflict intensity in the asylum country (UCDP): 0 = none, 1 = low (25–999 battle deaths), 2 = high ($\geq$1,000). In regressions, split into `a_conflict_high` (=1 if 2) and `a_conflict_low` (=1 if 1); no conflict is the reference category.	UCDP Armed Conflict Dataset
`a_conflict_bd`	Battle deaths in the asylum country (UCDP 'best' estimate). Available from 1989.	UCDP Battle-Related Deaths Dataset

Table 3: Codebook: Asylum-Country Variables (prefix: `a_`) *(continued)*

Variable	Description	Source
`a_forced`	Coercive exclusion policies in the asylum country at the dyad level: 0 = none; 1 = moderate (threats or encouragement to induce return without policy implementation, or implemented policies targeting <10% of the refugee stock); 2 = severe (implemented policies intended or expected to affect ≥10% of the refugee stock). Coding reflects the policy itself, not realized return. In regressions, split into `a_forced_high` (severe) and `a_forced_low` (moderate); no exclusion policy is the reference category. See Table 14 for example coding.	Hand-coded from WRS and USHR
`a_camp`	Ordinal share of the refugee population living in camps: 0 = no camps, 1 = <10% in camps, 2 = 10–50%, 3 = 50–95%, 4 = 95–100%.	Hand-coded from WRS and USHR
`a_camp_most`	Binary: 1 if most refugees live in camps (`a_camp` ∈ {3, 4}), 0 otherwise.	Derived from `a_camp`
`a_urban_z`	Standardized (z-score) measure of settlement type. Underlying ordinal variable: 3 if no camps or <10% in camps (`a_camp` ∈ {0, 1}), 2 if 10–50% in camps (`a_camp` = 2), 1 if 50–100% in camps (`a_camp` ∈ {3, 4}).	Derived from `a_camp`
`a_funding`	UNHCR funding allocated to the asylum country (USD, country-year level).	UNHCR ExCom (1980–2009); UNHCR Yearbooks (2010–2023)
`a_funding_pp`	UNHCR funding per refugee in the asylum country. In regressions, used in standardized log form (`a_funding_pp_zl`).	Derived: `a_funding` / asylum-country refugee stock
`a_rights`	Refugee rights index (inverse-covariance weighted) from DWRAP. In regressions, standardized (`a_rights_z`).	Blair et al. (2022)
`a_gdp_l`	Log GDP per capita of the asylum country.	World Bank WDI
`a_edu`	Mean years of schooling for the 20–24 age cohort of the asylum country's own resident population – its citizens, not the refugees it hosts (a near-completion measure of recent educational attainment), linearly interpolated from 5-year reporting intervals to annual frequency.	Wittgenstein Centre
`a_region`	World Bank regional classification of the asylum country.	World Bank

Table 4: Codebook: Origin-Country Variables (prefix: `o_`)

Variable	Description	Source
`o_conflict`	Conflict intensity in the origin country (UCDP): 0 = none, 1 = low (25–999 battle deaths), 2 = high (≥1,000). Matched to the specific conflict driving displacement. Missing for non-conflict displacement (e.g., Venezuela). In regressions, split into `o_conflict_high` and `o_conflict_low`; no conflict is the reference category.	UCDP Armed Conflict Dataset

Table 4: Codebook: Origin-Country Variables (prefix: `o_`) *(continued)*

Variable	Description	Source
`o_conflict_bd`	Battle deaths in the origin country (UCDP 'best' estimate). Available from 1989. Matched to the specific conflict producing refugees.	UCDP Battle-Related Deaths Dataset
`o_conflict_years`	Years elapsed in the origin-country conflict spell matched to this displacement. The counter increments in any year of armed conflict at either intensity, carries through a single lapse year, and resets to zero after two consecutive years without conflict; it is the age of the current spell rather than a cumulative total. Left-truncated at 1980, so spells beginning earlier are counted from the panel's first year. Because `o_conflict` is matched to the conflict driving each displacement, the counter can differ across the hosts of one origin in the same year. In regressions, used in standardized log form (`o_conflict_years_zl`), captured as $\log(\text{years} + 0.01)$ and standardized; 68 per cent of dyad-years are zero, so the standardized measure separates an active spell from none more sharply than it distinguishes long spells from short ones.	Derived from UCDP Armed Conflict Dataset
`o_funding_repat_pp`	UNHCR repatriation funding allocated to the origin country, divided by the origin's global refugee population (USD per refugee). In regressions, used in standardized log form (`o_funding_repat_pp_zl`).	UNHCR ExCom (1980–1999); UNHCR Yearbooks (2011–2023). 2000–2010 missing.
`o_eth_name`	Name of the refugee group's ethnic identity (e.g., "Kurds," "Tutsi," "Bosniaks/Muslims").	EPR Refugee Dataset
`o_eth_id`	EPR group identifier (`gwgroupid`) for the refugee group.	EPR Refugee Dataset; EPR Dataset
`o_eth_status`	EPR power-status category of the refugee group in the origin country (categorical: monopoly, dominant, senior partner, junior partner, powerless, discriminated, irrelevant, self-exclusion, state collapse).	EPR Dataset
`o_eth_high`	Binary: 1 if the refugee group holds high political status ('monopoly,' 'dominant,' 'senior partner,' or 'junior partner'), 0 otherwise.	EPR Dataset; EPR Refugee Dataset
`o_eth_low`	Binary: 1 if the refugee group holds low political status ('discriminated' or 'powerless'), 0 otherwise.	EPR Dataset; EPR Refugee Dataset
`o_eth_power`	Ordinal power score (0–6) derived from `o_eth_status`: 6 = monopoly, 5 = dominant, 4 = senior partner, 3 = junior partner, 2 = state collapse, 1.5 = discriminated, 1 = powerless, 0.5 = irrelevant, 0 = self-exclusion.	Derived from `o_eth_status`
`o_eth_cat`	Three-category collapse of `o_eth_status`: "In Power" (monopoly, dominant, senior/junior partner), "Excluded" (powerless, discriminated), or "Other" (irrelevant, self-exclusion, state collapse).	Derived from `o_eth_status`
`o_eth_inpower`	Binary: 1 if the refugee group is "In Power," 0 otherwise. Equivalent to `o_eth_high`.	Derived from `o_eth_status`
`o_eth_excluded`	Binary: 1 if the refugee group is "Excluded," 0 if "In Power," NA for "Other" categories. Differs from `o_eth_low` in coding "Other" as NA rather than 0.	Derived from `o_eth_status`

Table 4: Codebook: Origin-Country Variables (prefix: `o_`) *(continued)*

Variable	Description	Source
`o_eth_claim`	Binary indicator for whether the rebel group in the conflict producing refugees has made an exclusive claim to fight on behalf of an ethnic group (ACD2EPR).	ACD2EPR (Wucherpfennig et al.)
`o_civlib_z`	Freedom House Civil Liberties index for the origin country, standardized.	Freedom House
`o_terror_z`	Political Terror Scale for the origin country, standardized.	Political Terror Scale (Amnesty International, Human Rights Watch, U.S. State Department)
`o_genocide`	Binary indicator for genocide or politicide in the origin country.	PITF Genocide/Politicide Dataset
`o_pitf_any`	Binary indicator for any PITF political violence in the origin country (ethnic war or revolutionary war).	PITF Ethnic/Revolutionary War Datasets
`o_pagree`	Binary indicator for a peace agreement signed in the origin country in a given year.	UCDP Peace Agreement Dataset
`o_gdp_l`	Log GDP per capita of the origin country.	World Bank WDI
`o_edu`	Mean years of schooling for the 20–24 age cohort of the origin country's own resident population – citizens who did not flee, not the refugee population abroad (a near-completion measure of recent educational attainment), linearly interpolated from 5-year reporting intervals to annual frequency.	Wittgenstein Centre
`o_region`	World Bank regional classification of the origin country.	World Bank

Table 5: Codebook: Dyad-Level Variables (prefix: `d_`)

Variable	Description	Source
`d_distance`	Distance between origin and asylum countries (km). In regressions, used in log form (`d_distance_l`).	CEPII GeoDist
`d_contig`	Binary: 1 if the countries share a border, 0 otherwise.	CEPII GeoDist

1.3 Summary Statistics

Table 6 reports summary statistics for every variable in the main specification, followed by the between-dyad measures that enter the pooled column only. Statistics are computed on the estimation sample (dyad-years with more than 1,000 refugees in the previous year).

Figure 1 shows the distribution of the dependent variable. The left panel displays the full range of total repatriation counts with the y-axis capped to show detail among non-zero values. The right panel plots positive values on a log scale, showing a roughly log-normal distribution among dyad-years with nonzero repatriation. The heavy concentration of zeros, extreme right skew, and count nature of the dependent variable motivate the use of Poisson pseudo-maximum likelihood estimation with the log stock at risk as the offset.

Table 6: Summary statistics

Variable	Obs	Mean	Std. Dev.	Min	Max
Return and stock					
Total Repatriation	5109	6,513.72	48,844.54	0.00	1,700,000.00
Refugee Stock (Lagged)	5131	129,495.83	400,251.60	1,002.00	6,013,249.00
Return Rate (% of stock at risk)	5109	2.99	11.03	0.00	100.00
Exclusion policy					
Exclusion policy: severe	4415	0.03	0.18	0.00	1.00
Exclusion policy: moderate	4415	0.07	0.25	0.00	1.00
Conflict in COA: high	5131	0.07	0.26	0.00	1.00
Conflict in COA: low	5131	0.18	0.38	0.00	1.00
Existing explanations					
War Ending	5131	0.04	0.20	0.00	1.00
Conflict in COO: high	5131	0.12	0.32	0.00	1.00
Conflict in COO: low	5131	0.22	0.42	0.00	1.00
GDP (COO)	4845	-0.00	1.00	-2.98	2.22
GDP (COA)	5095	-0.00	1.00	-2.23	1.45
Civil Liberties Restrictions (COO)	5086	0.00	1.00	-2.82	1.44
Refugee Rights	4911	-0.00	1.00	-2.16	3.79
Conflict Duration	5131	0.00	1.00	-0.78	1.83
Measures entering the pooled specification only					
Distance (log)	5069	7.49	1.20	2.35	9.69
Contiguous	5069	0.49	0.50	0.00	1.00
Refugee Stock at Risk (log)	5131	9.92	1.86	6.91	15.61
Refugee Convention	5131	0.85	0.36	0.00	1.00
Common spoken language	5097	0.35	0.48	0.00	1.00
Religious similarity (z)	4956	-0.00	1.00	-1.48	1.63
Kin share of origin population (z)	5131	0.00	1.00	-0.52	3.24

Note: Computed on the filtered sample: dyad-years with a lagged refugee stock above 1,000.

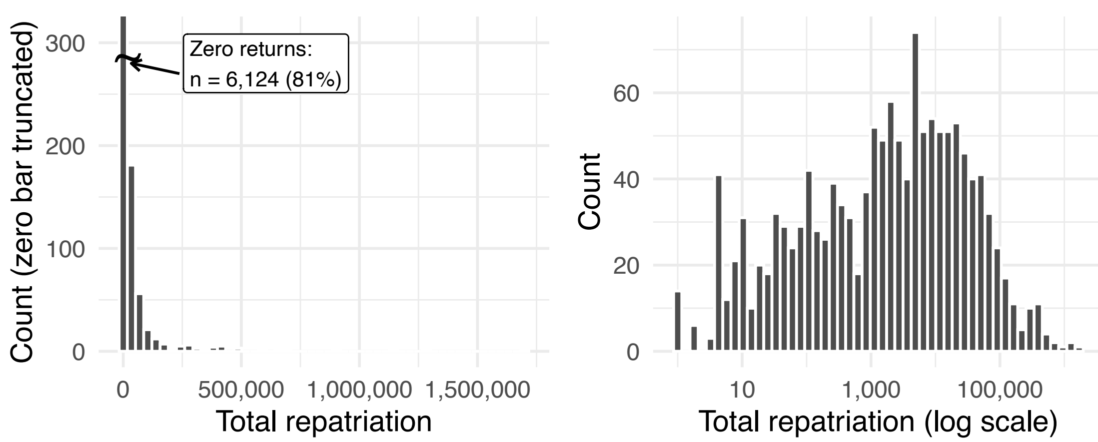

Figure 1: Distribution of the dependent variable (total repatriation). Left panel shows the raw distribution; right panel uses a log scale and excludes zeros.

1.4 Cases

The analysis includes 293 dyads. The tables below list the dyads with the most return, the largest single-year movements, and the largest populations with little or no return. They use the full GRR panel rather than the estimation sample, so they also include dyads omitted from the fixed-effects models.

Table 7: The twenty dyads with the most recorded return, 1980–2023

Dyad	Origin → asylum	Years	Total return	Peak stock	Return/peak (%)	% yrs COO conflict	Push ever
AFG-IRN	Afghanistan → Iran	1980–2023	7,507,327	6,352,317	118	100	Both
AFG-PAK	Afghanistan → Pakistan	1980–2023	7,018,397	3,997,133	176	100	Both
IRQ-IRN-KUR	Iraq (Kurds) → Iran	1980–2023	1,426,658	160,000	892	33	Both
RWA-COD	Rwanda → Congo - Kinshasa	1980–2023	1,190,071	1,000,000	119	49	Both
MOZ-MWI	Mozambique → Malawi	1985–2023	1,126,986	1,085,000	104	44	None
BDI-TZA	Burundi → Tanzania	1980–2023	1,049,147	443,700	236	53	Both
ETH-SOM	Ethiopia → Somalia	1980–2023	1,039,754	705,000	147	52	COA conflict
MMR-THA	Myanmar (Burma) → Thailand	1984–2023	735,024	1,653,500	44	72	Both
KHM-THA	Cambodia → Thailand	1981–2023	631,632	365,000	173	45	Both
SDN-SSD-UGA	Sudan → Uganda	1985–2023	623,004	1,037,898	60	97	COA conflict
SSD-SDN	South Sudan → Sudan	2014–2023	572,695	852,080	67	80	COA conflict
SYR-TUR	Syria → Turkey	2009–2023	559,246	3,737,369	15	87	COA conflict
SOM-ETH	Somalia → Ethiopia	1986–2023	532,300	419,000	127	81	COA conflict
UKR-POL	Ukraine → Poland	2002–2023	468,475	958,519	49	45	None
ETH-ERI-SDN	Ethiopia → Sudan	1980–2023	433,357	730,000	59	26	COA conflict
LBR-GIN	Liberia → Guinea	1988–2023	404,223	420,000	96	14	COA conflict
AGO-COD	Angola → Congo - Kinshasa	1980–2023	383,166	312,700	123	40	Both
UGA-SDN	Uganda → Sudan	1980–2010	382,250	250,000	153	95	COA conflict
MOZ-ZAF	Mozambique → South Africa	1984–2023	359,700	300,000	120	50	Both
CIV-LBR	Côte d'Ivoire → Liberia	2002–2023	328,376	128,067	256	18	COA conflict

Note: Return over peak stock can exceed 100 per cent where a population arrived and left more than once over the period.

Table 8: The forty largest single-year return episodes

Dyad	Year	Origin → asylum	Return	Stock at risk	Rate (%)	COO	COA	Forced
AFG-PAK	2002	Afghanistan → Pakistan	1,700,000	2,100,000	81.0	Low	Low	Moderate
IRQ-IRN-KUR	1991	Iraq (Kurds) → Iran	1,295,000	1,366,400†	94.8	Low	Low	None
AFG-IRN	2020	Afghanistan → Iran	1,151,789	2,701,142	42.6	High	Low	Severe
AFG-PAK	1992	Afghanistan → Pakistan	910,000	3,997,133	22.8	High	Low	None
RWA-COD	1996	Rwanda → Congo - Kinshasa	700,000	900,000	77.8	Low	High	None
AFG-IRN	2019	Afghanistan → Iran	668,209	2,701,142	24.7	High	Low	Severe
MOZ-MWI	1994	Mozambique → Malawi	600,000	700,000	85.7	None	None	None
AFG-IRN	2005	Afghanistan → Iran	489,600	1,702,800	28.8	High	Low	Severe
AFG-PAK	2005	Afghanistan → Pakistan	475,100	967,200	49.1	High	None	Severe
AFG-IRN	1993	Afghanistan → Iran	471,500	2,862,500	16.5	High	Low	Severe
AFG-IRN	2004	Afghanistan → Iran	419,500	1,600,000	26.2	Low	None	Severe
AFG-PAK	2004	Afghanistan → Pakistan	409,100	1,200,000	34.1	Low	Low	None
AFG-IRN	2008	Afghanistan → Iran	409,000	2,414,700	16.9	High	Low	Severe
UKR-POL	2022	Ukraine → Poland	403,000	1,361,519†	29.6	High	None	None
MOZ-MWI	1993	Mozambique → Malawi	400,000	1,085,000	36.9	None	None	None
SSD-SDN	2023	South Sudan → Sudan	386,793	796,831	48.5	None	High	Moderate
AFG-PAK	2016	Afghanistan → Pakistan	381,275	2,366,950	16.1	High	Low	None
AFG-IRN	2007	Afghanistan → Iran	379,650	2,190,400	17.3	High	Low	Severe
AFG-IRN	2002	Afghanistan → Iran	376,300	2,855,000	13.2	Low	None	Severe
AFG-PAK	1993	Afghanistan → Pakistan	358,800	2,075,000	17.3	High	None	None
AFG-PAK	2003	Afghanistan → Pakistan	343,537	1,500,000	22.9	Low	Low	Moderate
ETH-SOM	1984	Ethiopia → Somalia	320,554	700,000	45.8	None	Low	None
ETH-SOM	1991	Ethiopia → Somalia	320,000	355,000	90.1	None	High	None
AFG-IRN	1992	Afghanistan → Iran	300,000	3,030,555	9.9	High	None	None
AFG-IRN	2003	Afghanistan → Iran	285,400	2,500,000	11.4	Low	None	Severe
AFG-IRN	2018	Afghanistan → Iran	275,061	2,451,142	11.2	High	Low	Severe
AFG-IRN	2017	Afghanistan → Iran	274,291	2,451,149	11.2	High	Low	Severe
AFG-IRN	2006	Afghanistan → Iran	264,350	1,940,000	13.6	High	Low	Severe
AFG-IRN	2001	Afghanistan → Iran	263,501	1,982,000	13.3	High	Low	Severe
AFG-PAK	2008	Afghanistan → Pakistan	261,220	2,018,150	12.9	High	High	Severe
AFG-PAK	2007	Afghanistan → Pakistan	253,590	2,280,000	11.1	High	Low	Severe
KHM-THA	1992	Cambodia → Thailand	249,000	365,000	68.2	Low	None	None
AFG-IRN	1995	Afghanistan → Iran	245,972	1,645,500	14.9	High	None	Severe
AFG-IRN	2016	Afghanistan → Iran	234,298	3,951,175	5.9	High	Low	Severe

Table 8: The forty largest single-year return episodes *(continued)*

Dyad	Year	Origin → asylum	Return	Stock at risk	Rate (%)	COO	COA	Forced
AFG-IRN	1994	Afghanistan → Iran	227,387	1,925,000	11.8	High	None	Severe
ETH-SOM	1983	Ethiopia → Somalia	200,000	700,000	28.6	Low	Low	None
RWA-COD	1997	Rwanda → Congo - Kinshasa	189,000	200,000	94.5	High	High	Moderate
AFG-IRN	2000	Afghanistan → Iran	183,600	1,825,000	10.1	High	Low	Severe
SDN-SSD-UGA	2021	South Sudan → Uganda	180,356	889,054	20.3	Low	None	None
SYR-TUR	2018	Syria → Turkey	177,305	3,424,399	5.2	High	Low	Moderate

Note: *dagger* marks dyad-years in which recorded return exceeds the lagged stock, where the denominator is the year-end stock plus return rather than the lagged stock, so the rate is an upper bound. COO and COA give conflict intensity in the origin and asylum country that year; Forced gives the asylum country's exclusion policy.

Table 9: The non-return cases

Dyad	Origin → asylum	Years	Refugee-years (m)	Peak stock	Total return	% yrs COO conflict	Push ever
PSE-JOR	Palestinian Territories → Jordan	1981–2008	44.62	2,439,718	233	89	None
VNM-CHN	Vietnam → China	1980–2021	11.97	321,502	0	15	COA conflict
VEN-USA	Venezuela → Colombia	2009–2023	11.93	2,875,743	0	0	COA conflict
PSE-SYR	Palestinian Territories → Syria	1980–2016	10.62	592,200	233	83	COA conflict
PSE-LBN	Palestinian Territories → Lebanon	1981–2023	9.24	500,000	233	81	COA conflict
VEN-PER	Venezuela → Peru	2003–2023	6.98	1,542,004	0	0	COA conflict
ESH-DZA	Western Sahara → Algeria	1980–2023	5.89	165,000	0	21	COA conflict
COL-ECU	Colombia → Ecuador	1990–2023	4.71	674,500	5,102	85	COA conflict
CHN-IND	China → India	1984–2023	4.01	123,000	0	0	COA conflict
UKR-RUS	Ukraine → Russia	1996–2023	3.56	1,275,925	0	42	COA conflict
PHL-MYS	Philippines → Malaysia	1981–2023	2.70	136,075	0	91	COA conflict
BTN-NPL	Bhutan → Nepal	1991–2023	2.33	133,700	0	0	Both
RUS-USA	Russia → United States	1988–2023	2.22	245,238	0	0	None
ERI-ETH	Eritrea → Ethiopia	1993–2023	2.03	181,271	0	8	COA conflict
TUR-DEU	Turkey → Germany	2000–2023	1.98	181,344	0	96	None
IRN-TUR	Iran → Turkey	1982–2023	1.98	257,000	2,208	53	Both
CHN-USA	China → United States	1988–2023	1.96	173,873	0	0	Moderate only
VEN-BRA	Venezuela → Brazil	2011–2023	1.93	510,499	0	0	None
GTM-USA	Guatemala → United States	1988–2023	1.76	192,596	0	20	Moderate only
ARM-AZE	Armenia → Azerbaijan	1992–2013	1.56	229,000	0	38	COA conflict

Note: The twenty largest dyads whose cumulative return never reaches 1 per cent of peak stock. Ranked by refugee-years, the sum of the annual stock over the dyad's observed history, which measures how much displacement the dyad represents rather than how long it is observed. The three Palestinian dyads each record the same 233 returns in 1986, which is one reported figure carried across all three rather than three independent observations.

1.5 Limitations of GRR

Three limitations follow from the sources GRR is built on.

First, the *World Refugee Survey* trades numerical precision for breadth of coverage. Its country reports were written to document conditions in asylum rather than to compile a statistical series, and they frequently give rounded estimates, ranges, or several conflicting figures for the same population in the same year. GRR applies standardized coding rules to these cases, and the diagnostics in Section 1.6 show their signature: GRR holds a stock unchanged from the previous year more often than UNHCR does, and reports exact multiples of ten thousand more often, because a range repeated across reports is carried forward rather than resolved into movement between its bounds. The series is therefore more stable within dyads, and visibly less granular, than an administrative series would be, and a single dyad-year figure should be read as a contemporaneous estimate rather than as a count.

Second, coverage is uneven across asylum countries. It is least complete for some high-income asylum states, whose reports discuss the treatment of refugees in aggregate rather than by country of origin, so origin-specific stocks and returns often cannot be recovered. This is why the core dataset is restricted to asylum outside Europe and North America, and why the extended series takes UNHCR's figures as its stock and voluntary-return baseline from 2009 onwards rather than coding them independently. The comparison in Section 1.6 is correspondingly most informative for 1980–2008, the years in which the two series were produced independently of one another.

Third, there are no observations for 1989. The *World Refugee Survey* was not published in 1990, so every variable coded from it is missing for the preceding reference year. The gap is left missing rather than interpolated, and the lag and lead operators used throughout the analysis refuse to span it, so no within-dyad change is computed across the break.

None of this is unique to GRR: UNHCR's series rests on government and field estimates that carry the same rounding and the same reporting gaps, as the internal-consistency diagnostics in Section 1.6 show. GRR is best understood as an improved but still imperfect measure of movements that are inherently difficult to observe, and the analysis is reported on both the core and the extended panels so that the reader can see where the two disagree.

1.6 Comparison of GRR and UNHCR

I compare GRR's refugee-stock and return series with the corresponding UNHCR series. GRR's stock data include unregistered refugees, while its total-return measure includes both voluntary and forced return. The comparison uses the GRR core dataset and excludes populations that UNHCR does not record, such as Palestinians under UNRWA. Restricting the sample to dyad-years observed in both sources leaves 1,415 dyad-years across 96 dyads.

This section reports the comparison of the two return series in full and then extends it. I examine the stock data over the full panel, show the differences through dyadic plots, check the internal consistency of each series, and test whether replacing GRR with UNHCR changes the paper's main estimates.

1.6.1 Voluntary and Forced Return

Table 10 sets the two return series side by side. At the aggregate level the two datasets agree closely on voluntary return. UNHCR records 19.1 million returns across the comparison sample, compared with 19.5 million voluntary returns in GRR.

Table 10: Comparison of GRR and UNHCR return data

	1980s	1990s	2000s	All
Sample				
Dyad-years compared	319	563	533	1,415
Dyads	51	83	79	96
Aggregate calibration				
GRR voluntary return	1,770,932	10,050,798	7,643,847	19,465,577
GRR forced return	259,027	957,565	2,337,900	3,554,492
UNHCR recorded return	1,250,530	9,757,187	8,105,436	19,113,153
UNHCR / GRR voluntary	0.71	0.97	1.06	0.98
UNHCR / GRR total	0.62	0.89	0.81	0.83
Episode capture				
Forced-return episodes	21	61	75	157
invisible in UNHCR	14 (67%)	14 (23%)	21 (28%)	49 (31%)
token in UNHCR	0	2	5	7
Share of forced-return volume				
Forced-return volume in invisible episodes	71%	16%	34%	32%
in invisible or token episodes	71%	26%	53%	47%
where UNHCR ≈ GRR voluntary	60%	36%	72%	62%

Note: The comparison covers the GRR core dataset, 1980–2008, with asylum countries outside Europe and North America. It excludes dyad-years in which GRR's voluntary-return estimate is copied from or defined using UNHCR data. The 2000s column covers 2000–2008. An episode is a dyad-year in which GRR records forced return. *Invisible* means that UNHCR records no return in that year or either adjacent year; *token* means that UNHCR records a positive number smaller than one-tenth of the GRR forced-return estimate. *UNHCR ≈ GRR voluntary* denotes forced-return volume in dyad-years where UNHCR's figure is within 5% of GRR's voluntary-return estimate.

Once GRR's 3.6 million forced returns are included, however, UNHCR records only 83% of total realized return. Of the 157 dyad-years in which GRR identifies forced return, 49 (31%) are invisible in UNHCR: it records no return in that dyad-year, the preceding year, or the following year. Another seven appear only as token movements, where UNHCR's count across those three years is positive but less than one-tenth of GRR's forced-return estimate for that one dyad-year. Together, these episodes account for 47% of GRR's forced-return volume.

Moreover, 62% of forced-return volume occurs in dyad-years where UNHCR's figure closely matches GRR's voluntary-return estimate, which suggests that the forced component is omitted rather than reclassified. In other cases coercive movements appear in the UNHCR data as voluntary repatriation. GRR classifies Rohingya returns from Bangladesh between 1994 and 1997 as forced, for example, while UNHCR records approximately the same number as voluntary.

As Table 10 shows, the discrepancies are greatest in the 1980s but persist into later decades. In the 1980s UNHCR records 71% of GRR's voluntary-return total and two-thirds of GRR's forced-return episodes are invisible; the correspondence improves considerably thereafter, but forced-return episodes remain likely to be omitted through the 2000s and the two series never fully converge. Neither dataset provides an independent ground truth for every movement, but the concentration of disagreement in forced-return episodes is consistent with the coverage problems set out in Section 1.1.

1.6.2 Comparing Refugee Stocks

In this section, I compare GRR and UNHCR on stock data, in the same way as the analysis of returns above. This subsection uses the core dataset, covering 1980–2008 and excluding asylum in Europe and North America. Logged formal stocks correlate at 0.86 and are exactly equal in 7% of dyad-years. The two series therefore track each other closely, but they are not identical.

Within the core dataset, the larger difference concerns GRR's broad stock, which includes unregistered refugees absent from UNHCR's formal statistics. Table 11 compares registered and broad stocks in the same way that Table 10 compares voluntary and forced return. For registered refugees, UNHCR records 0.82 of GRR's total. For the broad stock, the ratio falls to 0.61, a difference of 90.6 million unregistered refugee stock-years. Of the 424 dyad-years in which GRR records an unregistered population, 108, or 25%, have no UNHCR stock at all, while another 62 have an UNHCR figure below one-tenth of GRR's.

The stock and return comparisons therefore show a similar pattern: populations missing from formal registration are also less likely to appear in recorded returns. The timing differs, however. Return coverage is weakest in the 1980s and improves thereafter, whereas coverage of GRR's broad stock is weakest in the 2000s, when the ratio falls to 0.44. This is also the decade in which GRR records the largest unregistered refugee population.

Table 11: Registered and unregistered refugee stocks, GRR against UNHCR

Quantity	1980s	1990s	2000s	All
Sample				
Dyad-years compared	371	638	605	1,614
Dyads	55	90	89	104
Aggregate calibration				
GRR registered stock	78,321,850	104,907,809	83,528,312	266,757,971
GRR unregistered stock	14,177,285	27,405,166	49,059,410	90,641,861
UNHCR recorded stock	73,605,396	86,767,560	58,795,551	219,168,507
UNHCR / GRR registered	0.94	0.83	0.70	0.82
UNHCR / GRR total	0.80	0.66	0.44	0.61
Population capture				
Dyad-years with unregistered population	98	162	164	424
invisible in UNHCR	38 (39%)	40 (25%)	30 (18%)	108 (25%)
token in UNHCR	5	26	31	62
Share of unregistered volume				
Unregistered stock-years in invisible dyad-years	35%	45%	10%	25%

Note: Core dataset only: 1980–2008, asylum outside Europe and North America. Stocks are summed over dyad-years, so the unit is a stock-year. A dyad-year is *invisible* when UNHCR records no stock for it and *token* when UNHCR's figure is below a tenth of GRR's broad stock. Built to the same row blocks as the comparison of voluntary and forced return above.

In the later years of the extended dataset, GRR and UNHCR are more closely aligned. That convergence partly reflects a change in how GRR is constructed. After the *World Refugee Survey* ends, GRR takes UNHCR's registered stock as its starting point, then adds unregistered refugee populations identified in State Department reports and other archival sources. The two series therefore share the same baseline after 2009, but GRR remains broader where those additional populations can be identified.

During the *World Refugee Survey* period, the two stock series are identical in 55.9% of dyad-years, while GRR exceeds UNHCR in roughly one-quarter. In the State Department period, they are identical in 95.3%, and GRR exceeds UNHCR in fewer than one dyad-year in twenty. Where GRR does exceed UNHCR, the median difference is 76%.

The return data show the same shift. GRR and UNHCR report identical return figures in 9.7% of dyad-years before 2009, compared with 84.0% after 2009. GRR's independent contribution is therefore concentrated before 2009. In later years, it is better understood as UNHCR's series supplemented with archival additions and corrections. The comparison between the two sources is therefore most informative for 1980–2008, when they were produced more independently.

1.6.3 Internal Consistency

Table 12: Internal-consistency diagnostics on the stock series

Series	Returns exceed prior stock (%)	Unexplained collapse (%)	V-shaped reversal (%)	Spike then collapse (%)	Unchanged from prior year (%)	Exact multiple of 10,000 (%)	Median abs. log change
GRR stock	0.365	3.28	0.848	0.348	8.37	7.49	0.111
UNHCR stock	0.437	3.54	0.941	0.282	4.15	1.82	0.113

Note: The first four columns are impossibilities or near-impossibilities; the last three are fingerprints of estimation rather than measurement.

The internal-consistency checks are similar across the two sources. Pairing GRR's returns with its stock puts 0.36% of dyad-years above a rate of one, and the corresponding share for UNHCR is similar. GRR holds a stock unchanged from the previous year more often than UNHCR does, and reports round multiples of ten thousand more often. This follows from the coding rule: when an archival range does not change, GRR carries the previous estimate forward rather than moving between bounds, and the underlying sources often report rounded figures. The result is a more stable within-dyad series that is also visibly less granular.

1.6.4 Replicating the Main Analysis

Table 13: Alternative return series and offsets

Term	(1) GRR returns GRR stock	(2) UNHCR returns GRR stock	(3) UNHCR returns UNHCR stock
Exclusion policy: severe	1.09*** (0.19)	0.51* (0.22)	0.30 (0.24)
Exclusion policy: moderate	0.48* (0.22)	-0.23 (0.16)	-0.31† (0.16)
Conflict in COA: high	0.59† (0.31)	0.11 (0.29)	-0.04 (0.29)
Conflict in COA: low	0.81*** (0.21)	0.14 (0.20)	0.18 (0.21)
War Ending	1.00*** (0.22)	1.32*** (0.21)	1.29*** (0.22)
Conflict in COO: high	0.11 (0.28)	0.65† (0.37)	0.80* (0.37)
Conflict in COO: low	0.30 (0.22)	0.60* (0.25)	0.64* (0.27)
GDP (COA)	-1.09† (0.59)	0.31 (0.66)	-0.69 (0.71)
GDP (COO)	0.39 (0.28)	-0.02 (0.30)	-0.04 (0.31)
Civil Liberties Restrictions (COO)	-0.51** (0.17)	0.10 (0.16)	0.20 (0.16)
Refugee Rights	-0.02 (0.17)	0.02 (0.24)	0.02 (0.22)
Conflict Duration	-0.03 (0.08)	0.04 (0.11)	-0.02 (0.12)
Dyad-years	2,327	2,297	2,126
Dyads	110	109	107

Note: Every column is the paper's specification — the right-hand side of the baseline table, dyad and year fixed effects, log stock at risk as offset, dyad-clustered standard errors — re-estimated with the dependent variable and the offset swapped for other measures. Column 1 is the paper's baseline, reproduced coefficient for coefficient. Samples differ across columns because a rung drops the dyad-years whose own offset is missing or zero. Significance: [dagger] 0.10, [*] 0.05, [**] 0.01, [***] 0.001.

This section replaces GRR's return and stock measures with UNHCR's and examines how the results change. In Table 13, Column 1 reproduces the main specification. Column 2 replaces the outcome with UNHCR's return series, while Column 3 also replaces the refugee stock, yielding a specification based entirely on UNHCR measures.

The results change substantially. Neither severe exclusion policies nor conflict in the asylum country is associated with higher return in the UNHCR specifications. The coefficient on severe exclusion policy falls from 1.09 in Column 1 to 0.51 when UNHCR's return series is used, and to 0.30 when UNHCR's stock measure is used as well. Neither UNHCR estimate is distinguishable from zero. The same is true of conflict in the asylum country, which is coded independently of either return series.

Other coefficients also move in less expected directions. Conflict in the origin is positively associated with return in both UNHCR specifications, even after including the covariates and fixed effects. In the full UNHCR specification, a moderate exclusion policy is associated with *lower* return, although only weakly significantly. At the same time, none of the economic or political conditions in either the origin or asylum country that are commonly thought to shape return is statistically significant in either UNHCR specification.

One possible explanation is that UNHCR's return series reflects where the organization is able to observe and record return most consistently. If coverage is systematically greater in some settings than others, those coverage patterns could enter the estimates along with the underlying return process. I do not have direct evidence that this is what produces the results, however, so I do not treat this explanation as a finding. More generally, neither GRR nor UNHCR provides a ground-truth measure of return.

1.6.5 Dyadic Return Patterns

Figures 2 and 3 plot refugee stocks and total returns over time for the forty largest dyads in the core dataset: the two largest asylum destinations for each of the twenty largest origin countries by displaced population. GRR is shown in black and UNHCR in red. Across most dyads, UNHCR records lower stocks and returns than GRR.

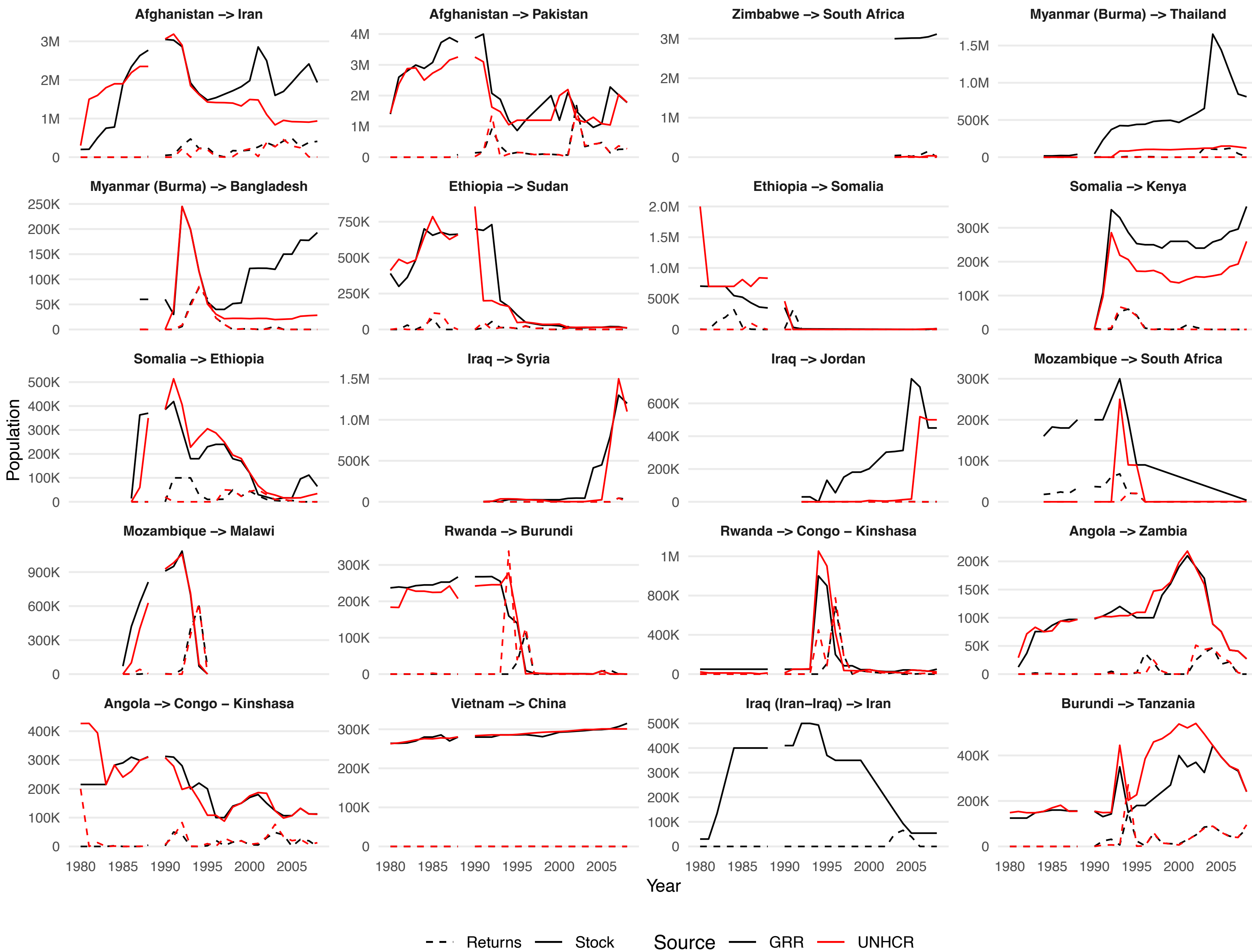

Figure 2: Refugee Stocks and Returns by Dyad (1 of 2)

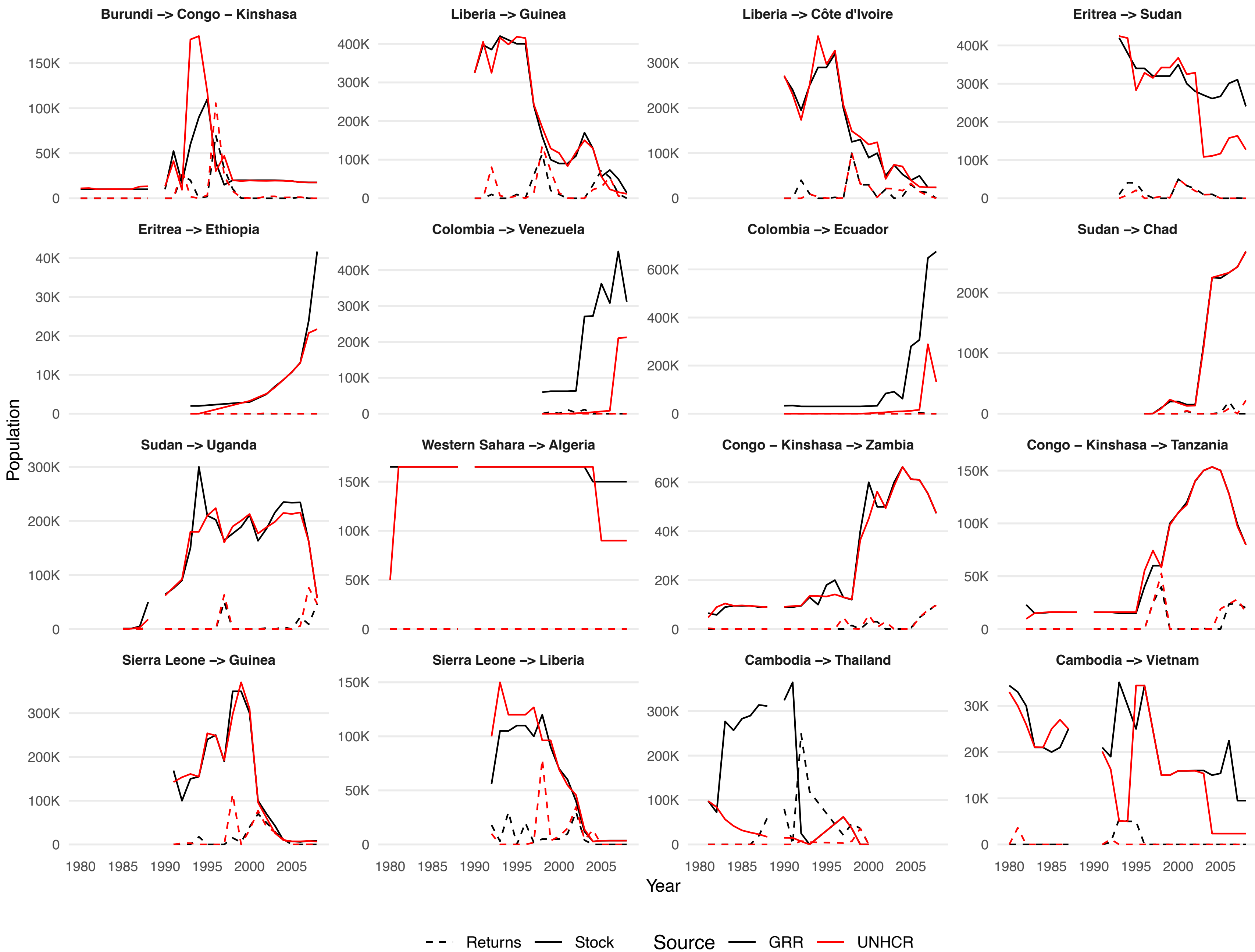

Figure 3: Refugee Stocks and Returns by Dyad (2 of 2)

Table 14: Examples of Exclusion Policies

Moderate exclusion policies	**Severe exclusion policies**
"In a move protested by UNHCR, authorities in Congo-Brazzaville forced 19 asylum seekers back to Congo-Kinshasa in April." (COD–COG, 2002)	"South Africa enacted new legislation in 1991, the Aliens Control Act, giving immigration authorities even more power to seek out and deport Mozambicans... South Africa has forcibly repatriated thousands of Mozambican refugees to a land of deprivation and death in recent years. This practice violates international standards and must cease," USCR stated. (MOZ–ZAF, WRS 1993)
"The Tibetan refugee leadership encourages most of those on pilgrimages and many who come to study to return to Tibet, both to counteract the overwhelming Chinese presence in Tibet and because the two dozen Tibetan refugee settlements in India cannot absorb so many people." (CHN–IND, 1999)	"In April, the Government announced that it would expel all Rwandans whose cases it rejected and the number of applicants declined substantially... Deterred by the low acceptance rates, some 2,000 never registered with the Government and more than 10,500 of those who registered dropped their claims and returned." (RWA–BDI, WRS 2006)
"There were scattered reports that the government intended to expel Palestinians who had not arrived in 1948, but who had come to Lebanon from other Arab countries, such as Egypt and Syria. If implemented, this could affect up to 50,000." (PAL–LEB, 1993)	"The events in Zaire produced strains in Congo that resulted in the Congolese authorities forcibly deporting some 10,000 Zaireans in November." (COD–COG, WRS 1992)
"UNHCR expressed concern that some refugees were being manipulated to repatriate prematurely to Congo-Kinshasa based on misinformation about conditions there. Some refugees reportedly repatriated and had to flee again to Rwanda because conditions in Congo-Kinshasa remained dangerous." (COD–RWA, 2000)	"Yemeni authorities forcibly returned 438 Somalis, at least 200 of whom were UNHCR-recognized refugees. They had been part of a group of 1,200 persons who had been rounded up in Aden as part of a sweep of the city for illegal immigrants." (SOM–YEM, 1996)
"In western Sudan, 45,000 Chadian refugees remain in UNHCR-assisted reception centers despite government and UNHCR attempts to persuade them to return to Chad."	"UNHCR protested when the government's roundup of 1,000 undocumented immigrants in January resulted in the detention of several refugees who possessed UNHCR permits. Persons judged to be undocumented were deported without UNHCR screening to determine claims to refugee status." (COD–ZMB, 1993)

1.7 Exclusion-Policy Measurement and Validation

This section sets out how the exclusion-policy measure is coded and discusses validation checks.

1.7.1 Measurement

The *Exclusion Policy* variable is hand-coded from the qualitative country narratives in the *World Refugee Survey* and, after the Survey ceased publication, the U.S. State Department's *Country Reports on Human Rights Practices*. Non-zero dyad-years are assigned to one of two categories, *moderate* or *severe*, using two criteria: (i) whether forced return is implemented or only threatened, and (ii) the share of the refugee stock the policy is intended or expected to affect.

A dyad-year is coded **severe** when the asylum government implements an exclusion policy intended or expected to affect at least 10% of the refugee stock. Severe policies include large-scale arrests and deportations, non-renewal of residence cards or refugee status, closures of camp schools and marketplaces, and cuts to food aid expressly intended to induce return. A dyad-year is coded **moderate** when the asylum government either (a) threatens or encourages forced return without policy implementation (e.g., political rhetoric, parliamentary debate, public ultimatums), or (b) implements an exclusion policy targeting fewer than 10% of the refugee stock. The 10% threshold separates broad coercive policies, which we expect to shift return rates, from narrower actions targeting specific individuals or groups but not intended to bring about widespread return.

Coding depends only on the policy itself, not on the return outcome. A policy that cuts food aid across all camps to induce return is coded as severe regardless of whether refugees subsequently repatriate. This rules out the most direct form of measurement endogeneity: the variable captures the policy environment refugees face rather than the return rate it is intended to predict.

Table 14 provides illustrative examples of moderate and severe coding decisions. The GRR codebook records a verbatim source quote for every non-zero observation, allowing external verification of all coding decisions.

1.7.2 Validation

In core GRR, exclusion policies and forced-return counts are coded from the same archival sources, raising the possibility of a mechanical association. I address this in three ways: by asking whether policies are coded in years with no return, whether the result persists when forced returns are removed from the outcome, and whether the measure agrees with an independently coded benchmark. The third check compares GRR's record of observed practice with DWRAP's coding of statutory protection against refoulement.

Table 15: Return in the years a policy is coded

Policy	Dyad-years	Zero return	Return below 1% of stock	Median return rate
None	3,988	76.3%	84.8%	0.00%
Moderate	290	35.9%	58.3%	0.35%
Severe	150	8.0%	22.0%	10.64%

Note: If the measure were read off the return, the zero-return column would be near empty.

Moderate policy is coded in 290 dyad-years, and 36% record no return at all. Severe policy is coded in 150 dyad-years, of which 8% record no return. This is possible because the policy measure captures threats, ultimatums, camp closures, deadlines, and other government actions, rather than return itself. Moreover, most returns are not coded from forced-return passages: 81% of all returns in GRR are voluntary, and forced returns occur in only 4.1% of dyad-years.

A remaining concern is that the association could still depend on the coding of forced returns. Table 18 below replicates the main analysis using only voluntary returns as the dependent variable. The severe-policy coefficient falls from 1.09 to 0.52 when forced returns are removed, or 47 per cent of the baseline, but remains statistically significant.

This pattern is reassuring for two reasons. First, if the baseline result were driven mainly by mechanical overlap in the coding of exclusion policy and forced-return counts, the association should largely disappear once forced returns are removed from the outcome. Second, its persistence for voluntary return suggests that coercive policies may also increase returns classified as voluntary, but undertaken under substantial pressure. The smaller coefficient is therefore consistent with severe policies producing both directly forced return and ostensibly voluntary return under coercive conditions.

1.7.3 De Jure Law and De Facto Practice

DWRAP provides an independent benchmark for the GRR exclusion-policy measure. It codes statutory refugee policy for 90 host countries, covering 95% of the panel. Its non-refoulement measure distinguishes statutes that permit refoulement, are silent or ambiguous, or explicitly protect against it. DWRAP measures national law, while GRR measures observed practice toward a particular refugee population.

Table 16: Exclusion policy by statutory position on refoulement

Statutory position	Dyad-years	Hosts	Any policy	Severe policy
Explicit provision permitting refoulement	509	34	15.3%	3.9%
Silent or ambiguous	698	29	21.3%	7.0%
Explicit right to non-refoulement	3,044	66	6.8%	2.5%

Note: Restricted to dyad-years in which both measures are coded. Exclusion policy is hand-coded from the archival sources; the statutory position is DWRAP's independent coding of national law.

Table 17: Exclusion policy and the statutory right to non-refoulement

Specification	Explicit right to non-refoulement	Dyad-years
Any, host + year FE	-1.15*** (0.34)	3,321
Any, dyad + year FE	-1.09* (0.48)	2,128
Severe, host + year FE	-1.29* (0.61)	2,268
Severe, dyad + year FE	-1.66* (0.83)	970

Note: Logit of hand-coded exclusion policy on the statutory right to non-refoulement, compared with the two remaining DWRAP positions combined. Standard errors clustered by dyad. The dyad-fixed-effects columns are identified from statutory changes within a host over time.

The main contrast is in the expected direction. Exclusion policy occurs in 6.8% of dyad-years in which the host grants an explicit right to non-refoulement, compared with 21.3% where the statute is silent or ambiguous. The statutory right is negatively associated with coercion in each specification in Table 17, including models with dyad and year fixed effects. This correspondence between an independently coded legal measure and observed practice provides evidence for the validity of the GRR measure.

The three statutory categories are not monotonic: hosts whose laws permit refoulement record less coercion than those whose laws are silent or ambiguous. Because the permitting category is small and concentrated among relatively few hosts, Table 17 compares an explicit right to non-refoulement with the other two positions combined.

Formal protection does not eliminate coercion. Among dyad-years in which the host grants an explicit right to non-refoulement, 206 record an exclusion policy and 77 record a severe policy, across 30 hosts. Statutory protection is associated with less coercion, but it remains distinct from state practice.

As a further check, I am constructing an independently coded measure of exclusion policy from Amnesty International reports. The coding and analysis is ongoing.

2 Robustness Checks

Unless otherwise noted, each robustness model changes one feature of column 1 in the main table and holds the remainder of the specification fixed.

The prose below describes what each check does and reports the estimates only where they change a conclusion or add something the table does not already show. Where nothing is said about a coefficient, it keeps the sign, significance and approximate magnitude it has in the main table.

2.1 Alternative Measures of Return

Because GRR return counts are hand-coded from archival sources, some annual variation may reflect reporting error. I therefore vary the dependent variable in several ways.

Model 1 of Table 18 reproduces the baseline, Model 2 uses voluntary return only, and Model 3 uses a centered three-year moving average of total return.

Two estimates move. Removing forced movements from the outcome reduces the severe-policy coefficient from 1.09 to 0.52 and turns the moderate-policy coefficient to -0.18, consistent with the outcome reflecting both forced movement and voluntary return under coercive conditions. The post-war coefficient falls from 1.00 to 0.59 under the moving average; unlike severe policy it rests on only 204 dyad-years and is sensitive to smoothing the outcome.

Table 18: Alternative Measures of the Dependent Variable

Dep. Var.:	Total Repatriation Total Returns	Voluntary Repatriation Voluntary Returns	Total Repat. (3-yr MA) 3-Year MA
Model:	(1)	(2)	(3)
Variables			
Exclusion policy			
Exclusion policy: severe	1.09***	0.52**	0.95***
	(0.19)	(0.17)	(0.21)
Exclusion policy: moderate	0.48*	-0.18	0.49*
	(0.22)	(0.15)	(0.20)
Conflict in COA: high	0.59†	0.38	0.39†
	(0.31)	(0.31)	(0.23)
Conflict in COA: low	0.81***	0.22	0.67***
	(0.21)	(0.15)	(0.16)
Existing explanations			
War Ending	1.00***	1.20***	0.59***
	(0.22)	(0.20)	(0.18)
Conflict in COO: high	0.11	0.16	0.27
	(0.28)	(0.29)	(0.31)
Conflict in COO: low	0.30	0.51*	0.27
	(0.22)	(0.23)	(0.20)
GDP (COO)	0.39	0.27	0.16
	(0.28)	(0.28)	(0.26)
GDP (COA)	-1.09†	-0.26	-0.97
	(0.59)	(0.57)	(0.59)
Civil Liberties Restrictions (COO)	-0.51**	-0.44**	-0.47***
	(0.17)	(0.17)	(0.14)
Refugee Rights	-0.02	0.06	-0.03
	(0.17)	(0.17)	(0.16)
Conflict Duration	-0.03	-0.05	-0.10
	(0.08)	(0.09)	(0.09)
Fixed-effects			
Dyad	Yes	Yes	Yes
Year	Yes	Yes	Yes
Fit statistics			
Observations	2,327	2,281	2,207
Squared Correlation	0.76	0.86	0.81
Pseudo R^2	0.80	0.79	0.83

Clustered (Dyad) standard-errors in parentheses
*Signif. Codes: ***: 0.001, **: 0.01, *: 0.05, †: 0.1*

Standard errors clustered by dyad in parentheses. All models: Poisson PML with log stock-at-risk offset and dyad and year fixed effects. Model 3 uses a centered 3-year moving average of total returns as the dependent variable.

Table 19: Discrete return episodes: seven codings of the dependent variable

Episode rule:	Returns (count)	Rate > 0.05	Rate > 0.10	Rate > 0.25	> 25,000	> 50,000	Mean + 1 SD	3× median
Model:	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Poisson	Logit	Logit	Logit	Logit	Logit	Logit	Logit
Variables								
Exclusion policy								
Exclusion policy: severe	1.09***	2.91***	2.75***	2.16***	3.43***	4.76***	2.17***	2.09***
	(0.19)	(0.34)	(0.33)	(0.53)	(0.40)	(0.77)	(0.38)	(0.49)
Exclusion policy: moderate	0.48*	0.71**	0.48†	0.51	0.76†	0.59	0.75*	0.58†
	(0.22)	(0.25)	(0.28)	(0.42)	(0.43)	(0.66)	(0.31)	(0.32)
Conflict in COA: high	0.59†	0.22	-0.03	-0.02	1.16	1.39	0.25	0.33
	(0.31)	(0.42)	(0.43)	(0.45)	(0.72)	(1.34)	(0.43)	(0.46)
Conflict in COA: low	0.81***	0.25	-0.17	-0.04	0.23	1.24	-0.15	-0.01
	(0.21)	(0.31)	(0.30)	(0.36)	(0.56)	(0.82)	(0.32)	(0.34)
Existing explanations								
War Ending	1.00***	1.13**	1.26**	0.61†	1.92**	3.29**	1.15***	0.97**
	(0.22)	(0.36)	(0.39)	(0.32)	(0.60)	(1.24)	(0.28)	(0.34)
Conflict in COO: high	0.11	-0.30	-0.42	-0.42	-0.24	1.54	-0.10	0.11
	(0.28)	(0.34)	(0.38)	(0.50)	(0.58)	(1.23)	(0.35)	(0.40)
Conflict in COO: low	0.30	-0.03	-0.14	-0.37	-0.04	0.98	-0.30	-0.16
	(0.22)	(0.27)	(0.28)	(0.36)	(0.52)	(0.65)	(0.28)	(0.33)
GDP (COO)	0.39	0.21	0.18	0.51	0.50	1.29	0.90*	0.49
	(0.28)	(0.34)	(0.32)	(0.44)	(0.67)	(1.44)	(0.40)	(0.44)
GDP (COA)	-1.09†	-0.09	-0.59	-0.30	-0.48	-1.19	-0.43	-0.13
	(0.59)	(0.60)	(0.66)	(0.93)	(0.78)	(2.13)	(0.64)	(0.66)
Civil Liberties Restrictions (COO)	-0.51**	-0.67**	-0.74**	-1.05***	-0.59†	-0.86*	-0.46*	-0.42†
	(0.17)	(0.21)	(0.27)	(0.29)	(0.32)	(0.43)	(0.20)	(0.22)
Refugee Rights	-0.02	0.34*	0.38*	0.28	0.45†	-0.16	0.10	0.33*
	(0.17)	(0.16)	(0.15)	(0.18)	(0.24)	(0.59)	(0.15)	(0.16)
Conflict Duration	-0.03	0.32*	0.37*	0.33	0.73**	0.36	0.41**	0.22
	(0.08)	(0.14)	(0.17)	(0.20)	(0.24)	(0.41)	(0.16)	(0.18)
Rule type	Count	Level	Level	Level	Level	Level	Change	Change
Dyads	110	93	84	68	55	32	107	72
Episode-years in sample	–	496	367	187	195	98	274	255
Fixed-effects								
Dyad	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Fit statistics								
Observations	2,327	2,023	1,834	1,530	1,240	732	2,300	1,743
Pseudo R^2	0.80	0.25	0.26	0.25	0.38	0.53	0.18	0.18

Clustered (Dyad) standard-errors in parentheses
*Signif. Codes: ***: 0.001, **: 0.01, *: 0.05, †: 0.1*

Column 1 is the paper's specification: Poisson PML on the return count with the log stock at risk as offset. Columns 2 to 8 are conditional logits on the episode codings named in the column headings. All models carry dyad and year fixed effects and cluster standard errors by dyad. Logit coefficients are log odds ratios and are not on the scale of column 1. The conditional logit drops dyads and years without variation in the outcome, which is why the observation and dyad counts fall below column 1.

Table 20: Discrete return episodes: estimators

Dep. Var.:	Rate > 0.10				Episode onset		Returns (count)
	Logit	LPM	LPM, logit rows	Onset logit	Onset hazard	Onset LPM	PPML, logit rows
Model:	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Logit	OLS	OLS	Logit	Logit	OLS	Poisson
Variables							
Exclusion policy							
Exclusion policy: severe	2.75***	0.36***	0.44***	2.71***	2.20***	0.24***	1.12***
	(0.33)	(0.05)	(0.05)	(0.53)	(0.31)	(0.05)	(0.20)
Exclusion policy: moderate	0.48†	0.05†	0.05	0.86**	0.66*	0.05*	0.54*
	(0.28)	(0.02)	(0.03)	(0.33)	(0.28)	(0.02)	(0.24)
Conflict in COA: high	-0.03	0.01	0.00	0.50	-0.01	0.03	0.55†
	(0.43)	(0.04)	(0.05)	(0.56)	(0.31)	(0.03)	(0.33)
Conflict in COA: low	-0.17	-0.01	-0.02	0.06	-0.22	0.01	0.83***
	(0.30)	(0.03)	(0.04)	(0.36)	(0.25)	(0.02)	(0.21)
Existing explanations							
War Ending	1.26**	0.14**	0.23**	1.25*	1.41***	0.06†	0.97***
	(0.39)	(0.04)	(0.07)	(0.53)	(0.40)	(0.03)	(0.23)
Conflict in COO: high	-0.42	-0.04	-0.06	0.69	0.45	0.03	0.05
	(0.38)	(0.03)	(0.05)	(0.57)	(0.42)	(0.03)	(0.29)
Conflict in COO: low	-0.14	-0.02	-0.02	0.20	0.20	0.01	0.27
	(0.28)	(0.03)	(0.04)	(0.44)	(0.38)	(0.03)	(0.23)
GDP (COO)	0.18	0.03	0.04	0.44	-0.20	0.03†	0.36
	(0.32)	(0.02)	(0.04)	(0.41)	(0.17)	(0.02)	(0.30)
GDP (COA)	-0.59	-0.12*	-0.08	-1.18†	-1.24***	-0.10**	-1.02
	(0.66)	(0.05)	(0.07)	(0.69)	(0.19)	(0.04)	(0.63)
Civil Liberties Restrictions (COO)	-0.74**	-0.05**	-0.09**	-0.43	-0.26*	-0.01	-0.56**
	(0.27)	(0.02)	(0.03)	(0.29)	(0.12)	(0.01)	(0.18)
Refugee Rights	0.38*	0.02	0.04*	0.44**	0.24**	0.02	0.00
	(0.15)	(0.02)	(0.02)	(0.17)	(0.08)	(0.01)	(0.18)
Conflict Duration	0.37*	0.03*	0.04*	0.17	0.12	0.01	-0.05
	(0.17)	(0.01)	(0.02)	(0.26)	(0.18)	(0.01)	(0.09)
Years Since Last Episode (log)					-0.35*		
					(0.14)		
Sample	Panel	Panel	Logit rows	At risk	At risk	At risk	Logit rows
Dyads	84	245	84	74	241	234	84
Fixed-effects							
Dyad	Yes	Yes	Yes	Yes		Yes	Yes
Year	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Fit statistics							
Observations	1,834	3,913	1,834	1,266	3,221	3,214	1,834

Clustered (Dyad) standard-errors in parentheses
*Signif. Codes: ***: 0.001, **: 0.01, *: 0.05, †: 0.1*

The dependent variable is the ten per cent rate rule throughout: an episode is a dyad-year in which return exceeds a tenth of the stock at risk. Columns 4 to 6 use episode onset, one in the first year of a run, estimated on the dyad-years at risk of starting an episode; interior years of a run are removed from the risk set rather than coded zero. Column 5 is the discrete-time hazard: year fixed effects and log years since the last episode, without dyad fixed effects, because duration is collinear with dyad and year effects together; its dyad count is the number of clusters rather than the number of absorbed effects. Column 7 refits the paper's count model, outcome and offset unchanged, on the rows the conditional logit retains. Standard errors clustered by dyad throughout. Logit, linear-probability and PPML coefficients are on three different scales and should not be compared across columns of different estimators.

A further way of varying the outcome is to discard the exact number returned. In Tables 19 and 20 I recode it as an indicator for whether a dyad experiences a substantial return episode, which tests whether the results hold for movements too large to be explained by minor coding differences. There is no unique threshold, so I use seven alternatives. The *level* rules apply common thresholds across dyads: return above five, ten, or twenty-five percent of the population at risk, or more than twenty-five or fifty thousand people. The *change* rules define an episode relative to the dyad's history: a return rate more than one within-dyad standard deviation above its mean, or at least three times its median positive return rate. Table 19 applies each definition to the main specification, with the count model shown for comparison; Table 20 holds the threshold at ten percent and varies the estimator. I also re-estimated at nine rate thresholds from one to thirty-three percent, not reported.

Two features of the recoding are worth recording. How much of the panel counts as an episode depends heavily on which rule is used, from 129 dyad-years under the fifty-thousand threshold to 559 above the five percent rate, or 2.5 to 10.9 percent of the panel. And the choice determines how much of the panel a conditional logit can use, since only dyads whose coded outcome varies contribute: from 38 dyads under the fifty-thousand rule to 122 under the within-dyad standard deviation, so the change rules rest on more dyads than their episode counts alone suggest. A substantial episode occurs in 51 percent of severe-policy dyad-years against 8 percent otherwise. Across the nine unreported thresholds the severe-policy coefficient stays between 2.01 and 3.07 and the war-ending coefficient between 0.67 and 0.89, both significant at every one. Those two ranges were computed under the war-ending coding used before 2026-08-30 and have not been re-estimated on the current one; the severe-policy range is unlikely to move, since that coefficient is stable across every coding tried, but the war-ending range should be treated as provisional until the nine thresholds are re-run. The only estimate that weakens is the post-war window when the outcome is episode onset rather than the presence of an episode.

2.2 Alternative Measures of Conflict and War Endings

2.2.1 Alternative Definitions of a War Ending

The post-war window has two halves and they must come from the same coding: a list of war endings, and a test for whether the origin is at war again that voids the window. The paper uses Fearon's (2017) civil-war list, which dates wars by definitive beginnings and endings rather than by an annual battle-death threshold, extended by this project and merged with inter-state wars coded to a refugee origin, then filtered to endings in 1981–2018 with at least 25,000 refugees in exile at the ending. The at-peace test is the merged Fearon episode spans. Fearon's original coding runs through 2014; the extended version carries thirty-one of those cases forward with revised end dates and adds seventeen.

The substantive alternative uses UCDP data. There an ending is the last year of a UCDP high-intensity spell at the origin, and a dyad-year counts as post-war only if the origin has not returned to high-intensity war, so a low-intensity war may be running. The two codings are not close. Only 21 per cent of UCDP endings have a Fearon ending within two years of them, and 44 per cent the other way, so they disagree about most of the years either calls post-war. UCDP records an ending whenever annual battle deaths fall below the threshold, including interruptions in conflicts that later resume; the Fearon lists use a single definitive ending for each named war.

Columns 3 to 6 hold the at-peace test on the Fearon spans and vary the ending list or add a term. Column 3 keeps only the cases Fearon coded and this project did not revise, so it shows whether the result depends on the author-coded extension. Column 4 removes the 25,000-refugee requirement, the one filter that selects endings on a quantity related to the outcome. Column 5 adds the ending year itself, so the window runs from year zero rather than year one. Column 6 turns to a formal rather than a battlefield definition, adding an indicator for a peace agreement at the origin in the previous year, drawn from the UCDP Peace Agreements Dataset and merged to the origin's active conflict via UCDP `conflict_id`.

Table 21: Robustness: Alternative Definitions of a War Ending

	Total Repatriation					
	Main	UCDP end to end	Fearon original	No size floor	+ Ending yr	+ Peace ag.
Model:	(1)	(2)	(3)	(4)	(5)	(6)
Variables						
Exclusion policy						
Exclusion policy: severe	1.09***	1.02***	1.10***	1.08***	1.09***	1.07***
	(0.19)	(0.21)	(0.19)	(0.19)	(0.19)	(0.18)
Exclusion policy: moderate	0.48*	0.50*	0.48*	0.47*	0.49*	0.52*
	(0.22)	(0.21)	(0.22)	(0.23)	(0.22)	(0.21)
Conflict in COA: high	0.59†	0.54†	0.57†	0.59†	0.59†	0.72*
	(0.31)	(0.30)	(0.31)	(0.31)	(0.31)	(0.30)
Conflict in COA: low	0.81***	0.82***	0.81***	0.81***	0.81***	0.84***
	(0.21)	(0.21)	(0.21)	(0.21)	(0.21)	(0.21)
Existing explanations						
War Ending	1.00***	0.62**	1.03***	0.99***	1.00***	0.99***
	(0.22)	(0.20)	(0.23)	(0.22)	(0.21)	(0.21)
War Ending (Year Of)					0.05	
					(0.25)	
Conflict in COO: high	0.11	0.13	0.08	0.11	0.11	0.08
	(0.28)	(0.35)	(0.28)	(0.28)	(0.28)	(0.27)
Conflict in COO: low	0.30	0.02	0.28	0.31	0.30	0.24
	(0.22)	(0.22)	(0.22)	(0.22)	(0.22)	(0.22)
GDP (COO)	0.39	0.37	0.36	0.38	0.39	0.43
	(0.28)	(0.29)	(0.29)	(0.28)	(0.28)	(0.28)
GDP (COA)	-1.09†	-1.15†	-1.09†	-1.12†	-1.09†	-1.16*
	(0.59)	(0.64)	(0.59)	(0.59)	(0.59)	(0.56)
Civil Liberties Restrictions (COO)	-0.51**	-0.55***	-0.54**	-0.53***	-0.51**	-0.52**
	(0.17)	(0.16)	(0.17)	(0.16)	(0.17)	(0.18)
Refugee Rights	-0.02	0.04	-0.02	-0.01	-0.02	0.02
	(0.17)	(0.16)	(0.17)	(0.17)	(0.17)	(0.17)
Conflict Duration	-0.03	-0.09	-0.02	-0.03	-0.03	0.01
	(0.08)	(0.09)	(0.08)	(0.09)	(0.08)	(0.07)
Peace Agreement (COO, t-1)						-0.44**
						(0.15)
All controls	Yes	Yes	Yes	Yes	Yes	Yes
Ending list	Fearon narrow	UCDP spells	Fearon original	No 25,000 floor	Fearon narrow	Fearon narrow
At-peace test	Fearon spans	UCDP high-int.	Fearon spans	Fearon spans	Fearon spans	Fearon spans
Endings	36	76	33	75	36	36
Fixed-effects						
Dyad	Yes	Yes	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes	Yes	Yes
Fit statistics						
Observations	2,327	2,327	2,327	2,327	2,327	2,214
Pseudo R^2	0.80	0.79	0.80	0.80	0.80	0.81

Clustered (Dyad) standard-errors in parentheses
*Signif. Codes: ***: 0.001, **: 0.01, *: 0.05, †: 0.1*

All columns are the specification of column 1 of the main table with the post-war window rebuilt; nothing else changes. Column 1 is the window the paper uses: the extended Fearon civil-war list merged with the coded inter-state wars, filtered to endings in 1981–2018 with at least 25,000 refugees in exile at the ending, with the at-peace test on the merged Fearon episode spans. Column 2 is the alternative CODING, varying both halves together: an ending is the last year of a UCDP high-intensity spell at the origin, and a dyad-year is post-war only if the origin has not returned to high-intensity war – a low-intensity war may be running. This is the window the paper used until 2026-08-30 and the two codings disagree about most years, so it is the substantive alternative rather than a variation on the baseline. Columns 3–6 hold the at-peace test on the Fearon spans. Column 3 keeps only the cases Fearon coded and this project did not revise, so it excludes both the post-2014 extension and the revised end dates. Column 4 removes the requirement that the origin have at least 25,000 refugees in the ending year, which is the one filter that selects endings on a quantity related to the outcome. Column 5 adds the ending year itself, so the window is identified from years zero to three rather than one to three. Column 6 adds an indicator for a peace agreement at the origin in the previous year, a formal rather than a battlefield definition of an ending. PPML with the log stock at risk as offset, dyad and year fixed effects, SEs clustered by dyad.

The post-war coefficient is positive under every definition. The paper's Fearon window gives 1.00, and the UCDP coding in Column 2 gives 0.62 – so the choice of coding matters for the magnitude of this coefficient and for its precision, and the paper reports the Fearon figure. It is the one row in the main table that is sensitive to this choice: the asylum-side coefficients and repatriation assistance move by less than 0.1 across all six codings tried (0_scripts/compare_ending_lists.R). Readers should treat the post-war estimate as a range rather than a point.

Within the Fearon coding the result is stable. Fearon's original cases alone give 1.03, so it does not depend on this project's extension of the list. Removing the size floor increases the list from 36 endings to 75 and leaves the window at 0.99; raising the floor to 100,000 moves it to 1.11 on 31 endings. Adding the ending year itself leaves that year not distinguishable from zero (0.05), consistent with the timing reported in the main text: return begins in the first full year of peace, not in the year the fighting stops.

The peace agreement in Column 6 enters negative and significant (-0.44). This is surprising as it does not align with the war ending result.

2.2.2 Alternative Measures of Origin-Country Conflict and Repression

In this section I vary the way that conflict in both the origin and the asylum country is coded. Model 1 reproduces the baseline. Model 2 replaces the ordinal conflict indicators with logged battle deaths, a continuous measure of intensity available from 1989. Model 3 dates origin-country conflict at $t-2$ to allow for a delayed response. Model 4 replaces the origin conflict levels with a de-escalation indicator equal to one when conflict falls from high-intensity to low or none, capturing transitions rather than levels. Model 5 adds the PITF measure of warfare and violence, combining its Ethnic War and Revolutionary War datasets, which ends in 2017.

I then also vary the way that political repression is coded. In the main analysis this is the Freedom House civil liberties measure. Model 6 adds two further measures alongside it. The Political Terror Scale (PTS) measures state-perpetrated physical integrity violations, including torture, political imprisonment, extra-judicial killings, and disappearances, on a 1–5 ordinal scale; I use the mean of three sources (Amnesty International, Human Rights Watch, and the U.S. State Department), standardized across the estimation sample (`o_terror_z`). The second is a binary indicator equal to one in any country-year with an ongoing episode of genocide or politicide, drawn from the Political Instability Task Force (PITF) Geno-Politicide event dataset (`o_genocide`). The PITF episode list ends in 2017, so observations after 2017 are coded as missing and Model 6 runs on a shorter panel.

The Political Terror Scale is indistinguishable from zero once civil liberties are controlled for (-0.01), so the repression result operates through the Freedom House measure rather than through physical-integrity violations specifically. Surprisingly, the genocide/politicide indicator enters positive and significant (0.81), opposite to expectation.

2.2.3 Ethnic Group Political Power

In the main table, I examine correlates of shared ethnic and social ties between refugees and hosts. In this section, I examine factors related to the political status of the refugees' ethnic group in both the origin and asylum country, where there are shared ethnic ties. A group that gains power at home might be more likely to go back, and one whose kin hold power in the host might be more likely to stay. I test both with the EPR data in Models 1 and 2 of Table 23. I find that refugees from politically dominant groups return at higher rates than those from excluded ones. The standing of co-ethnics in the host has no bearing on return: neither EPR-TEK indicator is distinguishable from zero (-0.29 for kin in power, 0.21 for kin excluded).

The ordinal EPR status variable (`o_eth_status`) takes nine values from 'self-exclusion' to 'monopoly'. Refugee groups are matched to EPR group identifiers via the EPR Refugee Dataset, which links populations of at least 2,000 refugees to EPR groups in neighboring or proximate asylum countries (within 950 km of the origin border). For dyads outside this coverage I code the refugee group manually from World

Table 22: Robustness: Alternative Measures of Conflict and Repression

	Total Repatriation					
	Main	Battle deaths	COO t-2	De-escalation	PITF war	+ PTS/Genocide
Model:	(1)	(2)	(3)	(4)	(5)	(6)
Variables						
Exclusion policy						
Exclusion policy: severe	1.09^{***}	0.88^{***}	1.06^{***}	1.09^{***}	1.10^{***}	1.05^{***}
	(0.19)	(0.19)	(0.20)	(0.19)	(0.20)	(0.20)
Exclusion policy: moderate	0.48^{*}	$0.44^{\dagger}$	0.54^{*}	0.50^{*}	0.50^{*}	$0.43^{\dagger}$
	(0.22)	(0.23)	(0.22)	(0.23)	(0.23)	(0.24)
Conflict in COA: high	$0.59^{\dagger}$		0.76^{*}	$0.52^{\dagger}$	$0.58^{\dagger}$	$0.60^{\dagger}$
	(0.31)		(0.31)	(0.30)	(0.32)	(0.32)
Conflict in COA: low	0.81^{***}		0.85^{***}	0.78^{***}	0.82^{***}	0.79^{***}
	(0.21)		(0.19)	(0.20)	(0.21)	(0.22)
Existing explanations						
War Ending	1.00^{***}	0.95^{***}	0.91^{***}	1.02^{***}	0.94^{***}	0.94^{***}
	(0.22)	(0.20)	(0.21)	(0.20)	(0.21)	(0.20)
Conflict in COO: high	0.11				0.06	0.02
	(0.28)				(0.28)	(0.29)
Conflict in COO: low	0.30				0.30	0.35
	(0.22)				(0.22)	(0.22)
GDP (COO)	0.39	0.39	$0.49^{\dagger}$	0.41	0.40	0.29
	(0.28)	(0.28)	(0.28)	(0.27)	(0.29)	(0.30)
GDP (COA)	$-1.09^{\dagger}$	-1.17^{*}	$-1.18^{\dagger}$	$-1.16^{\dagger}$	$-1.05^{\dagger}$	-0.67
	(0.59)	(0.46)	(0.61)	(0.60)	(0.59)	(0.59)
Civil Liberties Restrictions (COO)	-0.51^{**}	-0.47^{**}	-0.48^{**}	-0.50^{**}	-0.55^{**}	-0.56^{**}
	(0.17)	(0.16)	(0.16)	(0.16)	(0.17)	(0.19)
Refugee Rights	-0.02	0.18	0.07	0.00	-0.03	0.01
	(0.17)	(0.19)	(0.17)	(0.17)	(0.18)	(0.18)
Conflict Duration	-0.03	0.02	0.07	0.07	-0.09	-0.06
	(0.08)	(0.10)	(0.08)	(0.08)	(0.09)	(0.09)
Alternative conflict and repression measures						
Battle Deaths (COA, log)		0.13^{***}				
		(0.04)				
Battle Deaths (COO, log)		0.01				
		(0.04)				
Conflict in COO: high (t-2)			-0.19			
			(0.22)			
Conflict in COO: low (t-2)			-0.06			
			(0.11)			
De-escalation (COO)				-0.31		
				(0.23)		
PITF War/Violence (COO)					1.24^{*}	
					(0.58)	
Political Terror Scale (COO)						-0.01
						(0.13)
Genocide/Politicide (COO)						0.81^{*}
						(0.33)
All controls	Yes	Yes	Yes	Yes	Yes	Yes
Years	1980-2022	1990-2022	1982-2022	1980-2022	1980-2022	1980-2022
Fixed-effects						
Dyad	Yes	Yes	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes	Yes	Yes
Fit statistics						
Observations	2,327	2,134	2,128	2,327	2,238	1,994
Pseudo R^2	0.80	0.81	0.81	0.80	0.80	0.80

Clustered (Dyad) standard-errors in parentheses
*Signif. Codes: ***: 0.001, **: 0.01, *: 0.05, †: 0.1*

All models carry the full specification of the main table, including the post-war window and the Freedom House civil liberties measure, and differ only in how conflict and repression are measured. Columns 2 to 4 replace the ordinal conflict levels; columns 5 and 6 add measures alongside them. Column 6 adds the Political Terror Scale (the mean of the Amnesty International, Human Rights Watch and U.S. State Department ratings, standardised) and a binary indicator for an ongoing episode of genocide or politicide (PITF), whose episode list ends in 2017. PPML with the log stock at risk as offset, dyad and year fixed effects, SEs clustered by dyad.

Refugee Survey reports.[1] EPR coverage is unavailable for some refugee groups, reducing the estimation sample.

2.3 Alternative Explanatory Variable Measures

Table 24 tests sensitivity to alternative operationalizations and additional covariates. Model 1 reproduces the baseline. Model 2 replaces continuous origin and asylum GDP per capita with quartile indicators (Q1 reference) to relax linearity assumptions. Model 3 lags all independent variables by one year, addressing the possibility that contemporaneous asylum-country conflict partly follows refugee return or refugee mobilization.

Model 4 adds UNHCR funding per refugee in the asylum country. This measure is difficult to interpret, because it is a proxy for two things at once: what UNHCR spends on organising repatriation, and what it spends on supporting refugees where they are, including camps and health care. The first would be expected to raise return and the second to reduce the pressure to leave, and the funding series does not separate them. I therefore include it as a control rather than as a test of a mechanism, and the positive coefficient (0.95) should not be interpreted causally.

Models 5 and 6 add conditions in the asylum country. Model 5 includes educational attainment among the resident citizen populations of the origin and asylum countries, which is a national characteristic of each country rather than a property of the refugees themselves; no comparable schooling measure exists for refugee populations at this scale. Model 6 adds the asylum country's urban share and an indicator that most refugees live in camps.

Lagging every independent variable by a year raises severe exclusion policy (1.22 against 1.09) and lowers the post-war window (0.63 against 1.00).

2.4 Repatriation Assistance

Repatriation assistance enters the main table as one measure, UNHCR repatriation funding, in a column estimated on a smaller sample than the rest of the table. This section reports all three available measures of UNHCR involvement and separates what each measure does from what its sample does.

The funding series is hand-coded for 1980–1999 and 2011–2020 and is missing for every dyad-year in 2000–2010 and 2021–2023. That is a gap in the source record rather than a run of years without programmes, and it covers only 59% of the estimation frame. Adding the measure to the baseline therefore drops 949 of the baseline's 2,327 dyad-years, and the column that carries it is not the baseline plus a variable. Model 2 of Table 25 is the baseline specification refitted on exactly the rows Model 3 uses, with no assistance term, so that Model 3 can be read against it. Anything that differs between Models 1 and 2 is the loss of the uncoded years.

Most of the movement is the sample. Moderate exclusion policy falls from 0.48, SE = 0.22 in the baseline to 0.20, SE = 0.27 on the funding-coded rows before any assistance measure is included, and civil liberties restrictions in the origin country fall from -0.51, SE = 0.17 to -0.22, SE = 0.24 over the same step; neither is distinguishable from zero on the smaller sample. Severe exclusion policy is unaffected by the change of sample, moving only from 1.09, SE = 0.19 to 1.24, SE = 0.32. Repatriation funding itself is positive and significant at the 0.01 level (0.31, SE = 0.12), and the one coefficient it clearly moves is the post-war window, which falls from 1.29, SE = 0.40 without the measure to 1.05, SE = 0.38 with it. Part of what the window picks up in the baseline is therefore the assistance that accompanies a war ending rather than the ending itself, which is the mechanism H3 anticipates.

[1] The same ethnic group is carried forward as long as the refugee population is continuous; the assignment is set to missing when a discontinuity in the population could plausibly indicate a change in ethnic composition. Two case-specific adjustments warrant note: EPR classifies Somalis as 'irrelevant' within Somalia on grounds of ethnic homogeneity, and I reclassify this group as high-status; Peuhl/Goula refugees from the Central African Republic appear in the EPR Refugee Dataset but not in the EPR Power Dataset, and are coded as 'powerless' based on desk research. The 2021 EPR Refugee values are carried forward from 2020.

Table 23: Ethnic political status and refugee return

	Total Repatriation	
	Origin status (EPR)	Kin in host (EPR-TEK)
Model:	(1)	(2)
Variables		
Exclusion policy		
Exclusion policy: severe	1.32***	1.11***
	(0.23)	(0.20)
Exclusion policy: moderate	0.51*	0.54*
	(0.23)	(0.24)
Conflict in COA: high	0.47	0.63*
	(0.36)	(0.30)
Conflict in COA: low	1.00***	0.84***
	(0.20)	(0.22)
Existing explanations		
War Ending	0.68***	0.99***
	(0.19)	(0.22)
Conflict in COO: high	0.24	0.09
	(0.27)	(0.29)
Conflict in COO: low	0.57**	0.29
	(0.22)	(0.23)
GDP (COO)	0.21	0.38
	(0.23)	(0.28)
GDP (COA)	-2.22***	-1.10†
	(0.51)	(0.59)
Civil Liberties Restrictions (COO)	-0.47**	-0.52**
	(0.15)	(0.17)
Refugee Rights	0.07	-0.01
	(0.19)	(0.17)
Conflict Duration	-0.19†	-0.03
	(0.09)	(0.09)
Ethnic political status		
Ethnic Status: Monopoly (COO)	-1.51**	
	(0.53)	
Ethnic Status: Senior Partner (COO)	-0.53	
	(0.41)	
Ethnic Status: Junior Partner (COO)	-0.66	
	(0.46)	
Ethnic Status: Powerless (COO)	-0.65	
	(0.47)	
Ethnic Status: Discriminated (COO)	-0.93*	
	(0.44)	
Ethnic Status: Self-Excluded (COO)	-2.63***	
	(0.67)	
Ethnic Status: State Collapse (COO)	-1.50***	
	(0.40)	
Co-ethnic Kin in Power (COA)		-0.29
		(0.45)
Co-ethnic Kin Excluded (COA)		0.21
		(0.35)
All controls	Yes	Yes
Fixed-effects		
Dyad	Yes	Yes
Year	Yes	Yes
Fit statistics		
Observations	1,718	2,327
Pseudo R^2	0.82	0.80

Clustered (Dyad) standard-errors in parentheses
*Signif. Codes: ***: 0.001, **: 0.01, *: 0.05, †: 0.1*

Column 1 adds the ordinal EPR status of the displaced group in the origin, with Dominant omitted as the reference; EPR coverage is incomplete, which is why the sample is smaller. Column 2 adds the EPR-TEK indicators for whether the group's co-ethnics in the asylum country hold power or are excluded. Both carry the full specification of column 1 of the main table. PPML with the log stock at risk as offset, dyad and year fixed effects, SEs clustered by dyad.

Table 24: Robustness: Alternative Independent Variables

	Total Repatriation					
	Main	GDP Quartiles	Lagged IVs	+ Funding	+ Education	+ Urban/Camp
Model:	(1)	(2)	(3)	(4)	(5)	(6)
Variables						
Exclusion policy						
Exclusion policy: severe	1.09***	1.10***	1.22***	1.23***	1.12***	1.07***
	(0.19)	(0.20)	(0.29)	(0.22)	(0.19)	(0.20)
Exclusion policy: moderate	0.48*	0.50**	0.24	0.55*	0.51*	0.49*
	(0.22)	(0.19)	(0.25)	(0.23)	(0.22)	(0.22)
Conflict in COA: high	0.59†	0.86**	0.49*	-0.06	0.51	0.60*
	(0.31)	(0.27)	(0.24)	(0.34)	(0.33)	(0.28)
Conflict in COA: low	0.81***	0.67***	0.53**	0.90***	0.85***	0.79***
	(0.21)	(0.20)	(0.18)	(0.26)	(0.20)	(0.21)
Existing explanations						
War Ending	1.00***	0.94***	0.63*	0.76***	0.95***	0.99***
	(0.22)	(0.23)	(0.28)	(0.22)	(0.21)	(0.23)
Conflict in COO: high	0.11	0.06	-0.43	-0.23	0.14	0.11
	(0.28)	(0.28)	(0.28)	(0.32)	(0.28)	(0.29)
Conflict in COO: low	0.30	0.29	-0.14	0.04	0.33	0.31
	(0.22)	(0.22)	(0.20)	(0.25)	(0.22)	(0.23)
GDP (COO)	0.39		0.10	0.62†	0.44	0.37
	(0.28)		(0.29)	(0.33)	(0.28)	(0.28)
GDP (COA)	-1.09†		-0.26	-1.07	-1.23*	-1.04†
	(0.59)		(0.66)	(0.70)	(0.62)	(0.59)
Civil Liberties Restrictions (COO)	-0.51**	-0.55**	-0.27	-0.39*	-0.56***	-0.50**
	(0.17)	(0.19)	(0.16)	(0.17)	(0.17)	(0.16)
Refugee Rights	-0.02	0.05	0.10	0.04	-0.02	0.01
	(0.17)	(0.17)	(0.16)	(0.15)	(0.18)	(0.17)
Conflict Duration	-0.03	-0.04	0.11	-0.06	-0.08	-0.06
	(0.08)	(0.09)	(0.08)	(0.10)	(0.09)	(0.09)
Alternative and additional measures						
GDP (COA): Q2		-0.68*				
		(0.28)				
GDP (COA): Q3		-0.44				
		(0.50)				
GDP (COA): Q4		-1.11				
		(1.36)				
GDP (COO): Q2		-0.72**				
		(0.28)				
GDP (COO): Q3		-0.37				
		(0.35)				
GDP (COO): Q4		0.02				
		(0.40)				
UNHCR Funding p.p. (COA)				0.85**		
				(0.30)		
Education (COA)					-0.21	
					(1.25)	
Education (COO)					-0.23	
					(0.78)	
Urban						-0.30
						(0.25)
Majority in Camps						-0.21
						(0.51)
Fixed-effects						
Dyad	Yes	Yes	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes	Yes	Yes
Fit statistics						
Observations	2,327	2,327	2,164	1,928	2,202	2,325
Pseudo R^2	0.80	0.80	0.80	0.83	0.80	0.80

Clustered (Dyad) standard-errors in parentheses
*Signif. Codes: ***: 0.001, **: 0.01, *: 0.05, †: 0.1*

Poisson PML, log stock-at-risk offset, dyad + year FEs, SEs clustered by dyad. M1: column 1 of the main table. M2: GDP (COO/COA) quartiles (Q1 ref.). M3: all independent variables at t-1, including the war-ending window. M4: + UNHCR funding p.p. (z-log). M5: + educational attainment in the resident citizen populations of both countries. M6: + urban share and an indicator for the majority of the refugee population living in camps; the urban and camp series are missing for roughly 650 dyad-years, so M6 runs on a smaller sample.

The two dyadic measures raise a different problem, which is not the funding gap but a change of source part-way through the panel. Both are coded from archival reporting that changes character, and in both cases the estimate depends on whether the years either side of the change are pooled.

Model 4 adds the UNHCR repatriation programme flag over 1980–2008, the years for which the *World Refugee Survey* is the source. It is estimated on 1,225 dyad-years, positive in 281 of them across 60 switching dyads, and its coefficient is 0.67, SE = 0.23, significant at the 0.001 level. This is the estimate the main text reports and the one Panel A of the paper's predicted-rate figure draws. It is not a sample artefact: on the same rows without the flag, severe exclusion policy is 0.76, SE = 0.24 and the post-war window 1.10, SE = 0.27, both close to the baseline.

Model 5 pools the same flag across both eras, adding the years from 2009 in which the source is the State Department's country reports. The coefficient falls to 0.25, SE = 0.30, with $p = 0.41$, on 2,245 dyad-years. The two eras do not measure the same thing: the State Department reports are organised around state conduct rather than around UNHCR operations and record programmes far less systematically, so a zero after 2008 is often silence rather than absence. Model 5 is reported because the pooled null is what a reader who took the flag at face value would compute, and Model 4 is the estimate to use.

Model 6 adds a tripartite repatriation agreement in force over 1980–2002, the years for which the ExCom register is complete. It is estimated on 873 dyad-years, active in 32 of them across 16 switching dyads, and its coefficient is 0.66, SE = 0.27. This is the main text's tripartite estimate. Note that the sample matters here in a way it does not for the programme flag: on these rows without the agreement term, high-intensity conflict in asylum is 1.47, SE = 0.32 and severe exclusion policy 0.91, SE = 0.29, so Model 6's other coefficients should be read against that fit and not against Model 1.

Model 7 extends the tripartite measure past 2002 by splicing the ExCom register with the ExCom register spliced with Camarena (2022) after 2002, and sets post-2002 dyad-years outside that register's universe to missing rather than to zero. On 1,364 dyad-years, with 96 agreement-years across 23 switching dyads, the coefficient is 0.12, SE = 0.24 with $p = 0.62$. The tripartite association is therefore not robust to extending the register, and the extension is the version the event-study figure uses, so the two should be read together.

One thing is deliberately not a column. Running the ExCom measure on the full panel and reading every uncovered post-2002 dyad-year as recording no agreement returns 0.68, SE = 0.21, significant at the 0.05 level, off 35 agreement-years in 17 dyads. That is a statement about the missing-data assumption and not about agreements, and it is the one specification of the four that should not be reported as an estimate.

Taken together: the continuous funding measure is significant on its own sample; the programme flag is significant on the years where it means one thing and null when pooled across a source break; and the tripartite measure is significant on the years where its register is complete and null once the register is extended. Only the funding result survives every treatment of its own coverage.

Whether it survives the rest of this section's departures is a separate question, and Table 29 answers it: the funding coefficient holds under fifteen variations on the specification and fails under four, and the four that break it are informative rather than incidental.

2.5 Alternative Data Subsets

Table 26 tests whether the results hold across different subsets of the data. Model 1 reproduces the baseline on the extended GRR estimation panel. Model 2 restricts to the State Department-report period beginning in 2009.

Model 3 excludes dyads associated in the literature with the "refugee warrior" hypothesis, under which refugee populations, camps, or exile communities may become militarized and contribute to conflict spillover in host states (Zolberg 1989; Salehyan and Gleditsch 2006).[2] Excluding these cases assesses whether the main results are driven by settings in which refugee presence may be more plausibly endogenous to asylum-country conflict. Model 4 excludes asylum countries in Europe and North America. Model 5 restricts to dyads where refugees originally fled armed conflict rather than non-war forms of persecution. Model 6 restricts to the top

[2]The excluded dyads are: RWA–COD, RWA–TZA, AFG–PAK, KHM–THA, PSE–LBN, PSE–JOR, and NIC–HND.

Table 25: Robustness: Measures of Repatriation Assistance

	Total Repatriation						
	Main	Fund.rows	+Funding	Prog.WRS	Prog.all	Trip.02	Trip.spl
Model:	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Variables							
Exclusion policy							
Exclusion policy: severe	1.09***	1.24***	1.29***	0.57*	1.08***	0.89**	0.98***
	(0.19)	(0.32)	(0.31)	(0.28)	(0.22)	(0.28)	(0.21)
Exclusion policy: moderate	0.48*	0.20	0.13	0.19	0.44	0.46*	0.44*
	(0.22)	(0.27)	(0.28)	(0.26)	(0.27)	(0.20)	(0.19)
Conflict in COA: high	0.59†	0.96*	0.79†	0.86**	0.55†	1.41***	1.41***
	(0.31)	(0.43)	(0.43)	(0.27)	(0.30)	(0.32)	(0.29)
Conflict in COA: low	0.81***	0.93**	0.95**	0.65**	0.86***	0.95**	0.70*
	(0.21)	(0.34)	(0.36)	(0.22)	(0.20)	(0.36)	(0.30)
Existing explanations							
War Ending	1.00***	1.29**	1.05**	0.82**	0.87***	1.68***	1.24***
	(0.22)	(0.40)	(0.38)	(0.26)	(0.22)	(0.36)	(0.28)
Conflict in COO: high	0.11	0.57†	0.38	-0.08	-0.02	0.28	-0.07
	(0.28)	(0.34)	(0.35)	(0.32)	(0.27)	(0.34)	(0.31)
Conflict in COO: low	0.30	0.29	0.22	0.35	0.26	0.23	0.31
	(0.22)	(0.26)	(0.25)	(0.25)	(0.22)	(0.28)	(0.25)
GDP (COO)	0.39	0.73	0.80	0.18	0.33	0.31	0.50†
	(0.28)	(0.49)	(0.50)	(0.39)	(0.29)	(0.39)	(0.29)
GDP (COA)	-1.09†	-1.23	-1.40	-0.41	-1.33*	-0.14	0.31
	(0.59)	(0.83)	(0.86)	(0.64)	(0.53)	(0.67)	(0.62)
Civil Liberties Restrictions (COO)	-0.51**	-0.22	-0.10	-0.34†	-0.49**	-0.23	-0.27
	(0.17)	(0.24)	(0.26)	(0.19)	(0.19)	(0.22)	(0.21)
Refugee Rights	-0.02	-0.15	-0.19	-0.14	-0.05	-0.21	0.18
	(0.17)	(0.22)	(0.22)	(0.23)	(0.18)	(0.26)	(0.16)
Conflict Duration	-0.03	-0.07	-0.06	0.01	-0.02	0.08	0.02
	(0.08)	(0.13)	(0.12)	(0.11)	(0.09)	(0.14)	(0.11)
Repatriation assistance measures							
Repatriation Assistance			0.31**				
			(0.12)				
UNHCR Repatriation Programme				0.67**	0.25		
				(0.23)	(0.30)		
Tripartite Agreement						0.66*	
						(0.27)	
Tripartite Agreement (spliced)							0.12
							(0.24)
Fixed-effects							
Dyad	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Fit statistics							
Observations	2,327	1,378	1,378	1,225	2,245	873	1,364
Pseudo R^2	0.80	0.77	0.78	0.84	0.80	0.83	0.80

Clustered (Dyad) standard-errors in parentheses
*Signif. Codes: ***: 0.001, **: 0.01, *: 0.05, †: 0.1*

Poisson PML, log stock-at-risk offset, dyad and year fixed effects, standard errors clustered by dyad. Every column has a different sample, so no column should be read against M1 except M2. M1 is column 1 of the main table. M2 is not a robustness check: it is M1's specification refitted on exactly the rows M3 uses, with no assistance term, and it is the correct comparison for M3. The funding series is coded only for 1980–1999 and 2011–2020, so M2 and M3 drop 949 of M1's 2,327 dyad-years. M3 is column 2 of the main table. M4 and M5 add the dyadic UNHCR repatriation programme flag, M4 over 1980–2008, the years for which the World Refugee Survey is the source, and M5 pooled across both eras; the State Department reports used from 2009 record UNHCR operations far less systematically, so a zero after 2008 is often silence rather than absence, and M4 is the estimate to use. M4 is M3 of the paper's predicted-rate figure. M6 and M7 add a tripartite agreement in force, M6 over 1980–2002, the years for which the ExCom register is complete (and M4 of the paper's predicted-rate figure), and M7 over the whole period using the ExCom register spliced with Camarena (2022) after 2002, with post-2002 dyad-years outside that register's universe set missing rather than zero. Reading those years as zeros instead returns 0.68, SE = 0.21 on the full panel off only 35 agreement-years, which is a statement about the missing-data assumption rather than an estimate and is not reported as a column.

three asylum countries per origin by refugee stock, addressing concerns that refugees from some countries are more dispersed across multiple asylum destinations than others; because origins vary in the number of asylum countries they appear in, highly dispersed origins contribute more dyads and receive greater implicit weight when each dyad-year is weighted equally.[3] Model 7 is the core GRR panel: 1980–2008 with asylum outside Europe and North America. It is the intersection of Model 4's geographic restriction with the near side of Model 2's source split, and it is the version of the dataset built from the most detailed *World Refugee Survey* reporting. It was column 2 of the main table until the main table was reduced to three columns, and it is reported here so that the comparison the source split requires stays available.

Severe exclusion policy survives every subset, but the post-war window does not, and the source split is where it fails. Models 2 and 7 are the two halves of that split, and they do not agree. Severe exclusion policy is 0.71 in the *World Refugee Survey* years and 1.59 in the State Department years, significant in both, so the coercion result holds on either source and is markedly stronger on the later one. The window is 1.14 in the Survey years and 0.05, indistinguishable from zero, in the State Department years.

Two things are confounded in that contrast and this design cannot separate them. The later period has fewer war endings with large populations still in exile, so there may simply be less of the phenomenon to detect; the State Department subsample is 901 dyad-years against 1,056 in the core panel. The sources also differ in what they record: the Survey reports on return movements directly, whereas the State Department reports are organised around human rights conditions and give return figures less consistently, which is why GRR uses UNHCR as a baseline for those years. The war-ending estimate therefore rests mainly on the pre-2009 evidence. Separating period from source effects would require coding both sources for their overlapping years.

2.6 Alternative Specifications

Table 27 varies the model specification. Model 1 reproduces the baseline Poisson model. Models 2 and 3 are OLS on a transformed outcome, $\log(y+1)$ and $\operatorname{asinh}(y)$, the two transforms usually reached for when a count has many zeros. Model 4 estimates a negative binomial model, which relaxes the Poisson equidispersion assumption. Models 5 and 6 add two-way fixed effects on each side of the dyad in turn. Model 5 uses origin-by-year fixed effects, which absorb all time-varying origin-country confounders, including origin-country conflict, ethnic status, conflict years and the post-war window itself, and identify the asylum-side variables from variation across the hosts of one origin in one year. Model 6 uses asylum-by-year fixed effects, which do the same on the other side: they absorb everything common to a host in a given year, including its own conflict, its income and its statutory refugee-rights position, and identify the coefficients from variation across the origin groups that host receives in that year. The exclusion-policy measures are coded by dyad rather than by host, so they survive this second design.

Coefficient magnitudes are not comparable across estimators, and in Models 2 and 3 they are not interpretable on their own terms either. With 71 per cent of the dyad-years they use recording no return, neither $\log(y+1)$ nor $\operatorname{asinh}(y)$ is invariant to the units in which return is measured: counting returnees in thousands rather than in people moves the severe-policy estimate by about two-thirds in both columns, while leaving the Poisson estimate in column 1 exactly where it is. The two OLS columns establish that the association survives a different functional form; the proportional reading of the coercion result belongs to Model 1, where 1.09 is a 197 per cent increase in the return count.

Under origin-by-year fixed effects the severe-policy coefficient is larger than in the baseline (1.34 against 1.09), though the high-intensity asylum-conflict estimate is imprecise there. Under asylum-by-year fixed effects it is 1.70. The coercion result therefore holds both across the hosts of one origin in one year and across the origin groups inside one host in one year.

[3] Origins vary in the number of asylum destinations they appear in because the sample rule adds asylum countries for each origin until the cumulative displaced population reaches 75 per cent of the origin's total. The median origin appears in 3 asylum countries and the most dispersed in 10, and 33 of 84 origins appear in more than three. The restriction therefore leaves most origins untouched and retains 86 per cent of the estimating sample.

Table 26: Robustness: Alternative samples and periods

Dep. Var.:	Total Repatriation						
	Main	USHR	Excl.Warr	Excl.ENA	War.Orig	Top.3	Core.GRR
Model:	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Variables							
Exclusion policy							
Exclusion policy: severe	1.09***	1.59***	1.60***	1.04***	1.07***	1.06***	0.71**
	(0.19)	(0.27)	(0.31)	(0.19)	(0.18)	(0.19)	(0.24)
Exclusion policy: moderate	0.48*	0.25	0.36†	0.50*	0.48*	0.51*	0.39*
	(0.22)	(0.33)	(0.21)	(0.22)	(0.22)	(0.22)	(0.18)
Conflict in COA: high	0.59†	-0.94*	0.71**	0.64*	0.55†	0.64*	0.95***
	(0.31)	(0.46)	(0.23)	(0.30)	(0.32)	(0.31)	(0.29)
Conflict in COA: low	0.81***	0.64*	0.51*	0.79***	0.84***	0.81***	0.62**
	(0.21)	(0.31)	(0.22)	(0.21)	(0.21)	(0.22)	(0.24)
Existing explanations							
War Ending	1.00***	0.05	0.73**	0.97***	1.02***	1.06***	1.14***
	(0.22)	(0.52)	(0.23)	(0.23)	(0.22)	(0.23)	(0.28)
Conflict in COO: high	0.11	-0.08	-0.08	0.07	0.17	0.12	0.22
	(0.28)	(0.50)	(0.32)	(0.29)	(0.27)	(0.29)	(0.33)
Conflict in COO: low	0.30	-0.29	-0.03	0.28	0.32	0.31	0.41
	(0.22)	(0.36)	(0.22)	(0.23)	(0.22)	(0.23)	(0.26)
GDP (COO)	0.39	2.58***	0.46†	0.30	0.43	0.34	0.35
	(0.28)	(0.53)	(0.27)	(0.29)	(0.28)	(0.30)	(0.31)
GDP (COA)	-1.09†	-5.08**	-1.59***	-1.12†	-1.09†	-1.21*	-0.04
	(0.59)	(1.74)	(0.44)	(0.61)	(0.60)	(0.61)	(0.66)
Civil Liberties Restrictions (COO)	-0.51**	-0.33	-0.47**	-0.57**	-0.54**	-0.50**	-0.44*
	(0.17)	(0.36)	(0.17)	(0.18)	(0.17)	(0.17)	(0.19)
Refugee Rights	-0.02	0.01	0.06	0.02	-0.02	0.02	-0.06
	(0.17)	(0.28)	(0.15)	(0.17)	(0.19)	(0.18)	(0.22)
Conflict Duration	-0.03	0.22	0.02	-0.04	-0.02	-0.03	0.02
	(0.08)	(0.16)	(0.09)	(0.09)	(0.09)	(0.09)	(0.11)
All controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Source	Both	USHR	Both	Both	Both	Both	WRS
Years	1980-2022	2009-2022	1980-2022	1980-2022	1980-2022	1980-2022	1980-2008
Fixed-effects							
Dyad	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Fit statistics							
Observations	2,327	901	2,242	1,877	2,099	2,007	1,056
Pseudo R^2	0.80	0.82	0.79	0.80	0.80	0.80	0.82

Clustered (Dyad) standard-errors in parentheses
*Signif. Codes: ***: 0.001, **: 0.01, *: 0.05, †: 0.1*

All models: Poisson PML with log stock-at-risk offset, dyad and year fixed effects. Standard errors clustered by dyad. Model 1 is column 1 of the main table. Model 2 restricts to the years coded from State Department reports. Model 3 excludes dyads linked in the literature to the refugee warrior hypothesis. Model 4 excludes dyads with asylum countries in Europe and North America. Model 5 restricts to dyads where refugees originally fled armed conflict (o_pers_nonwar == 0). Model 6 restricts to the top 3 asylum countries per origin-year by refugee stock. Model 7 is the core GRR panel, 1980–2008 with asylum outside Europe and North America – the intersection of Model 4's restriction with the near side of Model 2's source split, and the version of the dataset with the most detailed World Refugee Survey reporting. Read Models 2 and 7 as the two halves of the source split. The checks that substitute UNHCR returns or the registered stock are reported in Table 13.

Table 27: Robustness: Alternative Specifications

Dep. Var.:	Total Repatriation	log(trepat+1)	asinh(trepat)	Total Repatriation		
	Main	OLS log	OLS asinh	Neg.Bin.	Origin-Yr	Asylum-Yr
Model:	(1)	(2)	(3)	(4)	(5)	(6)
	Poisson	OLS	OLS	Neg. Bin.	Poisson	Poisson
Variables						
Exclusion policy						
Exclusion policy: severe	1.09***	4.58***	4.87***	2.74***	1.34***	1.70*
	(0.19)	(0.42)	(0.45)	(0.51)	(0.34)	(0.78)
Exclusion policy: moderate	0.48*	1.64***	1.79***	1.23*	1.36***	0.94*
	(0.22)	(0.29)	(0.32)	(0.55)	(0.16)	(0.44)
Conflict in COA: high	0.59†	0.55	0.57	0.78	0.67	
	(0.31)	(0.47)	(0.50)	(0.51)	(0.51)	
Conflict in COA: low	0.81***	0.39	0.42	0.23	0.97***	
	(0.21)	(0.30)	(0.32)	(0.41)	(0.18)	
Existing explanations						
War Ending	1.00***	0.76*	0.79*	0.89*		1.32**
	(0.22)	(0.35)	(0.38)	(0.42)		(0.45)
Conflict in COO: high	0.11	-0.67	-0.73	0.56	-0.58	0.86†
	(0.28)	(0.42)	(0.46)	(0.52)	(0.78)	(0.46)
Conflict in COO: low	0.30	-0.24	-0.27	-0.13	0.05	0.77*
	(0.22)	(0.27)	(0.29)	(0.31)	(0.41)	(0.32)
GDP (COO)	0.39	0.31	0.32	0.23		-0.15
	(0.28)	(0.26)	(0.28)	(0.57)		(0.52)
GDP (COA)	-1.09†	-0.88	-0.95	-0.61	-2.14†	
	(0.59)	(0.57)	(0.62)	(1.04)	(1.11)	
Civil Liberties Restrictions (COO)	-0.51**	-0.49**	-0.53**	-0.48		-0.55*
	(0.17)	(0.18)	(0.19)	(0.33)		(0.26)
Refugee Rights	-0.02	0.21	0.23	0.31	0.10	
	(0.17)	(0.20)	(0.22)	(0.29)	(0.16)	
Conflict Duration	-0.03	0.28*	0.30*	0.46*	0.09	-0.13
	(0.08)	(0.13)	(0.14)	(0.21)	(0.18)	(0.20)
Log Stock at Risk		0.79***	0.84***			
		(0.10)	(0.11)			
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Estimator	Poisson	OLS	OLS	NegBin	Poisson	Poisson
DV	trepat	log(y+1)	asinh(y)	trepat	trepat	trepat
Fixed-effects						
Dyad	Yes	Yes	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes		
coo_iso-Year					Yes	
coa_iso-Year						Yes
Fit statistics						
Observations	2,327	3,913	3,913	2,327	1,397	1,013
Pseudo R^2	0.80	0.15	0.15	0.04	0.93	0.94

Clustered (Dyad) standard-errors in parentheses
*Signif. Codes: ***: 0.001, **: 0.01, *: 0.05, †: 0.1*

All models include dyad fixed effects and cluster standard errors by dyad; alternative clustering schemes are reported separately. Origin-Yr and Asylum-Yr are origin-by-year and asylum-by-year fixed effects, and y is the recorded return count. Models 2 and 3 are OLS on a transformed outcome and enter the log stock at risk directly in place of the offset, which is why they run on more dyad-years than the Poisson columns. Neither transform is invariant to the units of the outcome once the outcome has zeros, and 71 per cent of the dyad-years they use record no return: measuring return in thousands rather than in people moves the severe-policy estimate from 4.47 to 1.52 in Model 2 and from 4.76 to 1.80 in Model 3, on identical data, while leaving Model 1 unchanged. Columns 2 and 3 therefore test whether the sign and significance survive a different functional form, and their magnitudes should not be read as effect sizes or compared with column 1. Model 4 uses negative binomial. Model 5 adds origin-by-year and Model 6 asylum-by-year fixed effects; variables the fixed effects absorb are omitted, which is why the origin-side rows are empty in Model 5 and the asylum-country conflict, income and refugee-rights rows are empty in Model 6. The exclusion-policy measures survive Model 6 because they are coded by dyad and so vary across the origin groups one host receives in one year.

2.6.1 Alternative Clustering of the Standard Errors

The baseline clusters by dyad because the shocks used in the model — a host government's policy and a war ending at the origin — can persist within a dyad across years. Alternative groupings account for dependence among dyads that share an origin, a host, or a calendar year.

Table 28 reports the standard errors and p-values under each scheme; the point estimates are common to all of them. Clustering by year, or two-way by dyad and year, gives the largest standard errors, and even there the severe-policy estimate is significant at the 0.001 level and the war-ending estimate at the 0.01 level.

Table 28: Alternative clustering of the standard errors

		Exclusion policy: severe		War Ending	
Clustered by	Groups	SE	p	SE	p
Dyad (baseline)	110	0.19***	<0.001	0.22***	<0.001
Origin country	51	0.16***	<0.001	0.23***	<0.001
Asylum country	65	0.17***	<0.001	0.28***	<0.001
Origin + asylum, two-way	116	0.13***	<0.001	0.28***	<0.001
Year	41	0.26***	<0.001	0.20***	<0.001
Dyad + year, two-way	151	0.22***	<0.001	0.24***	<0.001
Heteroskedastic, no clustering	–	0.24***	<0.001	0.18***	<0.001

Note: The model is column 1 of the main table throughout, so the coefficients do not vary: severe exclusion policy is 1.09 and the war-ending window 1.00 in every row. Stars are on the coefficient's significance under that variance estimator. 'Groups' is the number of clusters, summed across dimensions for the two-way estimators. The dyad-plus-year row should be read with care: with 41 year groups the variance matrix is not positive semi-definite and fixest repairs it; the year-clustered row is the cleaner conservative benchmark.

2.7 Robustness of the Repatriation-Assistance Result

Table 29 applies the same robustness checks to the repatriation-funding result as to the asylum-side findings. The funding coefficient remains positive across alternative estimators, clustering strategies, and sample restrictions, and is statistically distinguishable from zero in 10 of the 15 specifications in which it can be estimated.

The main exceptions are informative. The estimate weakens when UNHCR voluntary-return counts are used, when the analysis is restricted to the post-2009 period, and when funding is measured only as the presence of any recorded assistance rather than the amount available per refugee. It also becomes small and imprecise when covariates are lagged and when asylum-by-year fixed effects are included. That specification retains fewer than half the dyad-years, and the estimate reflects the smaller sample rather than the cross-origin comparison: holding the fixed effects at the baseline and refitting on the same rows gives essentially the same coefficient. Origin-by-year fixed effects cannot be estimated at all, because the funding measure varies at the origin-country-year level and is collinear with them by construction.

These checks therefore support a robust contemporaneous association between the amount of repatriation funding and return, but not a causal interpretation. UNHCR may allocate more funding where it already expects substantial return, and the funding data also have important gaps in 2000–2010 and 2021–2023.

Table 29: Robustness: the repatriation-assistance coefficient

Departure	Repatriation assistance	Exclusion policy: severe	Dyad-years	Dyads
Baseline (main table col. 2)	0.31** (0.12)	1.29***	1,378	100
Outcome: UNHCR voluntary return	0.19 (0.13)	0.43	1,333	96
Outcome: 3-year moving average	0.26* (0.10)	1.00***	1,309	101
All covariates at $t-1$	0.10 (0.13)	1.37***	1,132	96
Asylum-by-year fixed effects	0.01 (0.14)	3.22**	592	60
Origin-by-year fixed effects	absorbed: the measure varies at the origin-year level	–	–	–
Negative binomial	0.58*** (0.17)	3.05***	1,378	100
Clustered by dyad and year	0.31** (0.12)	1.29***	1,378	100
Excl. refugee-warrior dyads	0.28* (0.11)	1.85***	1,318	96
Excl. asylum in Europe & N. America	0.37** (0.11)	1.20***	1,185	82
Top 3 asylum countries per origin	0.26* (0.11)	1.27***	1,221	95
Fled armed conflict only	0.28* (0.13)	1.20***	1,257	93
Core GRR panel (1980–2008, Global South)	0.35** (0.12)	1.13**	580	56
State Department years (2009–)	0.03 (0.25)	2.22***	593	67
Unstandardised: log funding per refugee	0.23** (0.08)	1.29***	1,378	100
Extensive margin: any funding recorded	0.14 (0.30)	1.24***	1,378	100

Note: Each row is the specification of column 2 of the main table – column 1 plus UNHCR repatriation funding per refugee, standardised – with one feature changed. PPML with the log stock at risk as offset, dyad and year fixed effects and standard errors clustered by dyad unless the row says otherwise. *Exclusion policy: severe* is reported alongside as a reference point and its standard errors are suppressed for space. Origin-by-year fixed effects are absorbed rather than omitted: the funding measure varies at the origin-country-year level, so a coo×year effect is collinear with it by construction. The unstandardised row changes only the scale of the measure, so its p-value matches the baseline; the extensive-margin row replaces the continuous measure with an indicator for any recorded funding, so its magnitude is not comparable with the rest of the column. Every row runs on the funding-coded years only and so inherits the coverage gap described in Section 2.4.

2.8 Event Study around Onsets of Severe Exclusion Policy

The main text reports a stacked event study around the onset of severe exclusion policies, which addresses the concern that host governments adopt coercive policies after return has already begun to rise. It finds no pre-trend two or three years before onset, a sharp rise in the onset year and a peak one year later, and notes that the design rests on few onsets and cannot rule out anticipation in the omitted year immediately before implementation.

This section reports the underlying table, for two- and three-year windows, and the linear version of the estimator. Severe exclusion policy begins 49 times and ends 51 times in the panel, so treatment is not absorbing; each onset therefore defines its own cohort with a window of $\pm K$ years, and controls are dyads observed throughout the same period whose host records no severe policy in any year of that window. The two-year window contains 34 cohorts across 24 dyads and the three-year window 23 cohorts across 17 dyads. Alongside PPML on the return count I estimate a linear model on return as a share of the stock at risk, which retains dyads with no within-dyad variation in return and so uses about 13,000 stacked dyad-years against about 5,300 under PPML.

The two estimators and the two windows give the same picture as the main text. The lead coefficients are small and not distinguishable from zero in any of the four: joint tests of the leads yield $p = 0.47$ for PPML and $p = 0.88$ for the return rate when $K = 2$, and $p = 0.55$ and $p = 0.11$ when $K = 3$.

Figure 4 plots the three-year window under both estimators. The vertical scales are not comparable, since the PPML coefficients are on the log scale and the linear-model coefficients are changes in the return rate. One further limitation is worth adding to those noted in the main text: the controls are dyads without severe coercion during the relevant window, not a randomly assigned comparison group, so the estimates address whether return was already rising before recorded policy onset rather than the full endogeneity of policy adoption.

Table 30: Event study around onsets of severe exclusion policy

Event year	PPML, $K = 2$	Rate, $K = 2$	PPML, $K = 3$	Rate, $K = 3$
$t - 3$	–	–	0.23 (0.21)	-0.03† (0.02)
$t - 2$	-0.31 (0.43)	-0.00 (0.02)	0.21 (0.32)	-0.00 (0.02)
$t - 1$	(ref.)	(ref.)	(ref.)	(ref.)
$t + 0$	0.71* (0.28)	0.11** (0.04)	0.96*** (0.24)	0.06* (0.03)
$t + 1$	0.83† (0.45)	0.07† (0.04)	1.53*** (0.29)	0.10* (0.05)
$t + 2$	0.34 (0.34)	0.03 (0.04)	0.88** (0.30)	0.06 (0.05)
$t + 3$	–	–	0.42 (0.31)	0.02 (0.02)

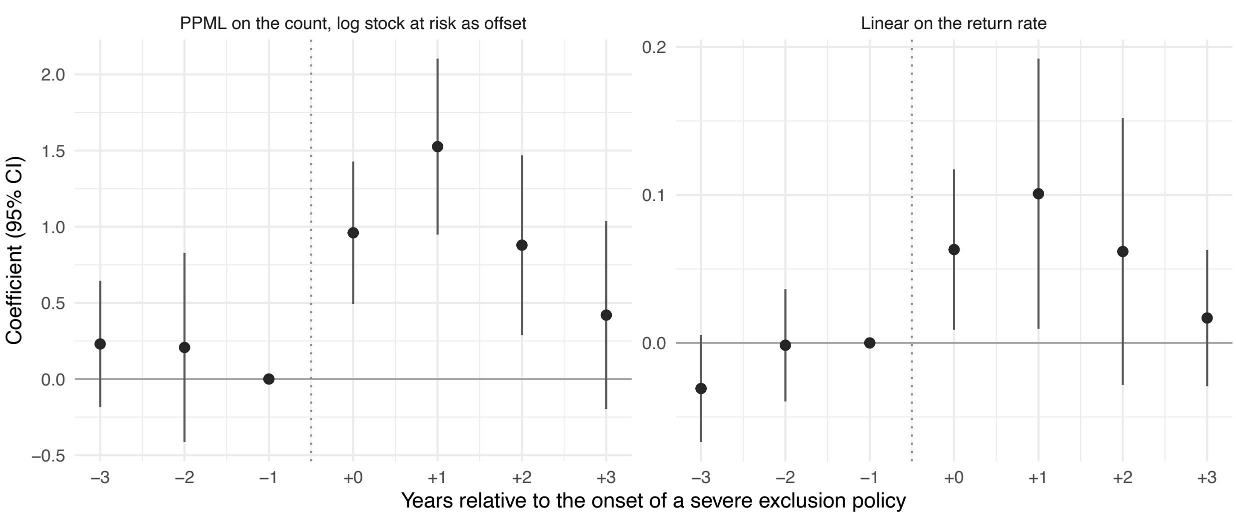

Figure 4: Event study around onsets of severe exclusion policy, three-year window, both estimators

2.9 Influential Cases

2.9.1 Dropping the Largest Cases

The top five percent of dyad-years account for 88.5 percent of recorded return. Table 31 tests whether this concentration drives the estimates by dropping the largest dyads or dyad-years and, separately, restricting the sample to the ten largest dyads.

The severe-policy coefficient is larger after each exclusion, not smaller: 1.44 after dropping the five largest dyads, 1.54 after dropping the largest one percent of dyad-years and 1.45 after dropping the largest five percent, against 1.09 in the baseline. Among the ten largest dyads alone it is smaller (0.86) on 275 dyad-years.

The post-war estimate is less stable. Dropping the five largest dyads moves it from 1.00 to 0.84 and dropping the largest one percent leaves it at 0.95; dropping the largest five percent gives 0.41 and among the ten largest dyads it is 0.87. The window contains only 204 dyad-years, so trimming the largest five percent removes much of its support. Low-intensity asylum-country conflict also becomes imprecise after excluding the largest cases, but is strongly associated with return among the ten largest dyads (1.24). Severe policy is therefore robust across the distribution, while the post-war window and low-intensity asylum conflict depend more on which large episodes are included.

2.9.2 Leave-One-Out Analyses

Figure 5 re-estimates the main model after excluding each origin country, asylum country, origin region, and asylum region in turn. Table 32 reports exclusions that change a coefficient's sign or move it by more than one.

No single origin, host, region or dyad accounts for the severe-policy result. Excluding Afghanistan–Pakistan moves the coefficient from 1.09 to 1.33, and excluding Rwanda–DR Congo to 1.29; both remain significant at the 0.001 level, as does every other single exclusion.

The larger shifts occur for coefficients that are weakly estimated in the baseline. High-intensity asylum-country conflict, significant only at the ten percent level (0.59), becomes negative when Sub-Saharan Africa is excluded as either an origin or an asylum region, although the resulting estimates are not significant, and it is most sensitive to Rwanda–DR Congo and the two Afghan dyads. The moderate-policy coefficient also changes when Asia and MENA are excluded, and origin-country conflict changes sign under several exclusions but is never distinguishable from zero in the baseline. The asylum-conflict coefficient is therefore the least stable of the main asylum-side results.

Table 31: Robustness: Concentration of Returns

Dependent Variable:	Total Repatriation				
	Full	Drop top 5 dyads	Drop top 1% yrs	Drop top 5% yrs	Top 10 dyads
Model:	(1)	(2)	(3)	(4)	(5)
Exclusion policy					
Exclusion policy: severe	1.09***	1.44***	1.54***	1.45***	0.86*
	(0.19)	(0.25)	(0.18)	(0.34)	(0.42)
Exclusion policy: moderate	0.48*	0.16	0.45**	0.58**	0.68
	(0.22)	(0.19)	(0.17)	(0.22)	(0.43)
Conflict in COA: high	0.59†	0.53*	0.51*	0.35	0.49
	(0.31)	(0.25)	(0.24)	(0.25)	(0.65)
Conflict in COA: low	0.81***	0.27	0.26	0.01	1.24***
	(0.21)	(0.26)	(0.25)	(0.17)	(0.37)
Existing explanations					
War Ending	1.00***	0.84**	0.95***	0.41*	0.87*
	(0.22)	(0.28)	(0.25)	(0.20)	(0.40)
Conflict in COO: high	0.11	0.07	0.06	-0.53*	-0.71***
	(0.28)	(0.31)	(0.29)	(0.27)	(0.18)
Conflict in COO: low	0.30	0.16	0.29	-0.25	-0.34
	(0.22)	(0.24)	(0.21)	(0.17)	(0.27)
GDP (COO)	0.39	0.58†	0.36	0.60**	1.63†
	(0.28)	(0.30)	(0.27)	(0.22)	(0.86)
GDP (COA)	-1.09†	-0.50	-0.59	0.12	-3.49*
	(0.59)	(0.58)	(0.48)	(0.57)	(1.39)
Civil Liberties Restrictions (COO)	-0.51**	-0.39*	-0.47*	-0.29	-0.89**
	(0.17)	(0.19)	(0.20)	(0.19)	(0.28)
Refugee Rights	-0.02	0.06	0.12	0.02	0.39*
	(0.17)	(0.15)	(0.15)	(0.13)	(0.18)
Conflict Duration	-0.03	0.04	0.01	0.07	0.36
	(0.08)	(0.10)	(0.09)	(0.09)	(0.31)
Fixed-effects					
Dyad	Yes	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes	Yes
Fit statistics					
Observations	2,327	2,199	2,289	2,081	275
Squared Correlation	0.76	0.62	0.59	0.33	0.89
Pseudo R^2	0.80	0.67	0.65	0.44	0.87

Clustered (Dyad) standard-errors in parentheses
*Signif. Codes: ***: 0.001, **: 0.01, *: 0.05, †: 0.1*

Table 32: Influential regions

Variable	Region Dropped	Drop Type	Main Est.	New Est.	Change
Conflict in COA: high	Sub-Saharan Africa	Origin Region	0.59+	-0.25	0.84
Conflict in COA: high	Sub-Saharan Africa	Asylum Region	0.59+	-0.18	0.77
Conflict in COO: high	Sub-Saharan Africa	Origin Region	0.11	-0.69	0.79
Conflict in COO: high	Sub-Saharan Africa	Asylum Region	0.11	-0.65	0.76
Conflict in COO: low	Sub-Saharan Africa	Asylum Region	0.30	-0.52+	0.82
Conflict in COO: low	Sub-Saharan Africa	Origin Region	0.30	-0.52+	0.82

Note: Regions whose exclusion causes a sign change or an absolute coefficient shift above 1. Stars indicate significance of each coefficient: + p<0.10, * p<0.05, ** p<0.01, *** p<0.001. Bold rows indicate sign reversals where the new coefficient remains statistically significant at p<0.05; sign flips on coefficients that fall back to non-significance after exclusion are shown but not bolded.

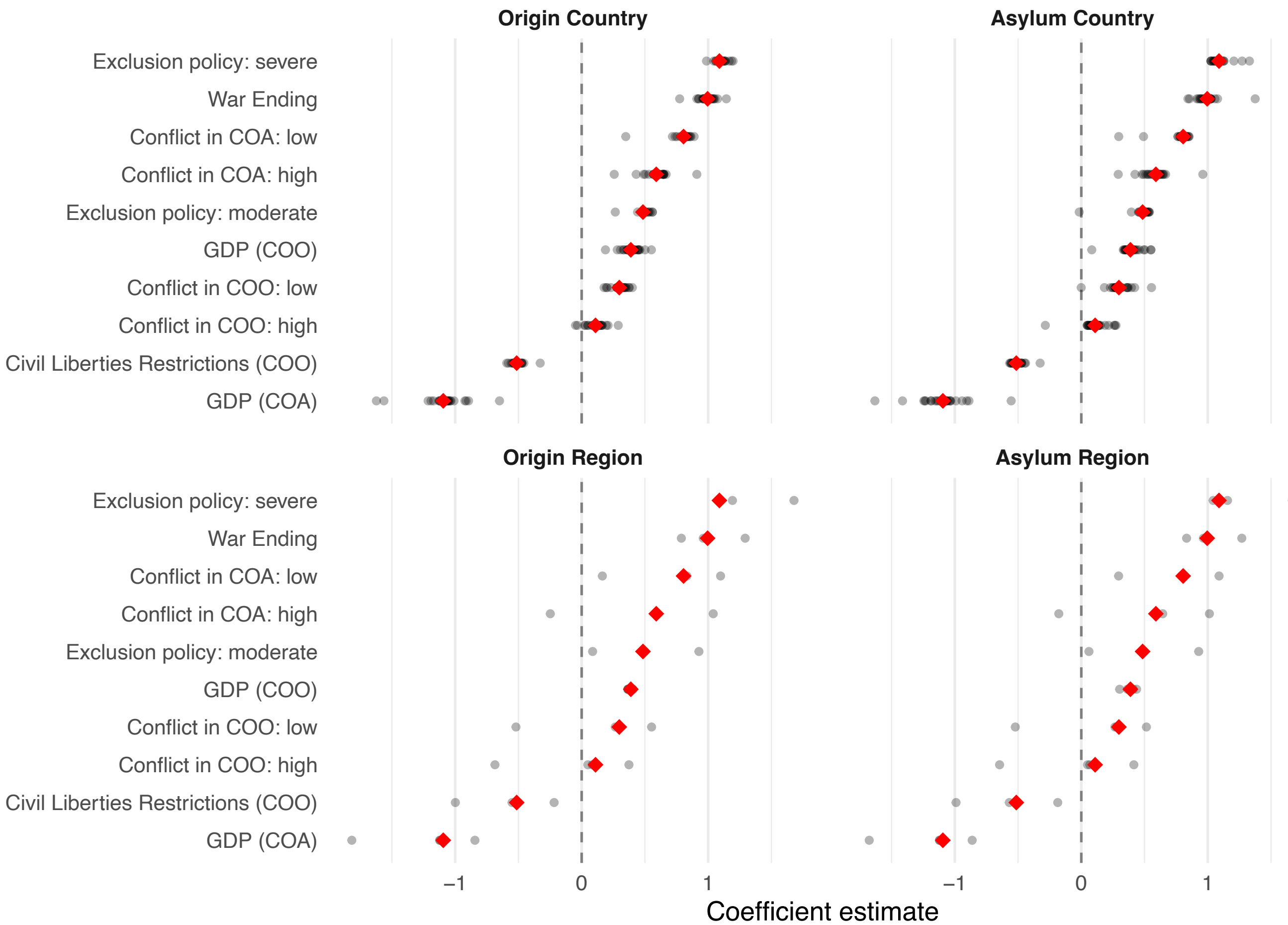

Figure 5: Coefficient stability across leave-one-out analyses

Table 33: Most influential dyads

Variable	Dyad	Main Est.	New Est.	Change
Conflict in COA: high	AFG-PAK	0.59+	0.96***	0.37
Conflict in COA: high	RWA-COD	0.59+	0.31	0.28
Conflict in COA: high	AFG-IRN	0.59+	0.34	0.25
Conflict in COA: high	SOM-ETH	0.59+	0.50	0.09
Conflict in COA: high	MOZ-MWI	0.59+	0.52+	0.07
Conflict in COA: low	AFG-IRN	0.81***	0.27	0.54
Conflict in COA: low	AFG-PAK	0.81***	0.49*	0.31
Conflict in COA: low	KHM-THA	0.81***	0.89***	0.08
Conflict in COA: low	MMR-THA	0.81***	0.73***	0.08
Conflict in COA: low	RWA-COD	0.81***	0.76***	0.05
Exclusion policy: severe	AFG-PAK	1.09***	1.33***	0.24
Exclusion policy: severe	RWA-COD	1.09***	1.29***	0.20
Exclusion policy: severe	AFG-IRN	1.09***	0.97***	0.12
Exclusion policy: severe	BDI-TZA	1.09***	1.19***	0.11
Exclusion policy: severe	MMR-BGD	1.09***	1.03***	0.06
Exclusion policy: moderate	AFG-IRN	0.48*	-0.01	0.49
Exclusion policy: moderate	AFG-PAK	0.48*	0.40*	0.09
Exclusion policy: moderate	BDI-TZA	0.48*	0.54*	0.06
Exclusion policy: moderate	ETH-SOM	0.48*	0.53*	0.05
Exclusion policy: moderate	RWA-COD	0.48*	0.53*	0.04
GDP (COA)	AFG-IRN	-1.09+	-0.54	0.56
GDP (COA)	KHM-THA	-1.09+	-1.63**	0.53
GDP (COA)	UGA-COD	-1.09+	-1.36*	0.26
GDP (COA)	UGA-SDN	-1.09+	-1.31*	0.21
GDP (COA)	MMR-BGD	-1.09+	-0.91	0.19
War Ending	AFG-IRN	1.00***	1.23***	0.24
War Ending	MOZ-MWI	1.00***	0.84***	0.15
War Ending	AFG-PAK	1.00***	0.86***	0.14
War Ending	LBR-GIN	1.00***	0.92***	0.08
War Ending	IRQ-IRN-IRN	1.00***	1.07***	0.07
Civil Liberties Restrictions (COO)	AFG-IRN	-0.51**	-0.39*	0.13
Civil Liberties Restrictions (COO)	ETH-ERI-SDN	-0.51**	-0.60***	0.09
Civil Liberties Restrictions (COO)	AFG-PAK	-0.51**	-0.44**	0.07
Civil Liberties Restrictions (COO)	MMR-THA	-0.51**	-0.45**	0.06
Civil Liberties Restrictions (COO)	IRQ-IRN-IRN	-0.51**	-0.45**	0.06
Conflict in COO: high	RWA-COD	0.11	-0.13	0.24
Conflict in COO: high	ETH-SOM	0.11	0.27	0.16
Conflict in COO: high	AFG-PAK	0.11	0.26	0.15
Conflict in COO: high	AGO-ZMB	0.11	0.20	0.09
Conflict in COO: high	SDN-SSD-UGA	0.11	0.20	0.09
Conflict in COO: low	RWA-COD	0.30	0.10	0.20
Conflict in COO: low	AFG-IRN	0.30	0.50*	0.20
Conflict in COO: low	ETH-SOM	0.30	0.42+	0.12
Conflict in COO: low	AFG-PAK	0.30	0.18	0.11
Conflict in COO: low	MOZ-MWI	0.30	0.37	0.07
GDP (COO)	KHM-THA	0.39	0.19	0.20
GDP (COO)	AFG-PAK	0.39	0.55+	0.16
GDP (COO)	MMR-BGD	0.39	0.55*	0.16
GDP (COO)	SOM-ETH	0.39	0.49+	0.11
GDP (COO)	MMR-THA	0.39	0.29	0.10

Note: For each key variable, the five dyads whose exclusion produces the largest absolute change in the coefficient. Stars indicate significance of each coefficient: + p<0.10, * p<0.05, ** p<0.01, *** p<0.001. Bold rows indicate sign reversals where the new coefficient remains statistically significant at p<0.05.

2.10 Aggregate Distribution and Model-Adjusted Contrasts

The main text shows how much return occurs around war endings, asylum-country push factors, and UNHCR repatriation assistance. This section first examines how these conditions overlap in observed return, then asks whether the main ordering persists after adjusting for observed differences across dyads. These comparisons are descriptive rather than causal.

2.10.1 Observed Return Across Contexts

Table 34 partitions dyad-years into mutually exclusive contexts defined by post-war periods, asylum-side shocks, and repatriation assistance. Panels A and B use the full descriptive sample and differ only in the definition of an asylum-side shock: Panel A includes either level of exclusion policy or asylum-country conflict, while Panel B restricts the definition to severe policy and high-intensity conflict. The similar ordering across the two panels shows that the descriptive pattern does not depend on the broader definition.

Panels C and D add two measures of UNHCR repatriation assistance. Panel C uses a dyad-level indicator for an active repatriation programme and is restricted to 1980–2008, when these programmes are recorded systematically. Panel D uses repatriation expenditure per refugee, measured at the origin-country-year level, for the years in which the funding series is available. Because the UNHCR measures cover different samples, shares should be compared within rather than across panels.

Table 34: Exposure and observed return by context, in levels

Context	Dyad-years	Stock-years at risk (m)	Return (m)	Share of return	Return rate
Panel A. Any exclusion policy or asylum conflict					
Neither	3,449 (67.5%)	238.38 (35.8%)	4.82	14.5%	2.02%
Post-war window only	148 (2.9%)	10.84 (1.6%)	2.30	6.9%	21.19%
Asylum shock only	1,456 (28.5%)	404.50 (60.8%)	23.33	70.1%	5.77%
Both	56 (1.1%)	11.83 (1.8%)	2.83	8.5%	23.95%
Panel B. Severe policy or high-intensity asylum conflict only					
Neither	4,418 (86.5%)	462.70 (69.5%)	13.25	39.8%	2.86%
Post-war window only	188 (3.7%)	17.33 (2.6%)	4.26	12.8%	24.61%
Asylum shock only	487 (9.5%)	180.18 (27.1%)	14.90	44.8%	8.27%
Both	16 (0.3%)	5.34 (0.8%)	0.87	2.6%	16.21%
All dyad-years	5,109 (100.0%)	665.55 (100.0%)	33.28	100.0%	5.00%
Panel C. UNHCR repatriation programme (dyadic, 1980–2008)					
Outside window, no programme	1,777 (81.9%)	184.75 (61.9%)	7.25	29.5%	3.92%
Outside window, programme	248 (11.4%)	94.08 (31.5%)	12.45	50.6%	13.23%
Post-war window, no programme	86 (4.0%)	5.40 (1.8%)	0.24	1.0%	4.39%
Post-war window, programme	60 (2.8%)	14.14 (4.7%)	4.65	18.9%	32.86%
All dyad-years, 1980–2008	2,171 (100.0%)	298.37 (100.0%)	24.58	100.0%	8.24%
Panel D. UNHCR repatriation funding (origin-year, 1980–99 and 2011–20)					
Outside window, no funding	2,118 (70.2%)	175.80 (43.0%)	2.40	11.6%	1.36%
Outside window, funding	743 (24.6%)	218.64 (53.5%)	15.95	76.9%	7.29%
Post-war window, no funding	75 (2.5%)	5.04 (1.2%)	0.26	1.3%	5.18%
Post-war window, funding	81 (2.7%)	9.33 (2.3%)	2.13	10.3%	22.86%
All dyad-years, funding window	3,017 (100.0%)	408.81 (100.0%)	20.74	100.0%	5.07%

Note: Dyad-years with more than 1,000 refugees at risk in the previous year, excluding 1989. Contexts are mutually exclusive within each panel and each panel's shares are computed within its own sample, so the table reports three totals. Panels A and B partition all 5,109 dyad-years. Panel C partitions the 2,171 dyad-years from 1980 to 2008 for which the dyad-level UNHCR repatriation programme flag is coded, which is 92% of that period's dyad-years and 99% of its recorded return. Panel D partitions the 3,017 dyad-years for which UNHCR repatriation expenditure on the origin population is coded, complete for 1980–1999 and 2011–2020; that measure varies at the origin-country-year level rather than the dyad level. The post-war window covers the first three peaceful years following the end of a high-intensity origin-country conflict. The return rate is total recorded return over total stock-years at risk within the row, not the mean of dyad-year ratios, which would weight a small refugee population like a large one.

2.10.2 Model-Adjusted Aggregate Contrasts

Table 35: Exposure and model-adjusted aggregate contrasts

Condition	Dyad-years present	Stock-years exposed	Reduction in fitted return
Panel A. Model including repatriation assistance (2,248 dyad-years, 219 dyads); assistance coefficient 0.46 (SE 0.12)			
Severe exclusion policy	132 (5.9%)	99.9m	23.2%
Post-war window (1–3 years)	189 (8.4%)	21.6m	11.0%
High-intensity asylum conflict	280 (12.5%)	55.0m	6.9%
Repatriation assistance	706 (31.4%)	160.2m	47.5%
Panel B. Full pooled sample, no assistance term (3,810 dyad-years, 249 dyads)			
Severe exclusion policy	132 (3.5%)	99.9m	24.3%
Post-war window (1–3 years)	189 (5.0%)	21.6m	13.6%
High-intensity asylum conflict	280 (7.3%)	55.0m	5.7%

Note: The final column uses the pooled PPML specification, which is Model 3 of the main paper, and reports the percentage reduction in aggregate fitted return after setting the listed condition to its reference value, holding the remaining covariates fixed. Panel A is the model the main text reports, which includes repatriation assistance and therefore runs only on the dyad-years for which UNHCR repatriation funding is coded. PANEL B IS A DIFFERENT MODEL ON DIFFERENT ROWS and the two panels must not be read against each other: funding is uncoded for 2000–2010 and 2021–2023, so carrying it costs 41 per cent of Panel B's dyad-years. The three shared conditions appear in both panels precisely so that the cost of that sample is visible rather than implied. Panel B retains 66.5 per cent of stock-years in the estimation frame. Repatriation assistance enters as the standardised log of UNHCR repatriation funding per refugee; it is switched off by setting that measure to the value it takes at zero funding, not to zero on the standardised scale, which is the sample mean. Its dyad-years-present count is the number with any funding recorded. Dyad-year counts and stock-years in Panel B are the full-sample exposure levels shown against Panel B's denominator. Conditions can overlap, so the rows are not additive within either panel. The final column is a model-implied descriptive contrast, not an identified causal effect.

The descriptive comparisons may partly reflect differences in the populations exposed to each condition. Table 35 therefore uses a pooled PPML model to compare fitted aggregate return with and without each condition, holding the remaining covariates fixed.

Panel A is the model the main text reports. Because repatriation funding is observed for only part of the panel, it is a separate model estimated on the 2,248 dyad-years for which the series is coded, and the other three conditions are recomputed there so that all four contrasts come from one model on one sample. Removing repatriation funding reduces fitted return by 47.5 per cent, the largest of the four, against 23.2 per cent for severe exclusion policy, 11.0 per cent for the post-war window and 6.9 per cent for high-intensity asylum conflict. Funding is switched off by setting the standardised log of funding per refugee to the value it takes at zero funding, not to zero on the standardised scale, which is the sample mean; the distinction matters, because switching it off at the mean instead gives 32.6 rather than 47.5 per cent.

Panel B drops the assistance term and returns to the full pooled sample of 3,810 dyad-years. The ordering among the three shared conditions is the same in both panels: removing severe exclusion policy reduces fitted return more than removing either the post-war window or high-intensity asylum conflict. That ordering reflects exposure as well as the estimated association, since severe policies occur in relatively few dyad-years but affect large refugee populations.

Three limits apply to all of these figures. The model is pooled and carries no fixed effects, so it compares dyads with one another as well as a dyad with itself. The conditions overlap, so their contributions cannot be added. And the funding sample is selective in a way that bears directly on the largest contrast: UNHCR may allocate more where substantial return is already expected, and nothing here separates that from an effect of the funding. These are model-implied descriptive contrasts showing that the descriptive ordering survives adjustment for observed covariates, not estimates of how much return each condition caused.

3 Extensions and Supporting Analyses

3.1 Correlates of Global Refugee Return

Figure 6 compares the share of recorded return that occurs under each condition with the share of the population at risk exposed to it. The conditions are not mutually exclusive. These are descriptive comparisons without controls or fixed effects.

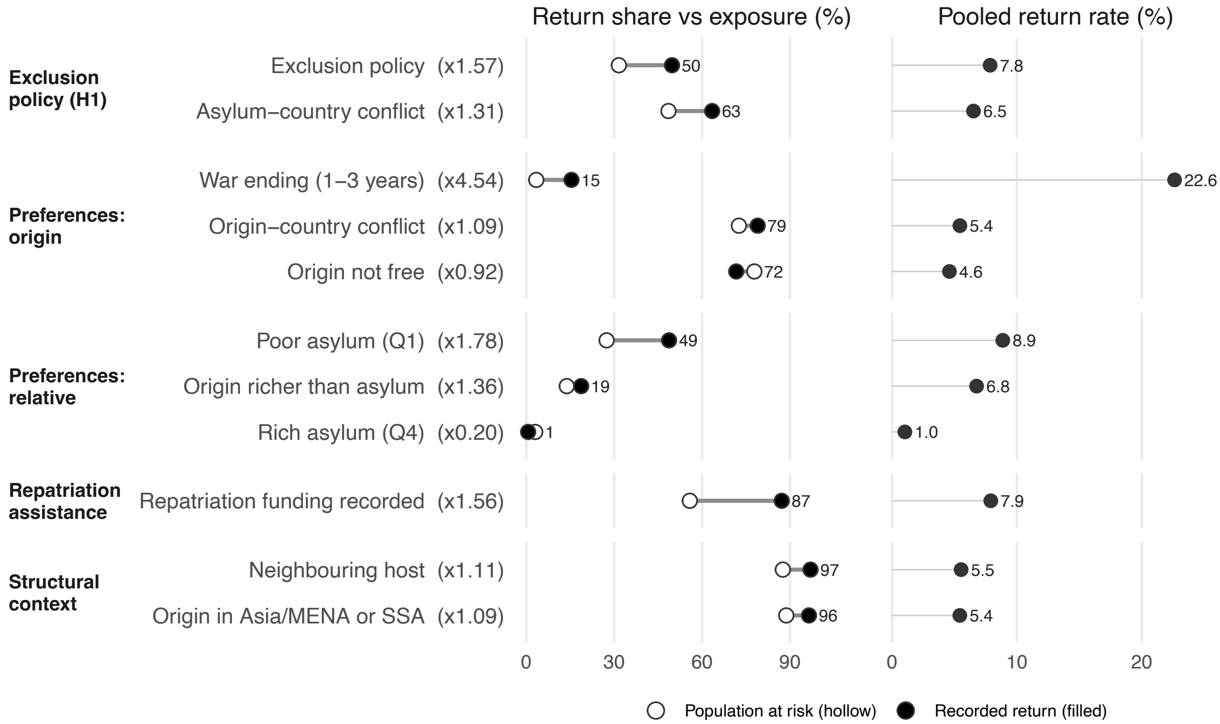

Figure 6: Correlates of global refugee return

3.2 Conflict, War Endings, and Exclusion Policy in the Largest Dyads

Figure 7 plots the twenty origin–asylum dyads with the most recorded return over time, showing origin-country conflict intensity alongside the asylum-side conditions the paper's descriptive figure aggregates. It is the disaggregated form of that figure: where the main text reports what share of return occurs under each condition, this shows where and when each condition actually held, and it keeps the distinctions the summary collapses — moderate against severe exclusion policy, and low- against high-intensity conflict in the country of asylum.

Two features of the figure carry the argument. Asylum-side shocks are common across these major displacement contexts rather than confined to a few, and they frequently occur while conflict in the origin country is still ongoing, which is visible as symbols sitting on shaded rather than unshaded stretches of a row. The rows are selected on the outcome, not on the treatment: they are the twenty dyads with the most recorded return, so an exclusion policy is free to appear or not, and it does both.

Row labels give the direction of return, from the country of asylum to the country of origin. Two dyads change origin partway through the period, at Eritrean and at South Sudanese independence, and are labelled with both origins.

3.3 The Implementation of Exclusion Policies

The main analysis treats exclusion policy as a constraint on refugees' ability to remain in asylum. This section asks which hosts and refugee populations are most likely to face that constraint, and when governments adopt it. Possible explanations concern the host government, characteristics of the refugee population, and the trajectory of the displacement crisis. Coercion may be more likely under authoritarian governments or weak

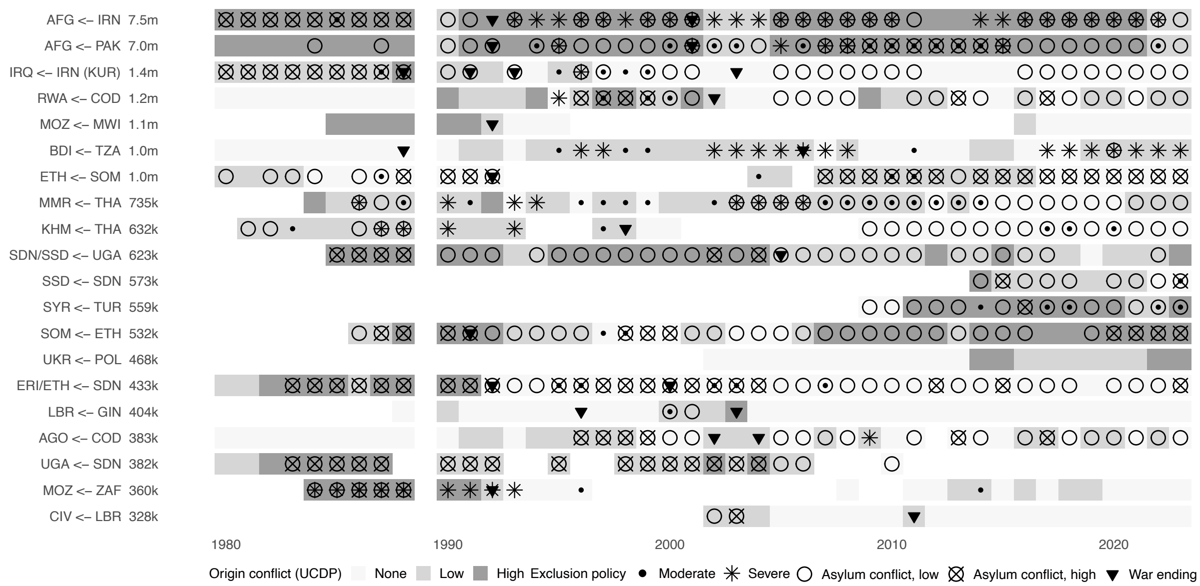

Figure 7: Conflict, war endings, and exclusion policy in the 20 dyads with the most return

Note: The twenty origin–asylum dyads with the most recorded return from 1980 to 2023, which together account for 78% of all recorded return. Row labels give the direction of return, from asylum to origin, followed by the dyad's total recorded return; rows are ordered by that total. Shading indicates origin-country conflict intensity from UCDP. Symbols indicate exclusion policy, asylum-country conflict, and war endings in the country of origin, as defined in the legend. Policy symbols are drawn over conflict symbols where both occur, so a dyad-year carrying both reads as a star or dot inside a circle. Of the twenty dyads, nine record a severe exclusion policy at some point and fifteen record one at any level, against 35% of dyads panel-wide that ever record a policy of any kind. GRR has no observations for 1989, which is the gap in every row. The war-ending markers are taken from the extended civil-war list merged with the coded interstate wars, which is a wider set of endings than the UCDP high-intensity spells used elsewhere in this appendix.

legal protections, against large, socially distant, or militarized populations, or as conflict and displacement persist.

The analysis uses the full GRR policy panel, retaining dyad-years with a coded exclusion-policy measure and more than 1,000 refugees in the preceding year. Severe policy appears in 150 dyad-years (3.4%), while either moderate or severe policy appears in 441 (9.9%). These are descriptive rather than causal estimates, but they show which associations persist across different comparisons.

Five tables follow. Table 36 enters each candidate correlate on its own, with no controls, to show what the intuitive explanations look like before composition is addressed. Table 37 then holds host-year, origin-year and dyad characteristics fixed in turn, so each column poses the question to a different comparison set: different refugee populations in the same host and year, different hosts of the same origin in the same year, and the same dyad over time. Table 38 repeats that ladder for social and ethnic ties between origin and host, and Table 40 for origin battle deaths and reported militarization, the two security explanations that most plausibly compete with population size. Table 39 turns from which populations to when: it reports the probability that a severe policy begins, by how long the dyad has been in the panel and by whether the population is above or below 100,000.

The clearest result, in Tables 36 and 37, is the importance of the size of the refugee population targeted by the policy. In the fully adjusted pooled model, a ten-fold increase in lagged dyad-specific stock is associated with 8.10 times the odds of severe policy. The relationship remains positive when comparing different refugee populations within the same host-year and when comparing the same dyad over time. It is also the dyad-specific population, rather than the host country's overall refugee burden, that matters: once both are included, the association with total host-country refugee population largely disappears.

Other proposed explanations are much less consistent. Democracy and refugee-rights protections are associated with less coercion in simple comparisons, but these relationships change sign or disappear after adjustment. Origin-country violence and reported refugee militarization are also associated with coercion in the bivariate data, but neither consistently predicts severe policy once refugee stock and other differences are taken into account. Nor is there consistent evidence that social proximity between refugees and hosts protects against coercion.

The timing of severe policy, in Table 39, reinforces the importance of population size but provides little evidence for a general "host fatigue" process. Severe policy often occurs during ongoing conflict in the origin, but the association with conflict weakens substantially once refugee stock is included. Similarly, the risk of severe-policy onset does not rise steadily as a crisis ages, nor is there a common population threshold at which coercion begins. Large refugee populations are instead exposed to severe policy at different points in the life of a crisis.

Taken together, the results suggest that severe exclusion policy is most consistently associated with the size of the particular refugee population a government is hosting. Large populations often arise in protracted, unresolved conflicts, but battlefield violence, crisis duration, host political institutions, militarization, and social ties do not independently predict coercion with the same consistency. The evidence therefore points less to a general threshold of "host fatigue" than to the pressures associated with accommodating large refugee populations during unresolved displacement crises.

Table 36: Correlates of exclusion policy, entered one at a time

Correlate	Severe policy	Any policy	Dyad-years
Scale of the refugee population			
Refugee stock in this dyad (log, lagged)	1.35*** (0.10)	0.91*** (0.05)	4,448
Total refugees hosted by the asylum country (log)	0.83*** (0.10)	0.53*** (0.06)	4,448
Camp residence (ordinal)	0.48*** (0.08)	0.39*** (0.05)	4,425
Refugees reported militarized	1.53*** (0.30)	1.60*** (0.21)	4,360
Host government and host pressure			
GDP p.c. in asylum (log)	-0.53*** (0.09)	-0.47*** (0.05)	4,412
V-Dem polyarchy (host)	-0.41*** (0.09)	-0.43*** (0.05)	4,448
Freedom House political rights (host)	0.36*** (0.09)	0.42*** (0.05)	4,111
DWRAP refugee-rights index (host)	-0.12 (0.08)	-0.24*** (0.05)	4,285
Conflict in asylum: high	-0.42 (0.37)	0.45** (0.16)	4,448
Conflict in asylum: low	1.03*** (0.17)	0.70*** (0.11)	4,448
Origin-country conditions			
Conflict in origin: high	1.33*** (0.18)	1.02*** (0.12)	4,448
Conflict in origin: low	$0.32^{\dagger}$ (0.18)	0.28* (0.11)	4,448
Battle deaths in origin (log)	0.63*** (0.08)	0.43*** (0.05)	4,113
GDP p.c. in origin (log)	-1.05*** (0.10)	-0.59*** (0.06)	4,179
Civil-liberties restrictions in origin	0.72*** (0.11)	0.56*** (0.06)	4,403
Cumulative years of origin conflict (log)	0.67*** (0.08)	0.45*** (0.05)	4,448
Dyadic relationship and duration			
Distance origin–asylum (log)	-0.36*** (0.07)	-0.42*** (0.05)	4,399
Origin and asylum contiguous	1.67*** (0.24)	1.47*** (0.13)	4,399
Composite cultural-tie index	0.39*** (0.08)	0.44*** (0.05)	4,274
Years since the dyad entered the panel	-0.12 (0.09)	-0.13* (0.05)	4,448

Note: Each cell is a separate logit of the policy indicator on that correlate alone, with no controls and no fixed effects; standard errors in parentheses. Continuous measures are standardised on the policy sample, so coefficients are per standard deviation. These are descriptive associations and are not to be read as effects.

Table 37: Predictors of exclusion policy, holding host-year, origin-year and dyad fixed

Variable	(1) Pooled, severe	(2) Pooled, any	(3) Same host-year, severe	(4) Same origin-year, severe	(5) Same dyad, severe	(6) Same dyad, any
Scale of the refugee population						
Refugee stock in this dyad (log, lagged)	1.71*** (0.32)	0.71*** (0.17)	4.23*** (1.01)	$2.20^{\dagger}$ (1.32)	1.19** (0.40)	0.21 (0.22)
Total refugees hosted by the asylum country (log)	-0.05 (0.24)	0.15 (0.21)	–	-0.74 (1.19)	-0.29 (0.46)	0.38* (0.19)
Camp residence (ordinal)	0.34 (0.24)	0.19 (0.18)	0.63 (1.39)	-0.86 (0.68)	0.67 (0.69)	0.20 (0.28)
Host government and host pressure						
GDP p.c. in asylum (log)	0.18 (0.34)	0.63* (0.28)	–	3.52* (1.42)	-1.26 (1.63)	-1.80* (0.78)
V-Dem polyarchy (host)	0.94** (0.36)	0.46* (0.21)	–	-1.03 (1.26)	0.55 (0.77)	0.23 (0.55)
DWRAP refugee-rights index (host)	0.40* (0.19)	-0.02 (0.16)	–	1.04* (0.48)	0.64* (0.32)	0.14 (0.31)
Conflict in asylum: high	-0.84 (0.56)	0.23 (0.47)	–	-1.39 (1.94)	-0.18 (0.83)	0.00 (0.45)
Conflict in asylum: low	0.48 (0.35)	0.32 (0.36)	–	-0.57 (1.08)	0.53 (0.51)	-0.01 (0.36)
Origin-country conditions						
Conflict in origin: high	0.04 (0.45)	0.11 (0.31)	4.00 (2.46)	–	0.03 (0.86)	-0.04 (0.39)
Conflict in origin: low	0.03 (0.48)	0.09 (0.30)	0.91 (1.90)	–	0.30 (0.46)	0.24 (0.35)
GDP p.c. in origin (log)	-0.85** (0.30)	-0.22 (0.18)	-2.12* (0.90)	–	-0.91 (0.99)	-0.21 (0.50)
Civil-liberties restrictions in origin	-0.17 (0.24)	0.11 (0.22)	-4.70*** (1.34)	–	0.56 (0.39)	0.52* (0.21)
Cumulative years of origin conflict (log)	0.10 (0.24)	0.14 (0.13)	1.19 (1.23)	–	0.10 (0.37)	0.15 (0.21)
Dyadic relationship and duration						
Distance origin–asylum (log)	-0.09 (0.26)	-0.26 (0.18)	-7.16** (2.68)	-0.43 (0.57)	–	–
Origin and asylum contiguous	-0.01 (0.58)	$1.21^{\dagger}$ (0.73)	$-4.57^{\dagger}$ (2.48)	6.94* (2.88)	–	–
Composite cultural-tie index	0.19 (0.21)	0.20 (0.15)	-0.78 (0.95)	-1.84 (1.36)	–	–
Years since the dyad entered the panel	0.15 (0.36)	0.13 (0.23)	$-1.84^{\dagger}$ (1.07)	-3.71* (1.58)	–	–
Specification						
Fixed effects	Year	Year	Asylum × year	Origin × year	Dyad, year	Dyad, year
Dyad-years	3,622	3,828	225	285	917	1,977

Note: Logit coefficients, standard errors clustered by dyad in parentheses. Column 1 and 2 are pooled with year effects. Column 3 compares different origin groups facing the same asylum government in the same year, so every host-level characteristic is absorbed by construction. Column 4 compares hosts of the same origin in the same year, so every origin-level characteristic is absorbed. Columns 5 and 6 compare a dyad to itself over time, so time-invariant dyadic measures drop mechanically. A dash means the design absorbs the variable; textitsep. means the coefficient is a report of a cell with no counter-examples rather than an estimate. All models converged.

Table 38: Social and ethnic ties as predictors of exclusion policy

Tie measure	(1) Year FE	(2) + region	(3) + geography	(4) + observables	(5) Origin × year	(6) Asylum × year
Severe exclusion policy						
Common official language	0.49 (0.54)	0.25 (0.58)	0.30 (0.64)	0.35 (0.39)	1.15 (1.60)	5.25 (4.93)
Common spoken language	0.22 (0.56)	0.07 (0.57)	0.10 (0.66)	0.19 (0.38)	0.49 (1.41)	-0.27 (1.26)
Religious similarity, broad tradition	0.36 (0.30)	0.19 (0.25)	0.14 (0.28)	0.27 (0.22)	-2.10 (1.30)	-1.45 (1.07)
Religious similarity, denominational	-0.06 (0.12)	-0.46* (0.22)	-0.65** (0.22)	-0.38 (0.28)	-2.10 (1.40)	-3.41† (1.77)
Any transnational ethnic kin	1.24* (0.61)	0.70 (0.69)	0.49 (0.72)	0.49 (0.66)	-1.54 (2.19)	sep.
Kin share of origin population	0.21* (0.10)	0.09 (0.14)	0.02 (0.15)	0.02 (0.14)	-0.33 (0.55)	-1.08 (0.75)
Colonial or common-state history	0.65 (0.54)	0.69 (0.52)	0.53 (0.58)	1.03* (0.45)	2.07 (1.77)	-4.41 (3.35)
Summary index of all ties	0.35 (0.26)	0.19 (0.28)	0.11 (0.35)	0.23 (0.22)	-1.45 (1.26)	-0.53 (1.00)
Any exclusion policy						
Common official language	0.40 (0.32)	0.20 (0.32)	0.17 (0.34)	0.20 (0.30)	0.56 (0.81)	3.15*** (0.79)
Common spoken language	0.07 (0.31)	-0.11 (0.31)	-0.20 (0.34)	-0.12 (0.29)	-0.23 (0.81)	0.97 (0.99)
Religious similarity, broad tradition	0.42* (0.18)	0.27 (0.19)	0.15 (0.18)	0.33† (0.18)	0.76* (0.35)	0.07 (0.54)
Religious similarity, denominational	0.13 (0.13)	-0.16 (0.18)	-0.39† (0.20)	-0.31 (0.24)	-0.47 (0.45)	-1.29 (0.79)
Any transnational ethnic kin	1.33*** (0.35)	1.08* (0.45)	0.71† (0.41)	0.94* (0.37)	0.76 (0.74)	2.00** (0.66)
Kin share of origin population	0.32** (0.11)	0.25* (0.12)	0.10 (0.16)	0.13 (0.15)	0.09 (0.42)	1.31*** (0.34)
Colonial or common-state history	0.25 (0.34)	0.21 (0.33)	-0.15 (0.42)	-0.28 (0.41)	-0.31 (0.84)	-0.24 (1.26)
Summary index of all ties	0.42** (0.14)	0.30* (0.14)	0.10 (0.16)	0.22 (0.16)	0.38 (0.58)	0.95† (0.53)

Note: The outcome here is exclusion policy, not return. Each cell is a separate logit with one tie measure entered on its own alongside the stated conditioning set, standard errors clustered by dyad. The measures are almost time-invariant, so there are no dyad fixed effects; column 5 identifies the tie from variation across the hosts of one origin in one year and column 6 from variation across the origin groups in one host in one year. Column 3 adds distance and contiguity, column 4 adds the full pooled observable set. textitsep. marks a cell with no counter-examples rather than an estimate. Conclusions from Section refapp:ties, which asks which hosts refugees return from, do not transfer here.

Table 39: Onset risk of severe policy by displacement age and population size

Years since the dyad entered the panel	All sizes	Under 100,000 refugees	100,000 refugees or more
0–1	0.00% (0/20)	0.00% (0/11)	0.00% (0/9)
2–3	3.98% (9/226)	3.64% (6/165)	4.92% (3/61)
4–5	1.53% (4/262)	1.04% (2/193)	2.90% (2/69)
6–10	1.36% (9/663)	0.83% (4/481)	2.75% (5/182)
11–15	1.16% (8/687)	0.60% (3/497)	2.63% (5/190)
16–20	1.16% (7/604)	0.00% (0/464)	5.00% (7/140)
21+	0.79% (11/1400)	0.18% (2/1106)	3.06% (9/294)

Note: Each cell is the probability that a new severe exclusion policy begins, with onsets over dyad-years at risk in parentheses. The risk set is dyad-years whose previous CALENDAR year is observed in the sample and carried no severe policy, so 1988 and 1990 are never treated as consecutive. Age is measured from the dyad's first year of observed refugee presence in the full panel.

Table 40: Origin battle deaths and reported militarization as predictors of exclusion policy

Specification	Severe policy	Any policy	N (severe)	N (any)
Origin battle deaths (log, per SD)				
Raw, year FE	0.69** (0.25)	0.47*** (0.13)	3,985	4,113
+ log refugee stock	0.21 (0.19)	0.16 (0.10)	3,985	4,113
+ stock and origin GDP p.c.	0.10 (0.18)	0.11 (0.10)	3,792	3,916
+ distance and contiguity	0.11 (0.18)	0.10 (0.11)	3,745	3,868
Full pooled covariates	0.45 (0.51)	-0.36 (0.42)	3,493	3,613
Same host-year	1.72* (0.73)	0.24 (0.24)	221	733
Host and year FE	0.30 (0.25)	0.09 (0.13)	2,000	2,946
Dyad and year FE	0.42 (0.33)	0.11 (0.15)	893	1,959
Violence in the host's other neighbours, host and year FE	-0.06 (0.23)	0.05 (0.14)	2,055	3,006
Refugees reported militarized				
Raw, year FE	1.41* (0.55)	1.51*** (0.40)	4,121	4,345
Full pooled covariates	0.81† (0.47)	1.14** (0.40)	3,576	3,778
Same host-year	sep.	5.34*** (1.58)	229	751
Host and year FE	0.72 (0.47)	1.74*** (0.48)	2,136	3,145
Same origin-year	0.58 (1.22)	1.61† (0.86)	287	802
Dyad and year FE, contemporaneous	-0.05 (0.50)	1.27* (0.53)	1,022	2,198
Dyad and year FE, one-year lag	-0.72 (0.45)	0.25 (0.41)	946	2,062

Note: Each column is a progressively more demanding comparison. Logit coefficients on the row variable, standard errors clustered by dyad. Battle deaths are UCDP counts and begin in 1989, so those models run on the smaller sample the N columns report. The militarization flag is spliced from two non-overlapping World Refugee Survey coding series and is sometimes read from the same archival narrative as the exclusion-policy code itself, which is why the contemporaneous and lagged within-dyad estimates are shown side by side. textit-sep. marks a cell with no counter-examples rather than an estimate.

3.4 Conditional Effects of the Post-War Window

A more demanding version of the existing preference-based account implies that the association between peace at home and return should itself depend on how attractive remaining in asylum is. The main text reports the substance of that test; Table 41 reports the models in full, including the association of each condition with return outside the post-war window, the number of post-war dyad-years on each side of the split, and the fit statistics.

Each column interacts the post-war window with a binary condition divided at the sample median: asylum-country income, origin-country income, large UNHCR repatriation funding, and civil-liberties restrictions in the origin. The war-ending row gives the coefficient when the condition is absent, the interaction shows how it changes when the condition is present, and their sum gives the corresponding conditional coefficient. All columns include the full main specification and the continuous measure from which the binary condition is constructed.

The repatriation-funding column is the direct test of H3, which holds that the increase in return after a war ending should be larger where the resources to move are available. It is the one column that cannot be estimated on the full panel: the funding series is coded only for 1980–1999 and 2011–2020, so it runs on 1,378 of the 2,327 dyad-years the other three use, and only 42 of its post-war dyad-years carry large funding. Its interaction should be read with that support in mind rather than as an estimate the data can settle.

Asylum-country income is the one moderator whose interaction is precisely estimated, at 1.42 with a standard error of 0.25. In wealthier asylum countries a war ending has no detectable association with return, at 0.03; in poorer ones it is 1.44. That is the paper's clearest piece of conditional evidence and it points the way the argument predicts: peace translates into return where remaining in asylum is least attractive.

None of the other three interactions is distinguishable from zero – origin income -0.59, repatriation funding 0.66, civil liberties 0.63. Outside the post-war window, large repatriation funding on its own is associated with more return (0.69) and a freer origin with less (-0.69); those main effects are distinct from whether either condition moderates the post-war association.

Table 41: Interactions between the post-war window and permissive conditions

Term	(1) Poor asylum	(2) Rich origin	(3) Large funding	(4) Free origin
Window and condition				
Condition absent means	Rich asylum	Poor origin	No large funding	Repressive origin
War ending, condition absent	0.03 (0.24)	1.16*** (0.26)	0.57 (0.42)	0.84*** (0.21)
Condition alone, no war ending	0.21 (0.34)	0.79** (0.30)	0.69** (0.24)	-0.69† (0.40)
War ending x condition	1.42*** (0.25)	-0.59 (0.42)	0.66 (0.56)	0.63 (0.40)
War ending, condition present	1.44*** (0.22)	0.57† (0.32)	1.23** (0.48)	1.47*** (0.44)
Support and fit				
Post-war dyad-years, condition absent	62	87	53	90
Post-war dyad-years, condition present	73	48	42	45
All main-specification variables	Yes	Yes	Yes	Yes
Dyad and year fixed effects	Yes	Yes	Yes	Yes
Dyad-years	2,327	2,327	1,378	2,327
Pseudo R2	0.808	0.802	0.783	0.801

The repatriation-funding column is the direct test of H3 and it does not settle it. The sign is the one H3 predicts – 0.57 where large funding is absent against 1.23 where it is present – but the interaction is 0.66 with a standard error of 0.56, so the difference between the two is not distinguishable from zero. It also rests on 42 post-war dyad-years with large funding, and it is the one row in this table that is sensitive to the war-ending coding: under the two o_conflict codings tried on 2026-08-30 it runs from -0.31 to 1.00 (0_scripts/compare_ending_lists.R). The continuous funding measure, which is robust to all of them, is the one to rely on, and Section 2.4 reports it.

The full sample contains 204 post-war dyad-years. The splits are 62 and 73 for asylum income, 87 and 48 for origin income, 53 and 42 for repatriation funding, and 90 and 45 for civil liberties. Only the asylum-income column has enough support on both sides for its interaction to be read as a test; the other three should be read for the ordering of their pairs rather than for the difference within them.

3.5 Asylum-Side Shocks and Ongoing Conflict at the Origin

Table 42 examines whether asylum-side shocks are associated with return differently when conflict is still underway in the origin country. I collapse origin conflict into a single indicator and interact it with severe exclusion policy and both levels of asylum-country conflict.

None of the interactions is statistically distinguishable from zero; the smallest p-value is 0.10, and 0 of the three reach the 5% level. The asylum-side associations are therefore similar whether the origin is at peace or still at war. The largest difference is for low-intensity asylum conflict, which is more strongly associated with return during ongoing conflict at the origin, but the estimate is imprecise.

Table 42: Asylum-side shocks interacted with ongoing conflict at the origin

Term	Origin at peace	Interaction	p (interaction)	Origin at war
Severe exclusion policy	1.19** (0.36)	-0.15 (0.32)	0.64	1.04 (0.18)
High-intensity asylum conflict	0.57 (0.36)	-0.05 (0.49)	0.92	0.52 (0.38)
Low-intensity asylum conflict	0.38 (0.31)	0.49 (0.30)	0.10	0.87 (0.19)

Note: Two models, both the specification of column 1 of the main table with the two origin-conflict indicators collapsed into a single indicator for the origin being in armed conflict at either intensity. The first row comes from a model interacting that indicator with severe exclusion policy; the second and third from a model interacting it with both asylum-conflict intensities at once, so those two rows are mutually adjusted. *Origin at peace* is the asylum-side coefficient when the origin is not in conflict, *Interaction* is what ongoing conflict adds, and *Origin at war* is their sum, with a standard error formed from the covariance of the two terms. PPML with the log stock at risk as offset, dyad and year fixed effects, standard errors clustered by dyad. Stars are omitted from the final column because it is a linear combination rather than an estimated parameter.

4 Literature Review

Figure 1 is based on a systematic search of nineteen journals from 2006 through 2026:

- **Political science** (twelve): *American Political Science Review*, *American Journal of Political Science*, *Journal of Politics*, *Comparative Political Studies*, *British Journal of Political Science*, *World Politics*, *International Organization*, *International Studies Quarterly*, *Journal of Peace Research*, *Journal of Conflict Resolution*, *Political Science Research and Methods*, and *PNAS.*
- **Migration and refugee studies** (five): *Journal of Refugee Studies*, *International Migration Review*, *Journal of Ethnic and Migration Studies*, *International Migration*, and *Comparative Migration Studies.*
- **Development economics** (two): *World Development* and the *Journal of Development Economics.*

I supplemented the journal search by tracing citations to and from core studies, reviewing the 2024 World Bank–UNHCR Joint Data Center review of refugee return, and searching for combinations of *refugee return*, *repatriation*, *return intentions*, *IDP return*, and *internally displaced return*.

I conducted an additional search focused on qualitative research. It combined return and repatriation terms with *interview*, *ethnography*, *focus group*, *life history*, *decision-making*, *home*, *belonging*, *waiting*, *refusal*, and *circular*, together with publisher-site searches and citation tracing from key qualitative studies. The qualitative search was targeted rather than a second journal-by-journal census.

A study is included if it (a) reports original empirical evidence, (b) examines refugees or internally displaced people displaced by conflict, and (c) analyzes qualitative return decisions, stated return intentions or preferences, or realized return. I exclude reviews, conceptual and normative work, studies of state policy or pressure to leave, and studies of post-return reintegration unless return itself is an outcome. Working papers and online-first articles are included.

I code three things. *Outcome* distinguishes qualitative evidence on return decisions, stated intentions or preferences, and realized return. *Scope* distinguishes studies of one displacement context from multi-context or cross-national studies. The first category includes studies of several origin populations in one host and of one origin population across several hosts, provided that the study concerns a single displacement context. Only studies spanning several distinct displacement contexts are coded as cross-national. *Population* is shown by marker shape: internally displaced people, Ukrainian refugees, Syrian refugees, and other populations.

Table 43: Studies plotted in Figure 1 of the paper

Outlet	Study	Year	Title	Population	Scope	Outcome	Marker
Journal of Peace Research 61(6)	Costalli, Stefano (2024)	2024	Do UN Peace Operations Help Forcibly Displaced People?	Multiple	Multi-context	Realized return	Other refugees
J Refugee Studies	Zakirova, Komila (2021)	2021	The Road Back Home is Never Long: Refugee Return Migration	Multiple	Multi-context	Realized return	Other refugees
Peace Economics, Peace Science and Public Policy	Arias, María Alejandra (2014)	2014	The Desire to Return during Civil War: Evidence for Internally Displaced Populations in Colombia	Colombian	One context	Intentions	IDPs
Development in Practice 30	Collado, Zenaida C. (2020)	2020	Living in Displacement Contexts: Return Intentions among Internally Displaced Persons in Marawi, Philippines	Marawi IDPs	One context	Intentions	IDPs
International Migration 64(3)	Malashkhia, Mariam (2026)	2026	Integration and Return Intentions among Displaced Individuals	Georgian	One context	Intentions	IDPs

Table 43: Studies plotted in Figure 1 of the paper *(continued)*

Outlet	Study	Year	Title	Population	Scope	Outcome	Marker
European Journal of Psychotraumatology 14(2)	Thompson, Peter Onah (2023)	2023	Posttraumatic Stress Moderates Return Intentions: A Factorial Survey Experiment with Internally Displaced Persons in Nigeria	Conflict-displaced Nigerians	One context	Intentions	IDPs
JEMS	Di Saint Pierre, Francesca (2015)	2015	Return Wishes of Refugees in the Netherlands: The Role of Integration, Host National Identification and Perceived Discrimination	Multiple (Afghan, Iranian, Iraqi, Somali)	One context	Intentions	Other refugees
Comparative Migration Studies	Tefera, Tadele Muche (2026)	2026	Conflict, Drought, and Displacement: Determinants of Return Intentions among South Sudanese Refugees	South Sudanese	One context	Intentions	Other refugees
Population, Space and Place 18(1)	Zimmermann (2012)	2012	Understanding Repatriation: Refugee Perspectives on the Importance of Safety, Reintegration, and Hope	Somali and Afghan	One context	Intentions	Other refugees
International Migration	Özer, Yeşim (2024)	2024	Back for Good? Return Aspirations of Immigrants in the Netherlands	Multiple immigrant groups	One context	Intentions	Other refugees
IMR	Al Husein, Nawras (2023)	2023	Determinants of Intended Return Migration among Refugees: A Comparison of Syrian Refugees in Germany and Turkey	Syrian	One context	Intentions	Syrian
BJPS	Alrababah, Ala (2023)	2023	The Dynamics of Refugee Return: Syrian Refugees and Their Migration Intentions	Syrian	One context	Intentions	Syrian
Political Studies	Gerver, Mollie (2025)	2025	Refugee Resettlement and Preferences	Syrian	One context	Intentions	Syrian
APSR	Ghosn, Faten (2021)	2021	The Journey Home: Violence, Anchoring, and Refugee Decisions to Return	Syrian	One context	Intentions	Syrian
Journal of Population Economics 36	Hannafi, Cyrine (2023)	2023	Social Integration of Syrian Refugees and Their Intention to Stay in Germany	Syrian	One context	Intentions	Syrian
Journal of Immigrant and Refugee Studies	Kaya, Serdar (2020)	2020	Prospects of Return: The Case of Syrian Refugees in Germany	Syrian	One context	Intentions	Syrian
Journal of Immigrant and Refugee Studies	Kayaoglu, Ayselin (2021)	2021	Return Aspirations of Syrian Refugees in Turkey	Syrian	One context	Intentions	Syrian

Table 43: Studies plotted in Figure 1 of the paper *(continued)*

Outlet	Study	Year	Title	Population	Scope	Outcome	Marker
Journal of International Migration and Integration 24(4)	Özkan, Zafer (2023)	2023	Intentions to Return and Migrate to Third Countries: A Socio-Demographic Investigation among Syrians in Turkey	Syrian	One context	Intentions	Syrian
CESifo Working Paper 12118	Adema, Joop (2025)	2025	What Drives Refugees' Return after Conflict?	Ukrainian	One context	Intentions	Ukrainian
Journal of Ethnic and Migration Studies	Hernes, Vilde (2024)	2024	Where Does the Future Lie? Initial Aspirations for Return among Newly Arrived Ukrainian Refugees in Norway	Ukrainian	One context	Intentions	Ukrainian
International Migration 64(3)	Kochaniak, Katarzyna (2026)	2026	Return Intentions under Temporary Protection: Theoretical Perspectives on Migrant Households	Ukrainian households	One context	Intentions	Ukrainian
PLOS ONE 18	Kohlenberger, Judith (2023)	2023	High Self-Selection of Ukrainian Refugees into Europe: Evidence from Krakow and Vienna	Ukrainian	One context	Intentions	Ukrainian
IBS Working Paper 01/2025	Lewandowski, Piotr (2025)	2025	The Role of Job Task Routinization in Shaping Return Intentions: Evidence from Ukrainian War Refugees in Poland	Ukrainian	One context	Intentions	Ukrainian
Population, Space and Place	Novotný, Josef (2026)	2026	Drivers of Return Plans among Ukrainian War Refugees in Czechia: Challenging Common Assumptions and Refining the Theory	Ukrainian	One context	Intentions	Ukrainian
Journal of Ethnic and Migration Studies, online first	Olivo Rumpf, Karelis (2026)	2026	The Interplay between Origin- and Receiving-Country Conditions: Experimental Evidence on Ukrainian Refugees' Return Intentions	Ukrainian	One context	Intentions	Ukrainian
International Migration Review, online first	Perelli-Harris, Brienna (2025)	2025	The Uncertainty of Forced Displacement: How Language and Violence Shaped Displacement Trajectories during Russia's Invasion of Ukraine	Ukrainian	One context	Intentions	Ukrainian

Table 43: Studies plotted in Figure 1 of the paper *(continued)*

Outlet	Study	Year	Title	Population	Scope	Outcome	Marker
International Migration 63(5)	Saijo, Harunobu (2025)	2025	Which Policy Attributes Affect Assisted Voluntary Return and Reintegration Uptake among Ukrainian Evacuees in Japan?	Ukrainian	One context	Intentions	Ukrainian
Journal of Behavioral and Experimental Economics	Vakhitov, Volodymyr (2025)	2025	Return Intentions of Ukrainian Refugees: The Role of National Identity and Pride	Ukrainian	One context	Intentions	Ukrainian
International Migration 62	van Tubergen, Frank (2024)	2024	Return Intentions among Ukrainian Refugees in Europe: A Cross-National Study	Ukrainian women	One context	Intentions	Ukrainian
J Refugee Studies	Žíla, Ondřej (2026)	2026	'Partial Returns': Displacement, Mobility, and Translocal Connections in Post-war Bosnia and Herzegovina	Bosnian	One context	Qualitative	IDPs
Journal of Refugee Studies 34(1)	Antwi-Boateng (2021)	2021	The Dilemma of Ghana-Based Liberian Refugees and the Challenges of Repatriation and Integration	Liberian	One context	Qualitative	Other refugees
Journal of Ethnic and Migration Studies	Durodola (2026)	2026	Return to What? Reimagining Home and Belonging after Conflict and Refugee Status Cessation among Liberians in Nigeria	Liberian	One context	Qualitative	Other refugees
International Migration	Eastmond, Marita (2006)	2006	Transnational Returns and Reconstruction in Post-war Bosnia and Herzegovina	Bosnian	One context	Qualitative	Other refugees
J Refugee Studies	Hardgrove, Abby (2009)	2009	Liberian Refugee Families in Ghana: The Implications of Family Demands and Capabilities for Return to Liberia	Liberian	One context	Qualitative	Other refugees
International Migration 52(6)	Harpviken, Kristian Berg (2014)	2014	Split Return: Transnational Household Strategies in Afghan Repatriation	Afghan	One context	Qualitative	Other refugees
Journal of Eastern African Studies 4(1)	Kaiser (2010)	2010	Dispersal, Division and Diversification: Durable Solutions and Sudanese Refugees in Uganda	Sudanese	One context	Qualitative	Other refugees
African Human Mobility Review 6(2)	Khan (2020)	2020	Is Voluntary Repatriation the Preferred Durable Solution? The View of Refugees in South Africa	Multiple refugee origins	One context	Qualitative	Other refugees

Table 43: Studies plotted in Figure 1 of the paper *(continued)*

Outlet	Study	Year	Title	Population	Scope	Outcome	Marker
Journal of Refugee Studies	Krupalija, Mustafa (2026)	2026	Rebuilding Home on Hot Ground: Forced Migration and Return to Srebrenica	Bosnian	One context	Qualitative	Other refugees
Journal of Refugee Studies 19(4)	Muggeridge (2006)	2006	Back Home? Refugees' Experiences of their First Visit back to their Country of Origin	Multiple refugee origins	One context	Qualitative	Other refugees
J Refugee Studies	Nandi, Sarah (2025)	2025	"We were together and we had our own family in each other": Refusing Repatriation and Forging Gendered Belonging as Hijra Refugees in Kolkata	Bangladeshi (Hijra)	One context	Qualitative	Other refugees
Transnational Social Review 6(1-2)	Omata (2016)	2016	Home-making during Protracted Exile: Diverse Responses of Refugee Families in the Face of Remigration	Liberian	One context	Qualitative	Other refugees
JEMS	Omata, Naohiko (2013)	2013	The Complexity of Refugees' Return Decision-Making in a Protracted Exile: Beyond the Home-Coming Model and Durable Solutions	Liberian	One context	Qualitative	Other refugees
International Migration 55(5)	Porobić, Selma (2017)	2017	Daring Life-Return Projects to Post-Dayton Bosnia and Herzegovina	Bosnian returnees	One context	Qualitative	Other refugees
International Migration	Stefansson, Anders H. (2006)	2006	Homes in the Making: Property Restitution, Refugee Return, and Senses of Belonging in a Post-war Bosnian Town	Bosnian	One context	Qualitative	Other refugees
Geoforum 43(1)	Tete (2012)	2012	'Any Place Could Be Home': Embedding Refugees' Voices into Displacement Resolution and State Refugee Policy	Liberian	One context	Qualitative	Other refugees
Comparative Migration Studies	Van Houte, Marieke (2016)	2016	Deconstructing the Meanings of and Motivations for Return: An Afghan Case Study	Afghan	One context	Qualitative	Other refugees
Refugee Survey Quarterly 45(1)	Diab (2026)	2026	Prolonging the Temporary: Female-Headed Syrian Refugee Households on Displacement, Return and Waiting post-Assad	Syrian women heading households	One context	Qualitative	Syrian

Table 43: Studies plotted in Figure 1 of the paper *(continued)*

Outlet	Study	Year	Title	Population	Scope	Outcome	Marker
International Migration 64(3)	Dinçşahin (2026)	2026	From Protracted Liminality to Fractured Homecoming: Intrafamilial Bargaining and the Limits of Voluntary Repatriation for Syrians in Turkey	Syrian	One context	Qualitative	Syrian
Journal of Refugee Studies 39(1)	Enna (2026)	2026	'The Threat of Forced Return Is the Government's Last Resort': Structural, Cultural, and Direct Violence towards Syrian Refugees in Lebanon	Syrian	One context	Qualitative	Syrian
International Migration Review, Online-First	Masterson (2026)	2026	Circular Refugee Migration: Understanding Protracted Displacement Beyond a Refugee-Returnee Binary	Syrian	One context	Qualitative	Syrian
International Migration Review 57(4)	Müller-Funk, Lea (2023)	2023	I Will Return Strong: The Role of Life Aspirations in Refugees' Return Aspirations	Syrian	One context	Qualitative	Syrian
Comparative Migration Studies 14:32	Kokonowskyj, Larissa (2026)	2026	The Role of Family Dynamics in (Return) Migration Aspirations among Ukrainian Protection Holders in Germany	Ukrainian	One context	Qualitative	Ukrainian
Comparative Migration Studies 13:66	Ash, Konstantin (2025)	2025	Who Leaves and Who Returns? IDPs and Returnees after the Russian Invasion of Ukraine	Ukrainian	One context	Realized return	IDPs
JCR	Bove, Vincenzo (2025)	2025	What it Takes to Return: UN Peacekeeping and the Safe Return of Displaced People	South Sudanese	One context	Realized return	IDPs
AJPS	Camarena, Kara Ross (2020)	2020	When Do Displaced Persons Return? Postwar Migration among Christians in Mount Lebanon	Lebanese (Christians)	One context	Realized return	IDPs
International Migration 55(5)	Joireman, Sandra F. (2017)	2017	Ethnic Violence, Local Security and Return Migration: Enclave Communities in Kosovo	Kosovar displaced	One context	Realized return	IDPs
J Refugee Studies	Stefanovic, Djordje (2015)	2015	Home is Where the Heart Is? Forced Migration and Voluntary Return in Turkey's Kurdish Regions	Kurdish	One context	Realized return	IDPs

Table 43: Studies plotted in Figure 1 of the paper *(continued)*

Outlet	Study	Year	Title	Population	Scope	Outcome	Marker
Journal of Conflict Resolution 69(1)	Tellez, Juan F. (2025)	2025	Social Cohesion, Economic Security, and Forced Displacement in the Long Run: Evidence from Rural Colombia	Colombian IDPs	One context	Realized return	IDPs
Working Paper	Weber, Sigrid (2022)	2022	Property Rights and Post-conflict Recovery: Theory and Evidence from IDP Return Movements in Iraq	Iraqi (Yazidi and Sunni Muslim)	One context	Realized return	IDPs
Journal of Refugee Studies 23(1)	Huttunen (2010)	2010	Sedentary Policies and Transnational Relations: A 'Non-sustainable' Case of Return to Bosnia	Bosnian	One context	Realized return	Other refugees
J Refugee Studies	Klinthäll, Martin (2007)	2007	Refugee Return Migration: Return Migration from Sweden to Chile, Iran and Poland 1973-1996	Chilean, Iranian, Polish	One context	Realized return	Other refugees
International Migration 55(5)	Kutlay, Mustafa (2017)	2017	The Turks of Bulgaria: An Outlier Case of Forced Migration and Voluntary Return	Turks of Bulgaria	One context	Realized return	Other refugees
Journal of Ethnic and Migration Studies 51(1)	Omata (2025)	2025	Returning to Fund Refugeehood: Dispersal and Survival between Uganda and South Sudan	South Sudanese	One context	Realized return	Other refugees
International Migration 55(5)	Stefanovic, Djordje (2017)	2017	Peaceful Returns: Reversing Ethnic Cleansing after the Bosnian War	Bosnian	One context	Realized return	Other refugees
Middle Black Sea Journal of Communication Studies 11	Altınışık (2026)	2026	From Forced Migration to Voluntary Return: Psychosocial Factors Influencing Syrian Refugees' Decisions to Return to Their Homeland After the War	Syrian	One context	Realized return	Syrian
J Development Economics	Beaman, Lori (2022)	2022	When Do Refugees Return Home? Evidence from Syrian Displacement in Mashreq	Syrian	One context	Realized return	Syrian
International Review of Sociology	Bimay (2026)	2026	Between Hope and Hesitation: Perceptions and Experiences of Refugee Women Returning from Turkey to Syria	Syrian women	One context	Realized return	Syrian
Working Paper (EBRD)	Adema, Joop (2026)	2026	Refugee Return	Ukrainian	One context	Realized return	Ukrainian

Note: Drawn from the literature audit of 23 August 2026. Scope and Outcome are the two axes of that figure; Marker is the population class the figure shows by shape. Inclusion follows the Figure decision and Strict figure columns of the audit's two inventory tabs; two studies logged there as out of scope are excluded.